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1 Backround

For a function $f : \mathbb{R}^m \to \mathbb{R}^n$ the Jacobi-Matrix at point x_0 is given by

$$\frac{\partial}{\partial x} f(x_0) := \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x_0) & \cdots & \frac{\partial}{\partial x_m} f_1(x_0) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_n(x_0) & \cdots & \frac{\partial}{\partial x_m} f_n(x_0) \end{pmatrix} \in \mathbb{R}^{n \times m}$$

and the derivative of f at x_0 is then the mapping

$$f'(x_0) : \mathbb{R}^m \to \mathbb{R}^n, x \mapsto \frac{\partial}{\partial x} f(x_0) \cdot x.$$

Definition 1.1

We call the mapping

$$\phi : (0, \infty) \times \mathbb{R}/(2\pi\mathbb{R}) \to \mathbb{R}^2, \quad (r, \theta) \mapsto \phi(r, \theta) := (r \cos(\theta), t \sin(\theta))$$

polar coordinates.

It holds at point (r_0, θ_0) that

$$|det(\frac{\partial}{\partial x} \phi(r_0, \theta_0))| = r_0.???$$

Definition 1.2

For a set M and an equivalenz relation $\sim$ we call

$$[m] := \{n \in M \mid n \sim m\} \subset M$$

an equivalenz class. We call $M/\sim$ the set of equivalenz classes from M wrt. $\sim$. It is also called quotient set.

Note that

$$M = \bigcup_{[a] \in M/\sim} [a]$$

and that the mapping

$$M \to M/\sim, \quad m \mapsto [m]$$

is canonical, meaning that every element in M is mapped to a unique equivalenz class.

Remark 1.1

One can show that the mapping
$$\mathbb{R} \to \mathbb{R}/(2\pi\mathbb{Z})$$
is a group homomorphism and its kernel is $2\pi\mathbb{Z}$. Further, one can show that $\mathbb{R}/(2\pi\mathbb{Z})$ is isomorphic to $S^1 := \{x \in \mathbb{R}^2 \mid ||x||_2 = 1\}$.

Theorem 1.3: Inverse function Theorem

Let X, Y be two Banach spaces and $U \subset X$ open and $f : U \to Y$. If f' at $x_0 \in U$ is invertible and continuous then it follows that $\exists V \subset U$ and $W \subset Y$ open subsets of x_0 and $f(x_0)$ respectively such that
$$f\big|_V : V \to W,$$
is bijective and $\exists f^{-1} : W \to V$ is differentiable. Further,
$$(f^{-1})'(y) = (f'(f^{-1}(y)))^{-1}, \quad y \in W.$$

Theorem 1.4: Implicit Function Theorem

Let X and Y be Banach spaces, $U \subset X \times Y$ open, $f : U \to Y$ differentiable. If f' in $(x_0, y_0) \in U$ continuous and $\frac{\partial}{\partial y} f(y_0, y_0)$ invertible, then there exists $V \subset X, W \subset Y, O \subset U$ open neighborhoods of $x_0, f(x_0, y_0)$ and (x_0, y_0) repectively and $g : V \times W \to Y$ differentiable such that
$$\forall z \in W : \quad f^{-1}(\{z\}) \cap O = Graph(g(\cdot, z))$$

Definition 1.5

A set X and a system of sets $\mathcal{O} \subseteq \mathcal{P}(X) = \{A \mid A \subseteq X\}$ with

- $\emptyset, X \in \mathcal{O}$
- $\mathcal{O}' \subset \mathcal{O} \Rightarrow \bigcup_{U \in \mathcal{O}'} U \in \mathcal{O}$
- $U, V \in \mathcal{O} \Rightarrow U \cap V \in \mathcal{O}$

is called topology on X and $(X, \mathcal{O})$ is called topological space.

Definition 1.6

We call a topological space $(X, \mathcal{O})$ second countable if there exists a family $\mathcal{B} \subset \mathcal{O}$, that is a basis of $\mathcal{O}$, i.e.
$$\forall U \in \mathcal{O} \forall x \in U \exists B \in \mathcal{B} : \quad x \in B \subset U.$$

A topological space $(X, \mathcal{O})$ is called Hausdroff, if

$$\forall x, y \in X \text{ with } x \neq y \exists U_x, U_y \text{ open areas around } x \text{ and } y \text{ repectively:} \quad U_x \cap U_y = \emptyset.$$

Definition 1.7

A C^r-Atlas of dimension m on a Set X is a family of Maps $(U_i, \varphi_i)_{i\in\mathbb{N}}$ with $U_i \subset X$ and $\bigcup_{i\in\mathbb{N}} U_i = X$ and homöormorphismen $\varphi : i : U_i \to \Omega \subset \mathbb{R}^n$ open such that for all $i, j \in \mathbb{N}$ with $U_i \cap U_j \neq \emptyset$ we have that

$$\varphi_i \circ \varphi_j^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$$

is in $C^{1,r} := \{u \in C^r \mid D^\alpha u \in C^r, |\alpha| \leq r\}$.

Definition 1.8

A $C^{r,1}$ Manifold X of dimension m is a Hausdorff, second countable topological space with a at most $C^{1,r}$-Atlas $\{(U_i, \varphi_i)_{i\in\mathbb{N}}\}$ where $\varphi_i : U \to \Omega \subset \mathbb{R}^m$ are Homöomorphism are and

Definition 1.9

A topological space $(X, \mathcal{O})$ is called connected if the only subsets of X, that are open and closed are $\emptyset$ and X itself. Local Connected means that for every $x \in X$ every ball around x contains a connected area of X.

Theorem 1.10

1. Non empty subsets X of $\mathbb{R}$ are connected $\iff$ X is a bounded or unbounded intervall.
2. $\mathbb{R}$ is connected and also locally connected.

Proof. 1. Let $X \subset \mathbb{R}$ be connected and not empty. Assume that there exists an intervall in X, where there is an element, that is not in X, i.e.

$$\exists a, b \in X \text{ with } a < b : \quad x \in (a, b) \text{ with } x \notin X.$$

Then we have that

$$U := (-\infty, x) \cap X = (-\infty, x] \cap X, \quad \text{and} \quad V := (x, \infty) \cap X = [x, \infty) \cap X.$$

Both equalities above hold, because we assumed that $x \notin X$. Because U and V are open and closed at the same time we get that we can make a disjoint decomposition of open sets: $X = U \cup V$, which leads to a contradiction, because X is no longer connected!
This yields that the above constructed $x \notin X$ cannot exist. Thus, all elements in $(a, b) \subset X$ have to be in X.
2. Global:
Choose $\inf A = -\infty$ and $\sup A = \infty$. Then

$$\forall x \in (\inf X, \sup X) \exists a, b \in X : \quad x \in (a, b)$$

i.e. its always contained in an interval. This yields

$$(\inf X, \sup X) \subset X \subset [\inf X, \sup X]$$

which leads to the conclusion that X has to be either $(\inf X, \sup X)$, $[\inf X, \sup X)$, $(\inf X, \sup X]$ or $[\inf X, \sup X]$.
2. Local:

We can use that a interval I is connected $\iff$ $I = A \cup B$ of non empty subsets where at least one is not open.
Let $a \in A, b \in B$ with $a < b$ and define $c := \inf B \cap [a, b]$. If A is open then $c \in B$ and thus $[a, c) \subset A$. Because $c \notin A$ we get that B is not open and thus I is connected. □

Definition 1.11

A mapping $f : V \to U$ between two open vector spaces is called Diffemorphism if

- Bijective
- everywhere continuous differentiable
- the inverse f^{-1} everywhere continuous differentiable.

Definition 1.12

A Homöorphism is a bijective mapping φ that is continuous and its inverse φ^{-1} is continuous.

Definition 1.13

Let X be a topological space, then we call a Homöormorphism $\phi : U \to V$, where $U \subset X$ open and $V \subset \mathbb{R}^n$ open, a Map. We call U definition space and n the dimension of the Map.

Intuition: A Map assigns to every point $x \in U$ a unique coordinate $\phi(x) := (x_1, ..., x_n) \in V \subset \mathbb{R}^n$ and for every coordinate we can reconstruct a unique point.

Definition 1.14

On a topological space $(X, \mathcal{O})$ we call two maps $\phi_1 : U_1 \to V$ and $\phi_2 : U_2 \to V$, with $U_1, U_2 \subset X$ open and $V \subset \mathbb{R}^n$ open compatible, if

$$(\phi_1 \circ \phi_2^{-1})\big|_{U_1 \cap U_2} \text{ is a diffeomorphism, i.e. bijective, continuous and its inverse is continuous.}$$

The reasoning behind the diffeomorphism (???) is that we dont want information to be lost.

Remark 1.2

Note that $(\phi_1 \circ \phi_2^{-1})^{-1} = \phi_2 \circ \phi_1^{-1}$, i.e. no matter the order $\phi_1 \circ \phi_2^{-1}$ and $\phi_2 \circ \phi_1^{-1}$ we get a continuous bijective mapping from on of the definition spaces U_1 to the other U_2, because

$$\phi_1 \circ \phi_2^{-1} : U_1 \to U_2 \text{ and } \phi_2 \circ \phi_1^{-1} : U_2 \to U_1.$$

Definition 1.15

For a topological space $(X, \mathcal{O})$ we call a family of maps $\phi_i : U_i \to \mathbb{R}^n$ such that $X \subset \bigcup_{i=1}^n U_i$ is a cover of X an Atlas, if the maps are pairwise compatible.

Remark 1.3

We only want to consider Hausdorff spaces, because ???.
The topological space

$$X := \bigcup_{i \in \mathbb{N}} O_i \cup (\{0^*\} \cup 0) \cup (\{0^*\} \cup (O \setminus \{0\}))$$

where $\bigcup_{i \in \mathbb{N}} O_i$ is an open cover of $\mathbb{R}$ and 0^* is some element, that is not in $\mathbb{R}$, and $O \subset \mathbb{R}$ is an open area around 0. The two maps

$$\phi_1 : \mathbb{R} \setminus \{0^*\} \to \mathbb{R}, x \mapsto 1 \text{ and } \phi_2 : (\mathbb{R} \setminus \{0^*\}) \cup \{0^*\} \to \mathbb{R}, x \mapsto \mathbf{1}_{\mathbb{R} \setminus \{0\}} + 0\mathbf{1}_{\{0^*\}}$$

are an Atlas of X, as their definition space is a cover of X.
This is space, that we do not want to consider, because its not a Hausdorff space, because for arbitrary $V \subset \mathbb{R}$ open area of 0 and $W \subset X$ open area of 0^* there exists $O \subset \mathbb{R}$ open area of 0 with $0^* \in O$.

Definition 1.16: Manifold

We call a topological Hausdorff space with an Atlas differentiable Manifold.

The Idea here is that using an open cover of the topological space X, for each open set, we map using a map to an open subset of euklidean space, i.e. this what we call locally euklidean.

CHAPTER 1

FIRST ORDER PDES

Definition 0.1

On $\Omega \subset \mathbb{R}^n$ the function $F : \Omega \times \mathbb{R}^{k+1} \to \mathbb{R}$ satisfying the equation

$$F(D^k u(x), ..., Du(x), u(x), x) = 0, \quad k \in \mathbb{N}_0^n$$

is called partial differential equation of order $|k| := \sum_{i=1}^n k_i$ where $D^k u := (\partial^{k_1} \cdots \partial^{k_n})u$ are all partial derivatives of at most order k.

Definition 0.2

u is called classical solution of the PDE if it is k-times differentiable and satisfies the PDE.

Remark 0.1

Some Notation:
We write $\dot{u} := \frac{\partial}{\partial t}u$ and if $u : \Omega \times \mathbb{R}_+ \to \mathbb{R}$, where $\Omega \subset \mathbb{R}^d$ we write $(1, b)\nabla u = \dot{u} + b\nabla_x u$.
We also often write $b\nabla u := \langle b, \nabla u \rangle$.

1 Homogeneous Transport

For $u : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$, $b \in \mathbb{R}^n$ one of the simplest partial differential equations is the homogeneuos transport equation

$$\dot{u} + b \cdot \nabla u = 0,$$

where $\cdot$ denotes in the entire script the dot product and ∇ the sum of all partial derivatives wrt. x only. We begin by assuming that u is differentiable, then define

$$\forall (x_0, t_0) \in \mathbb{R}^n \times \mathbb{R} : \quad z(s) := u(x_0 + sb, t_0 s),$$

is a differentiable function wrt. $s \in \mathbb{R}$. The first derivative is

$$z'(s) = b\nabla_x u(x_0 + sb, t_0 + s) + \underbrace{\frac{\partial}{\partial t} u(x_0 + sb, t_0 + s)}_{=\dot{u}(x_0+sb,t_0+s)} \cdot 1 = b \cdot \nabla(x_0 + sb, t_0 s) + \dot{u}(x_0 + sb, t_0 + s) \overset{Def.\ PDE}{=} 0.$$

Therefore, we conclude that u is constant along all lines in the direction $(b, 1)$. Because these lines make up the entire space $\mathbb{R}^n \times \mathbb{R}$ u is completely determined by these lines.

Definition 1.1: Initial Value Problem (Cauchy Problem)

We seek to find a solution $u : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ of the linear homogenuous transport equation

$$\dot{u} + b \cdot \nabla u = 0, \quad u(x, 0) = g(x)$$

where $g : \mathbb{R}^n \to \mathbb{R}$ is some function.

The one dimensional homogeneous transport equation $\dot{u} + \nabla f(u) = 0$ is also called conservation law and f is the flux (stems from inflow and outflow) function.

For initial values (x_0, t_0) we can now solve the PDE, this is because the solution u is constant on $x(t)$ and thus

$$u(x(t), t) = u(x_0 + tb, t) \stackrel{const}{=} u(x_0, 0) = g(x_0) = g(x - tb)$$

where we used that $x = x_0 + tb$ is equivalent to $x_0 = x - tb$. Thus, if g is differentiable on $\mathbb{R}^n$ then the above solves the PDE. If g is not differentiable there does not exist a solution in a classical sense! Now lets generalize.

2 Inhomogenous linear transport Equation

For $u, f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$, $b \in \mathbb{R}^n$ we call

$$\dot{u} + b \cdot \nabla u = f,$$

linear inhomogeneous transport equation. In a similar way as before, we define

$$\forall (x_0, \underbrace{t_0}_{=0}) \in \mathbb{R}^n \times \mathbb{R} : \quad z(s) := u(x_0 + sb, s),$$

which is differentiable wrt. s and satisfies

$$z'(s) = b \cdot \nabla u(x_0 + sb, s) + \dot{u}(x_0 + sb, s) \stackrel{PDE\ Def}{=} f(x_0 + sb, s).$$

Note that the right hand side is only a function of s. Moreover, $z(0) = u(x_0, 0) = g(x_0)$ is known. This is exactly an ODE. THe solution might not be constant along the lines $(x_0 + sb, s)$, but it is only dependend on f. I.e. for every initial value $(x_0, 0)$ we have an ODE along the parallel lines. This means that u is completely determined by them. We can solve this ODE in the same way as in ODE theory:

$$\begin{aligned} u(x, t) = z(t) = z(0) + \int_0^t z'(s) ds &= g(x_0) + \int_0^t f(x_0 + sb, s) ds \\ &= g(x - tb) + \int_0^t f(x - tb + sb, s) ds. \end{aligned}$$

3 Scalar Conservation Law

We now consider a one dimensional non-linear transport equation called scalar conservation law: Find $u : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ such that

$$\dot{u}(x, t) + \frac{\partial}{\partial x} f(u(x, t)) = \dot{u}(x, t) + f'(u(x, t)) \frac{\partial}{\partial x} u(x, t) = 0,$$

for a smooth function $f : \mathbb{R} \to \mathbb{R}$. For any compact intervall $[a, b] \subset \mathbb{R}$ we have

$$\frac{d}{dt}\int_a^b u(x,t) = \int_a^b \dot{u}(x,t) = -\int_a^b \frac{\partial}{\partial x} f(u(x,t))dx = f(u(a,t)) - f(u(b,t)).$$

This means that the change of "mass" on the intervall $[a, b]$ over time is soley dependend on the boundary $\partial[a, b] = \{a, b\}$.
We now want to apply the method of characteristics here. However, now we cannot presume that the characteristic curves $(x(s), s)$ are parallel strate lines. Define $z(s) := u(x(s), t(s))$, then

$$z'(s) = \frac{\partial}{\partial x} u(x(s), t(s))x'(s) + \frac{\partial}{\partial t} u(x(s), t(s))t'(s).$$

Choose $x'(s) := f'(u(x(s), t(s)))$ and $t'(s) = 1$, then we have

$$z'(s) = \frac{\partial}{\partial x} u(x(s), s)f'(u(x(s), s)) + \dot{u}(x(s), s) = 0,$$

Thus, along these curves the solution is constant. We now need to answer the following questions

- How to determine z?
  We assume the characteristics begin at $(x_0, 0)$, i.e. $x(0) = x_0$. Because we have shown that z is constant along $(x(s), s)$ it follows
  $$z(s) = u(x(0), 0) = u(x_0, 0) = g(x_0),$$
  which answers the question.

- Does $x(s)$ exists?
  First, note that
  $$x'(s) = f'(u(x(s), s)) = f'(z(s)) = f'(g(x_0)),$$
  which implies that
  $$x(s) = \underbrace{x(0)}_{=x_0} + sf'(g(x_0)) = x_0 + sf'(g(x_0)).$$
  Thus, $x(s)$ exists.

Is u uniquely determined by the characteristic lines? This is a difficult question, whose answer will be answered in the remainder of this section. First, see that if $\mathbb{R} \to \mathbb{R}, s \mapsto x(s)$ is surjective we know that every point (x, t) has at least one line going through assigning at least one initial value. In that case u would be completly determined by the initial values, however not unqiue, because we can have two initial values x_1 and x_2 such that

$$g(x_1) \overset{constant}{=} u(x_1 + sf'(g(x_1)), t) = u(x_2 + sf'(g(x_2)), t) \overset{constant}{=} g(x_2).$$

If additionally this mapping is injective, then there is exactly one initial value dtermining the solution at $u(x(t), t)$. However, bijectivity alone does not suffice. ??? We need further that this map and its inverse is continuously differentiable.???

Theorem 3.1

Let $f \in C^2(\mathbb{R}, \mathbb{R})$, $g \in C^1(\mathbb{R}, \mathbb{R})$ and $f''(g(x))g'(x) > -\alpha$ for all $x \in \mathbb{R}$ and for some $\alpha \geq 0$ then it follows that there exists a unique $u \in C^1(\mathbb{R},\mathbb{R})$ called C^1 solution of the initial value problem

$$\frac{\partial}{\partial t}u(x,t) + f'(u(x,t))\frac{\partial}{\partial x}u(x,t) = 0, \quad u(x,0) = g(x)$$

on $\mathbb{R} \times [0,\alpha^{-1})$ if $\alpha > 0$ and on $\mathbb{R} \times [0,\infty)$ if $\alpha = 0$.

Proof. We have already shown that the solution on $(x(t),t)$, where $x(t) = x_0 + tf'(g(x_0))$, is constant and equal to $g(x_0)$, i.e.

$$u(x(t),t) = g(x_0).$$

- Let $t \geq 0$ such that $1 - t\alpha > 0$:
  Then it follows

  $$\frac{\partial}{\partial x_0}x(t) = \frac{\partial}{\partial x_0}(x_0 + tf'g(x_0)) = 1 + tg'(x_0)f''(g(x_0)) \overset{Assumption}{\geq} 1 - t\alpha > 0.$$

  This implies that $x(t)$ is strictly increasing in x_0 and by that is injective. Further, because $f,g \in C^1$ the term $f'(g(x_0))$ is bounded for all $x_0 \in \mathbb{R}$ and implying

  $$\lim_{x_0 \to \pm\infty} x_0 + tf'(g(x_0)) = \pm\infty.$$

  This with property with the strictly increasing condition wrt. x_0 implies surjectivity. Thus, x is bijective. We now want to apply the inverse function theorem. For that we need to show that $x'(t)$ is invertible and continuous wrt. x_0.
  Invertibility: Because we have shown that $\frac{\partial}{\partial x_0}x(t)$ is positive, it is also invertible (compare positive definite).
  $x(t)$ is continuous wrt. x_0, as it is a composition of functions in C^1 and C^2.
  Thus, with the inverse function theorem, it follows that x^{-1} exists locally (on open subsets). Due to the fact that $x(t)$ is a bijection wrt. x_0 it follows that this existence extends globally. We have now shown that $x(t)$ is a C^1 diffeomorphism from $\mathbb{R}$ to $\mathbb{R}$. This implies

  $$\forall x \in \mathbb{R} \exists! x_0 \in \mathbb{R}: \quad x(t) = x_0 + tf'(g(x_0)).$$

  So finally, $u(x(t),t) = u(x_0 + tf'(g(x_0)),t) = u(x_0,0) = g(x_0)$ solves the equation for all $t \in [0,\infty)$.

- Let $t \geq 0$ such that $1 - t\alpha \leq 0$:
  All arguments are the same, except for the positivity of the derivative of the characteristic:

  $$\frac{\partial}{\partial x_0}x(t) \overset{like\ above}{\geq} 1 - t\alpha > 0 \quad \Longleftrightarrow \quad t \leq \frac{1}{\alpha} \quad \Longleftrightarrow \quad 1 - t\alpha \leq 0.$$

  Thus, with the above arguments $u(x(t),t) = u(x_0 + tf'(g(x_0)),t) = u(x_0,0) = g(x_0)$ solves the equation for all $t \in [0,\alpha^{-1})$.

$\square$

Remark 3.1

For $f(u) := \frac{1}{2}u^2$ the Burgers equation is

$$\dot{u}(x,t) + \frac{\partial}{\partial x} f(u(x,t)) = \dot{u}(x,t) + \underbrace{f'(u(x,t))}_{=u(x,t)} \frac{\partial}{\partial x} u(x,t) = 0,$$

is called Burgers erquation. Because $f'(u)) = u$ it follows $x(t) = x_0 + g(x_0)t$. Thus, if the initial value g is continuously differentiable and monotonically increasing if follows

$$f''(g(x))g'(x) \geq 1 \cdot \underbrace{g'(x)}_{\geq 0} > -\alpha, \quad \alpha \geq 0,$$

which implies the solution of the initial value problem is

$$u(x + tg(x), t) = g(x),$$

on $\mathbb{R} \times [0, \infty)$ or $\mathbb{R} \times [0, \alpha^{-1})$.

4 Weak Solutions

In the above we required the solution to be differentiable. Further, we had cases where there were no or non-unique solutions. In this section we take scalar conservation laws and discuss a more general approach for solutions. We focus on the intervall $\Omega = [a, b]$, $a < b \in \mathbb{R}$. The scalar conservation law $\frac{\partial}{\partial t} u + \frac{\partial}{\partial x} f(u) = 0$ integrated on $[a, b]$ is

$$\int_a^b \frac{\partial}{\partial t} u(x,t) dx = -\int_a^b \frac{\partial}{\partial x} f(u(x,t)) dx = f(u(a,t)) - f(u(b,t)).$$

Thus, if we assume that u is not differentiable we can formulate this initial value problem as

$$\frac{\partial}{\partial t} \int_a^b u(x,t) dx = f(u(a,t)) - f(u(b,t)).$$

In the last theorem the classical solution was formulated in such a way that it existed either up to time α^{-1} or ∞ depending on $\alpha \geq 0$. After time α^{-1} this solution does not exist in classical sense. Here we want to formulate a weak solution, that exists for all $t > 0$. As the characteristic curves intersect beyond this time discontinuities will appear. Let $\{(x,t) \mid x = y(t)\}$ be such a discontinuity, that moves along this graph, where we assume that y is a C^1 function. We solve for the solution by splitting the intervall $[a, b] = [a, y(t)] \cup [y(t), b]$, i.e.

$$
\begin{aligned}
&f(u(a,t)) - f(u(b,t)) \\
&\overset{above}{=} \frac{d}{dt}\int_a^b u(x,t)dx = \frac{d}{dt}\int_a^{y(t)} u(x,t)dx + \frac{d}{dt}\int_{y(t)}^b u(x,t)dx \\
&\overset{Leipnitz}{=} y'(t) \lim_{x\nearrow y(t)} u(x,t) - 0 \cdot \lim_{x\searrow y(t)} u(x,t) + \int_a^{y(t)} \frac{\partial}{\partial t} u(x,t)dx \\
&+ 0 \cdot \lim_{x\nearrow y(t)} u(x,t) - y'(t) \lim_{x\searrow y(t)} u(x,t) + \int_{y(t)}^b \frac{\partial}{\partial t} u(x,t)dx \\
&= y'(t) \underbrace{\lim_{x\nearrow y(t)} u(x,t)}_{=:u_l(y(t),t)} - y'(t) \underbrace{\lim_{x\searrow y(t)} u(x,t)}_{=:u_r(y(t),t)} + \int_a^{y(t)} \frac{\partial}{\partial t} u(x,t)dx + \int_{y(t)}^b \frac{\partial}{\partial t} u(x,t)dx \\
&\overset{PDE}{=} y'(t)(u_l(y(t),t) - u_r(y(t),t)) - \int_a^{y(t)} \frac{\partial}{\partial x} f(u(x,t))dx - \int_{y(t)}^b \frac{\partial}{\partial x} f(u(x,t))dx \\
&= y'(t)(u_l(y(t),t) - u_r(y(t),t)) + f(u(a,t)) - f(u(b,t)) + \lim_{x\searrow y(t)} f(u(x,t)) - \lim_{x\nearrow y(t)} f(u(x,t)) \\
&= y'(t)(u_l(y(t),t) - u_r(y(t),t)) + f(u(a,t)) - f(u(b,t)) + f(u_r(y(t),t)) - f(u_l(y(t),t))
\end{aligned}
$$

The above is equivalent to

$$y'(t) = \frac{f(u_l(y(t),t)) - f(u_r(y(t),t))}{u_l(y(t),t) - u_r(y(t),t)}.$$

This is called the Rankine-Hugonoit condition, where $y'(t)$ is the speed of the discontinuity, also called shock speed.

Example 4.1

Consider the Burgers equation

$$
\begin{aligned}
\dot{u} + \frac{\partial}{\partial x}\frac{1}{2}u^2 = \dot{u} + u\frac{\partial}{\partial x}u = 0, \qquad \text{on } \mathbb{R} \times \mathbb{R}^+ \\
u(x,0) = g(x) = \begin{cases} 1 & , x \leq 0 \\ 1-x & , 0 \leq x < 1 \\ 0 & , 1 \leq x \end{cases}
\end{aligned}
$$

As $f'(u) = u$ it follows that the characteristics are given by

$$x(t) = x_0 + tf'(g(x_0)) = x_0 + tg(x_0) = \begin{cases} x_0 + t & , x_0 \leq 0 \\ x_0 + t(1-x_0) & , 0 < x_0 < 1 \\ x_0 & , 1 \leq x_0 \end{cases}$$

We see that at time $t = 1$ the characteristics intersect. Its inverse is given by

$$x_0 \mapsto \begin{cases} x - t & , x_0 \leq 0 \iff x - t \leq 0 \\ \frac{x-t}{1-t} & , 0 < x_0 < 1 \iff t < x < 1 \\ x & , 1 \leq x_0 \iff 1 \leq x \end{cases}$$

We have shown that the solution of this type of PDE is constant along the characteristic curvers, i.e.

$$u(x(t),t) = g(x_0) = \begin{cases} 1 & , x_0 \leq 0 \iff x \leq t \\ 1 - \frac{x-t}{1-t} = \frac{1-x}{1-t} & , 0 < x_0 < 1 \iff t < x < 1 \\ 0 & , 1 \leq x_0 \iff 1 \leq x \end{cases}$$

is the solution for $t < 1$. One can show that for $t > 1$ a solution of the form exists of the form

$$u(x,t) = \begin{cases} 1 & , x < y(t) \\ 0 & , x > y(t) \end{cases}$$

where $y(t)$ is the discontinuity that arises after $t = 1$. Note that this solution exists if and only if the Rankine-Hugonoit condition is satisfied. The speed of the shock is the Rankine-Hugonoit condition

$$y'(t) = \frac{f(u_l(y(t),t)) - f(u_r(y(t),t))}{u_l(y(t),t) - u_r(y(t),t)} = \frac{\frac{1}{2} \cdot 1^2 - 0}{1 - 0} = \frac{1}{2}.$$

The characteristic on which the shock moves is a linear function going through $(1,1)$ and having slope $1/2$, i.e.

$$y(t) = \frac{1}{2}(t-1) + 1.$$

Example 4.2

Consider the Burgers equation

$$\dot{u} + \frac{\partial}{\partial x}\frac{1}{2}u^2 = \dot{u} + u\frac{\partial}{\partial x}u = 0, \qquad \text{on } \mathbb{R} \times \mathbb{R}^+$$

$$u(x,0) = g(x) = \begin{cases} 0 & , x < 0 \\ 1 & , x > 0 \end{cases}$$

Then,

$$u(x,t) = \begin{cases} 0 & , x < y(t) \\ 1 & , y(t) > x \end{cases} \quad \text{is a solution} \quad \iff \quad y'(t) = \frac{f(u_l(y(t),t)) - f(u_r(y(t),t))}{u_l(y(t),t) - u_r(y(t),t)} = \frac{1}{2}.$$

This time the shock starts in $(0,0)$ which implies that $y(t) = \frac{t}{2}$. However, now there exists another continuous solution that also satisfies the Rankine-Hugonoit condition:

$$u(x,t) = \begin{cases} 0 & , x \leq 0 \\ \frac{x}{t} & , 0 < x < t \\ 1 & , t \leq x \end{cases}$$

Which solution is now the correct one?

Since the goal is to maximize regularity, we only accept a discontinuous weak solution if there are not continuous solutions. Thus in the above case, we would choose the continuous solution. Can we specify this?:

Definition 4.1: Lax Entropy Condition

A discontinuity of a weak solution along a C^1 path $t \mapsto y(t)$ satisfies the Lax Entropy condition if

$$f'(y_l(y(t),t)) > \dot{y}(t) > f'(u_r(y(t),t)).$$

A weak solution with discontinuities along C^1 paths is called an admissible solution, if it satisfies both the Rankine Hugonoit condition and the Lax Entropy condition.

So we choose a weak solution with discontinuities to be the "physically correct" one, if it satisfies additionally to the Rankine Hugonoit condition the Lax Entropy condition. In the following we show the uniqueness and existence of the admissible solution.

Theorem 4.2: Uniqueness

Let $f \in C^1(\mathbb{R}, \mathbb{R})$ be convex and u and v two admissible solution of

$$\dot{u}(x,t) + f'(u(x,t))\frac{\partial}{\partial x}u(x,t) = 0, \quad u \in L^1(\mathbb{R}),$$

then $t \mapsto \|u(\cdot,t) - v(\cdot,t)\|_{L^1(\mathbb{R})}$ is monotonically decreasing.

Proof. ???Skipped □

As the norm is non negative it follows that it reaches zero eventually??? This implies uniqueness, if of the admissible solution if it exists.

Theorem 4.3: Existence

For $f \in C^2(\mathbb{R}, \mathbb{R})$ is strictly convex and $g \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$, then

$$\exists! u \text{ admissible solution of } \quad \dot{u} + f'(u)\frac{\partial}{\partial x}u = 0, \text{ on } \mathbb{R} \times \mathbb{R}^+, \text{ and } u(x,0) = g(x), x \in \mathbb{R}.$$

5 Non-characteristic Hypersurfaces

Until now we had boundary conditions at time $t = 0$ and on the entire domain. To generalize consider $\Sigma \subset \Omega \subseteq \mathbb{R}^n$ where Ω is our domain and on the level set of φ given by $\Sigma := \{x \in \Omega \mid \varphi(x) = \varphi(x_0)\}$ we define the boundary condition. We assume that φ is a C^1 function.
Consider the first order PDE

$$F(\nabla u(x), u(x), x) = 0, \quad u(y) = g(y), \quad \forall y \in \Sigma,$$

where $F : W \subset \mathbb{R}^n \times \mathbb{R} \times \Omega \to \mathbb{R}$ and $g : \Omega' \to \mathbb{R}$. First, we show that locally (on open neighborhoods of x_0) that the boundary condition of every Cauchy problem (i.e. the above) can be brought into the following form:

$$u(y) = g(y), \quad \forall y \in \Omega \cap H, \quad \text{where } H := \{x \in \mathbb{R}^n \mid x^T e_n = x_0^T e_n\}.$$

Here e_n is the unit vector. We show that H is unique in x_0 and orthogonal to e_n.
Assume $\nabla\varphi(x_0) \neq 0$, i.e. there exists at least one $i \leq n$ such that $\frac{\partial}{\partial x_i}\varphi(x_0) \neq 0$. Simply permuting the coordinates gives us that $\frac{\partial}{\partial x_n}\varphi(x_0) \neq 0$. This condition will be now used:
Define $\Phi : \Omega \to \mathbb{R}, \quad x \mapsto (x_1, ..., x_{n-1}, \varphi(x))$, then its Jacobian is given by

$$D\Phi(x) = \begin{pmatrix} Id_{n-1\times n-1} \\ \nabla\varphi(x_0)^T \end{pmatrix}.$$

Thus Φ' is continuous and invertible(due to the non negativity assumed above), which implies with the inverse function theorem that in a neighborhood of x_0 there eixsts $x = \Phi^{-1}(y)$.
Due to $\Phi(x) = y$ it gollows that

$$\varphi(x_0) = \varphi(x) = y_n = y \cdot e_n \quad\iff\quad \varphi(x_0) = \varphi(x).$$

Next, we show that H is orthogonal to e_n. For every tangent vector $w \in H$ consisting of the difference of two elements $w := p - q$, where $p, q \in H$, it follows

$$w^T e_n = p^T e_n - q^T e_n = \varphi(x_0) - \varphi(x_0) = 0.$$

So the boundary straightens at x_0. We just transformed the boundary using Φ. This also has to be done with the solution. Define $u := v \circ \Phi$. It satisfies

$$\nabla u(x) = \nabla v(\Phi(x))\Phi'(x) = \nabla v(y)\Phi'(\Phi^{-1}(y)).$$

If u satisfies the PDE

$$F(\nabla u(x), u(x), x) = 0,$$

then it is equivalen to

$$F(\nabla u(x), u(x), x) = F(\nabla v(y)\Phi'(\Phi^{-1}(y)), u(\Phi^{-1}(x)), \Phi^{-1}(y)) = 0.$$

Thus, we can indeed assume locally that the boundary condition is a hyperplane, if we adjust the PDE respectively.
Next, we ask ourselves the question what information we can gather about u using the Hyperplane H? We can determine some partial derivatives of u

$$\forall i = 1, ..., n-1 : \frac{\partial}{\partial x_i}u(x_0) = \lim_{h\to 0} \frac{u(x_0 + he_i) - u(x_0)}{h} = \lim_{h\to 0} \frac{g(x_0 + he_i) - g(x_0)}{h} = \frac{\partial}{\partial x_i}g(x_0).$$

However

$$\frac{\partial}{\partial x_n}u(x_0) = \lim_{h\to 0} \frac{u(x_0 + he_n) - u(x_0)}{h}$$

cannot be determined, as $x_0 + he_n$ is not in H.
In the remainder of this section, we can show that the PDE on this simplified hyperplane H only depends on F and the initial condition g. We do this by substituting some variables. For $x_0 \in H$ it holds

$$F(\nabla u(x_0), u(x_0), x_0) = F(\left(\frac{\partial}{\partial x_1}g(x_0), ..., \frac{\partial}{\partial x_{n-1}}g(x_0), \frac{\partial}{\partial x_n}u(x_0)\right), g(x_0), x_0) = 0,$$

which has only free variable.
We want the characteristics curves $x(s)$ to move away from H to Ω, i.e. transversal to H, which is formally

$$\dot{x}(s)^T e_n \neq 0 \quad\iff\quad 0 \neq \dot{x}(s)^T e_n = \nabla_p F(p(x), u(x), x)^T e_n = \frac{\partial}{\partial p_n}F(p(x), u(x), x)$$

although the equivalence will be shown in the next section. This property is required for the method of characteristics to completely determine u on the entire domain. Thus, if a Hyperplane does not lie in such a way that the characteristics move away from it, we call it non-characteristic:

Definition 5.1

Consider the PDE as a function of $2n+1$ variables given by

$$F(p, z, x) = 0, \quad x, p \in \mathbb{R}^n, z \in \mathbb{R}.$$

Suppose that (p_0, z_0, x_0) is a solition, then the hyperplane $H := \{x \in \Omega \mid x_n = x_{0,n}\}$ is called non-characteristic at $x_0 := (x_{0,1}, ..., x_{0,n})$ if

$$\frac{\partial}{\partial p_n} F(p_0, z_0, x_0) \neq 0.$$

Example 5.1

Consider the PDE

$$F(p_1, p_2, z, x_1, x_2) = \frac{\partial}{\partial x_1} u(x,t) = p_1 = 0, \quad u(x_1, 0) = g(x_1), \quad (x,t) \in \mathbb{R}^2 \times \mathbb{R}.$$

Here we have $\frac{\partial}{\partial x_2} p_1 = 0$, i.e. not the non characteristic Property. We can now construct characteristics that lie inside the Hyperplane. Let $(x_1(s), x_2(s))$ be that characteristic along which we evaluate the solution $z(s) := u(x_1(s), x_2(s))$. Differentiation yields

$$z'(s) = \frac{\partial}{\partial x_1} u(x(s)) x_1'(s) + \frac{\partial}{\partial x_2} u(x(s)) x_2'(s).$$

If we choose $x_1'(s) = 1$ and $x_2'(s) = 0$ we have

$$\frac{\partial}{\partial x_1} u(x(s), s) = 0 = z'(s),$$

meaning the method of characteristics is applicape???
However, these characteristics lie inside the Hyperplane for all $s > 0$:

$$\begin{pmatrix} x_1(s) \\ x_2(s) \end{pmatrix} = \begin{pmatrix} x_{0,1} + s \cdot 1 \\ x_{0,2} + 0 \end{pmatrix} \in H.$$

Lemma 5.2

Let $F : W \to \mathbb{R}$ and $g : H \to \mathbb{R}$ continuous differential, $x_0 \in \Omega \cap H$ and $z_0 = g(x_0)$, $p_{0,1} = \frac{\partial}{\partial x_1} g(x_0), ..., p_{0,n-1} = \frac{\partial}{\partial x_{n-1}} g(x_0)$. If $\exists p_{0,n}$ such that $\frac{\partial}{\partial p_n} F(p_0, z_0, x_0) \neq 0$ (non characteristic) and $F(p_0, z_0, x_0) = 0$ then forall x_0 in an open neighborhood $\Omega_{x_0} \subset \Omega$ of x_0 there exists a $x \in \Omega_x \cap H$, a unique solution q of

$$F(q(x), g(x), x) = 0, \quad q_i(x) = \frac{\partial}{\partial x_i} g(x), \quad i = 1, ..., n-1, \quad q(x_0) = p_0.$$

Proof. Define $l : (x, q_n) \mapsto F(q_1(x), ..., q_{n-1}(x), q_n, g(x), x)$. By construction at $(x_0, p_{0,n})$ it takes 0 as a value. To apply the implicit function theorem, we need that l' is continuous in $(x_0, q_{0,n})$ and $\frac{\partial}{\partial q} l(x_0, q_{0,n})$ invertible. This follows from the definition of g and the non characteristic assumption of the Hyperplane/ $\frac{\partial}{\partial p_n} F(p_0, z_0, x_0) \neq 0$. Thus, the preimage of the differential equation wrt. q_n can be represented as a graph of a continuous function. This implies uniqueness of the the solution. $\square$

This Lemma shows that the gradient $q(x) = \nabla u(x)$ uniquely exists on $\Omega_{x_0} \cap H$ and on this set solves the PDE. However, we need to show that on Ω_{x_0} this gradient exists and solves the PDE. For that we use the method of characteristics.

6 Method of Characteristics

Our goal in this section is to show that the method of characaterstics exists.
Let $x(s)$ be a curve in the domain $\Omega \subset \mathbb{R}^n$ and $u(s) := u(x(s))$ the value of the solution along that curve. We now also consider the gradient of the solution along the curve $p(s) := \nabla u(x(s))$. Question: How should the curve $s \mapsto x(s)$ be chosen? This we will now derive.

$$\frac{d}{ds} p_i(s) = \frac{d}{ds} \frac{\partial}{\partial x_i} u(x(s)) = \sum_{j=1}^{n} \frac{\partial}{\partial x_j} \frac{\partial}{\partial x_i} u(x(s)) x_j'(s) = \sum_{j=1}^{n} \frac{\partial^2}{\partial x_j \partial x_i} u(x(s)) x_j'(s)$$

The total derivative of the PDE $F(\nabla u(x), u(x), x) = 0$ wrt. x_i along the curve $x(s)$ is

$$\begin{aligned}
0 &= \frac{d}{dx_i} F(\nabla u(x(s)), u(x(s), x(s)) \\
&= \sum_{j=1}^{n} \frac{\partial}{\partial p_j} F(p(s), z(s), x(s)) \frac{\partial}{\partial x_i} p_j(s) + \frac{\partial}{\partial z(s)} F(p(s), z(s), x(s)) \frac{\partial}{\partial x_i} z(s) \\
&+ \frac{\partial}{\partial x_i} F(p(s), z(s), x(s))
\end{aligned}$$

With commutativity of the partial derivative, the above is equivalent to

$$\begin{aligned}
&\sum_{j=1}^{n} \frac{\partial}{\partial p_j} F(p(s), z(s), x(s)) \frac{\partial^2}{\partial x_j \partial x_i} u(x(s)) \\
&= -\frac{\partial}{\partial z(s)} F(p(s), z(s), x(s)) \frac{\partial}{\partial x_i} u(x(s)) - \frac{\partial}{\partial x_i} F(p(s), z(s), x(s))
\end{aligned}$$

This we can plug into the first equation

$$\begin{aligned}\frac{d}{ds}p_i(s) &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}u(x(s))x_j'(s) \\ &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}u(x(s))\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)).\end{aligned}$$

Finally, it holds

$$\frac{d}{ds}z(s) = \frac{d}{ds}u(x(s)) = \sum_{j=1}^{n} \frac{\partial}{\partial x_j}u(x(s))x_j'(s) = \sum_{j=1}^{n} p_j(s)\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)).$$

In total we have derived the ODEs

$$\begin{aligned}\frac{d}{ds}x_i(s) &= \frac{\partial}{\partial p_i}F(p(s), z(s), x(s)), \quad i = 1, ..., n \\ \frac{d}{ds}p_i(s) &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}u(x(s))\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)), \quad i = 1, ..., n \\ \frac{d}{ds}z(s) &= \sum_{j=1}^{n} p_j(s)\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)).\end{aligned}$$

which is a $2n + 1$ dimensional system and closed (as it only depends on u). Something is now surpirsing??? In the following theorem it will be shown that under the condition that the characteristic curves $x(s)$ exist as the solution of the n dimensional ODE, then we can solve the other two ODEs systems.

Theorem 6.1

Let F be a real differentiable function on an open subset $W \subset \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$ and $\Omega \subset \mathbb{R}^n$ open and $u : \Omega \to \mathbb{R}$ a twice differentiable solution of $F(\nabla u(x), u(x), x) = 0$. THen if a solution $s \mapsto x(s)$ of the ODE

$$x_i'(s) = \frac{\partial}{\partial p_i}F(p(s), z(s), x(s)), \quad i = 1, ..., n$$

exists, then $p_i(s) := \frac{\partial}{\partial x_i}u(x(s))$ and $z(s) := u(x(s))$ solve the following ODEs

$$\begin{aligned}\frac{d}{ds}p_i(s) &= \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}u(x(s))\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)), \quad i = 1, ..., n \\ \frac{d}{ds}z(s) &= \sum_{j=1}^{n} p_j(s)\frac{\partial}{\partial p_j}F(p(s), z(s), x(s)).\end{aligned}$$

This Theorem reduces the problem of finding a unique solution ot the PDE as a question of finding a unique solution to an ODE system. We now want to show that a solution exists. Due to crossing characteristics global existence will not be possible. In the following Theorem we skow local existence.

Theorem 6.2

Let $F : W \to \mathbb{R}$, $g : H \to \mathbb{R}$ three times differentiable functions and suppose $(p_0, z_0, x_0) \in W$ such that

$$F(p_0, z_0, x_0) = 0, \quad z_0 = g(x_0), \quad p_{0,1} = \frac{\partial}{\partial x_1} g(x_0), ..., p_{0,n-1} = \frac{\partial}{\partial x_{n-1}} g(x_0).$$

Additionally assume H is non characteristic at x_0, i.e. $\frac{\partial}{\partial p_n} F(p_0, z_0, x_0) \neq 0$.
Then it follows that in $\Omega_{x_0} \subset \Omega$ open neighborhood of x_0 there exists a unique solution u of the boundary value proplem

$$F(\nabla u(x), u(x), x) = 0, \quad x \in \Omega_{x_0}, \quad u(y) = g(y), \quad y \in \Omega_{x_0} \cap H.$$

Proof. ??? □

We now explore the connection between the ad hoc versions of the method of characteristics from the previous sections to the one introduced here. THe point is that they are the same mehtod, but only simplified special cases, where they do not depend on $p = \nabla u$.

- Inhomogeneous linear transport equation:
  We define $(p_1, ..., p_{n+1}) := (\frac{\partial}{\partial x_1} u(x(s)), ..., \frac{\partial}{\partial x_n} u(x(s)), \frac{\partial}{\partial s} u(x(s)))$, then the PDE is

  $$0 = F(p, z, (x, t)) = \frac{\partial}{\partial t} u(x, t) + b \cdot \nabla u(x, t) - f(x, t) = b_1 p_1 + ... + b_n p_n + p_{n+1} - f(x, t).$$

  Plugging into two of the three ODEs yields

  $$\begin{aligned} x_i'(s) &= \frac{\partial}{\partial p_i} F(p(s), z(s), x(s)) = b_i, \quad x_{n+1}'(s) = t'(s) = \frac{\partial}{\partial p_{n+1}} F(p(s), z(s), x(s)) = 1 \\ z'(s) &= \sum_{j=1}^{n} \frac{\partial}{\partial p_j} F(p(s), z(s), x(s)) + \frac{\partial}{\partial p_{n+1}} F(p(s), z(s), x(s)) p_{n+1}(s) \\ &= b \cdot \nabla u(x, t) + \frac{\partial}{\partial t} u(x, t) = f(x, t) \end{aligned}$$

  These systems do not depend on the $p'(s)$ ODE, meaning we can solve for $x(s)$ and $z(s)$ independently from $p(s)$. In fact, we do not even have to solve for $p(s)$ due this fact.

- Homogeneous linear conservation Law:
  This case is even simpler as before. We have

  $$\dot{u}(x, t) + f'(u(x, t)) \nabla u(x, t) = 0, \quad u(x, 0) = g(x), \quad \forall x \in \mathbb{R}^n$$

  where $g : \mathbb{R}^n \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}^n$.. Again, set $x_{n+1} = t$, then

  $$0 = F(p, z, (x, t)) = \dot{u}(x, t) + f'(z) \cdot (p_1, ..., p_n) = p_{n+1} + f'(z) \cdot (p_1, ..., p_n).$$

  This gives

  $$x_i'(s) = f_i'(z), \quad x_{n+1}'(s) = t'(s) = 1, \quad z'(s) = \frac{\partial}{\partial t} u(x, t) + f' \cdot (p_1, ..., p_n) = 0.$$

  Because the initial value of the second ODE is $z(0) = g(x)$ it follows that the characteristic is $x_i(s) = x + s f_i'(g(x))$ where $t(s) = s$. Because $z'(s) = 0$ the solution is constant along the characteristc. This finally yields the same form we have derived before:

  $$u(x + t f'(g(x)), t) = g(x), \quad \forall (x, t) \in \mathbb{R}^n \times \mathbb{R}.$$

CHAPTER 2

GENERAL CONCEPTS

1 Divergence Theorem

We will do here a generalization of the fundamental theorem of calculus to a higher dimension. First, we generalize partial integration. Second in what sense higher dimensional scalar convervation laws describe conserved quantity???

Definition 1.1

The graph of a function $\lambda : U \subset \mathbb{R}^k \to \mathbb{R}^{n-k}$ is the k-dimensional subset

$$graph(\lambda) := \{(x, y) \in U \times \mathbb{R}^{n-k} \mid y = \lambda(x)\} \subset \mathbb{R}^n.$$

We allow reordering of the components of the graph, i.e. both $\{(x, y, x^2+y^2) \in \mathbb{R}^3\}$ and $\{(x, x^2+z^2, z) \in \mathbb{R}^3\}$ are graphs.

Definition 1.2

A subset $A \subset \mathbb{R}^n$ is called k-dimensional submanifold if it is a k-dimensional graph locally, i.e.

$$\forall x \in A \exists O_x \subset \mathbb{R}^n \text{ open: } A_x := A \cap O_x \text{ is a k-dimensional graph.}$$

We say that a graph or submanifold is in C^l, if locally the graph of $\lambda \in C^l$.

Example 1.1

The circle $S^1 := \{(x, y \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ is an example for a submanifold, that is not a graph globally, but only locally. This is because $y = \pm\sqrt{1 - x^2}$ is not a function. However we can write the circle as the union of four graphs

$$\begin{aligned} &S^1 = \{(x, \sqrt{1 - x^2}) \mid x \in (-1, 1)\} \cup \{(x, -\sqrt{1 - x^2}) \mid x \in (-1, 1)\} \\ &\{(\sqrt{1 - y^2}, y) \mid y \in (-1, 1)\} \cup \{(-\sqrt{1 - y^2}, y) \mid y \in (-1, 1)\}. \end{aligned}$$

A parametrization will be a toll that can cover more of a graph of a submanifold, which allows us to do less work.

Definition 1.3

A continuously differentiable injection $\Phi : U \subset \mathbb{R}^k \to A \subset \mathbb{R}^n$ is called a parametrisation of a submanifold A.
It is called regular if the Jacobian of $\Phi' := D\Phi$ has rank k at every point in U.

Because the Jacobain of Φ is a $n \times k$ matrix, its rank cannot be larger than k. This is why we call a refular parametrisation of full rank.

Remark 1.1

A graph is a parametrisation $\Phi : U \to graph(\lambda), x \mapsto (x, \lambda(x))$ and its Jacobian is

$$\frac{\partial}{\partial x}\phi(x_0) = \begin{pmatrix} 1 & 0 & \cdots & & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & & \cdots & & 1 \\ \frac{\partial}{\partial x_1} f_1(x_0) & & \cdots & & \frac{\partial}{\partial x_m} f_1(x_0) \\ \vdots & & \ddots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(x_0) & & \cdots & & \frac{\partial}{\partial x_m} f_n(x_0) \end{pmatrix}$$

which is of full rank (rank k).

Example 1.2

The parametrisation $(x, y) \mapsto (x, 0, 0)$ of the submanifold of the x axis in $\mathbb{R}^3$ is not regular, because

$$\Phi' = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

is not full rank. The coordinate y does not really play a role.

Definition 1.4

Let $A \subset \mathbb{R}^n$ be a subset with a regular parametrisation $\Phi : U \to A$ and $f : A \to \mathbb{R}$ continuous, then define

$$\int_A f d\sigma := \int_U f \circ \Phi \sqrt{\det((\Phi')^T \Phi')} d\mu_{\mathbb{R}^k}.$$

The parametrisation maps an infinitisimal cube in U, i.e. euklidean space, to a parallelotop, i.e. $\{c_1 v_1 + ... + c_k v_k \mid c_i \in [0, 1]\}$ a generalisation of a parallelolgram, wehere the k column vectors in Φ' are the vectors spanning the prallelotop. This distorsion is denoted in $\sqrt{\det(...)}$.
Does the value of the integral $\int_A f d\sigma$ depend on the regular parameterisation?

Lemma 1.5

Let $A := graph(\lambda)$, then $\int_A f d\sigma$ is independent of the choice of the regular parametrization.

Proof. First, consider the parametrization of the graph, i.e.

$$\phi : U \to graph(f), x \mapsto (x, f(x)), \quad U \subset \mathbb{R}^m, graph(f) \subset \mathbb{R}^{n+m}.$$

and second another Parametrization

$$\psi : V \to graph(f), \quad V \subset \mathbb{R}^m.$$

We then define $\Upsilon := \phi^{-1} \circ \psi : U \to V$, which we need to show is continuously differentiable, which is not trivial (*). Then we get that

$$\begin{aligned}
\int_A f d\sigma &= \int_U f \circ \phi \cdot det((\phi')^T \phi') d\mu \\
&\overset{\substack{Jacobi \\ Trafo}}{=} \int_V det(\Upsilon') \left((f \circ \phi)\sqrt{det((\phi')^T \phi')}\right) \circ \Upsilon d\mu \\
&= \int_V (f \circ \phi \circ \Upsilon)\sqrt{det((\phi')^T \phi')} det(\Upsilon') d\mu \\
&= \int_V (f \circ \phi \circ \Upsilon)\sqrt{det((\phi')^T \phi') det(\Upsilon')^2} d\mu \\
&= \int_V (f \circ \phi \circ \Upsilon)\sqrt{det((\phi')^T \phi') det((\Upsilon')^T \Upsilon')} d\mu \\
&= \int_V (f \circ \phi \circ \Upsilon)\sqrt{det((\phi')^T \phi' \cdot (\Upsilon')^T \Upsilon')} d\mu \\
&= \int_V (f \circ \phi \circ \Upsilon)\sqrt{det(((\phi \circ \Upsilon)')^T (\phi \cdot \Upsilon)')} d\mu \\
&= \int_V (f \circ \psi)\sqrt{det((\psi')^T \psi')} d\mu
\end{aligned}$$

In the Jacobi transformation we need that Υ is continuously differentiable and bijective on open sets and Υ' is invertible.
Continuously differentiable: For $\Upsilon := \phi^{-1} \circ \psi$ the continuously differentiability it is not clear, because ϕ^{-1} is only defined on A which is non-euclidean space, i.e. the chain rule does not apply directly. Hence, define

$$\Pi : \mathbb{R}^n \to \mathbb{R}^k, (x, y) \mapsto x$$

then it holds that $\Pi \circ \phi(x) = \Pi(x, \lambda(x)) = x$, i.e. $\Pi \circ \phi = Id$ which implies that

$$\Pi\big|_A = \phi^{-1}, \qquad \text{as } \Phi^{-1} \text{ is only defined on } A.$$

which in turn implies that $\Upsilon = \phi^{-1} \circ \psi = \Pi_A \circ \psi$ which is by definition a composition of continuous differentiable functions. $\square$

Many submanifolds cannot be covered by a single parametrisation.

Example 1.3

We cannot find one parametrisation of the sphere, as it is compact, but parametrisations are defined on open sets. However, we can choose two parametrisations that cover the sphere.

The problem with choosing multiple parametrisations to cover a submanifold is that the open sets can overlap. We do not want to integrate over an area twice. The solution is the following

Definition 1.6

Let $\Omega \subset \mathbb{R}^n$ be covered by a countable familiy $(U_i)_{i\in\mathbb{N}}$ of open subsets of $\mathbb{R}^n$, i.e. $\bigcup_{i\in\mathbb{N}} U_i = \Omega$. We call a smooth partition of unity a countable family $(h_i)_{i\in\mathbb{N}}$ of smooth functions $h_i : \Omega \to [0,1]$ that satisfy

- $\forall x \in \Omega\ \exists O_x \subset \Omega$ neigborhood of x on which all but finitely many h_i vanish
- $\forall x \in \Omega: \quad \sum_{i=1}^{\infty} h_i(x) = 1$
- $\forall h_i$ vanishes outside of a compact subset of U_i.

Theorem 1.7

For every countable family of open subsets of $\mathbb{R}^n$ a partition of unity exists.

Definition 1.8

Let $A \subset \mathbb{R}^n$ be a compact k-dimensional submanifold. Because A is compact and $A \subset \bigcup_{x\in A} O_x$ only finite many parametrisations are needed to cover it.
Choose the partition of unity $(h_i)_{i\in\mathbb{N}}$ such that the support is completely contained within the corresponding O_i. For continuous $f : A \to \mathbb{R}$ we define

$$\int_A f d\sigma := \sum_{i\leq n} \int_{A_i} h_i f d\sigma.$$

The idea here is that each $h_i f$ is zero outside of A_i, which yields a intefration only over A_i. We assume A to be compact so that we avoid any issues with convergence.

Lemma 1.9

The integral $\int_A f d\sigma$ neither depends on the choice of the partition of unity nor the parametrisation.

Proof. Let $(A_i)_{i\in\mathbb{N}}$ and $(B_i)_{i\in\mathbb{N}}$ with $A = \bigcup_{i\in\mathbb{N}} = A_i = \bigcup_{i\in\mathbb{N}} B_i$ and two partitions of unities $(h_i)_{i\in\mathbb{N}}$ and $(g_i)_{i\in\mathbb{N}}$. The product $(g_i h_i)_{i\in\mathbb{N}}$ is a again a partition of unity on $(C_{ij} := A_i \cap B_i)_{i\in\mathbb{N}}$. Note that we can parametrize C_{ij} by restricting $\Phi^{-1}\big|_{A_i\cap B_j}$. Then

$$\sum_{i\in\mathbb{N}}\int_{A_i} h_i f d\sigma = \sum_{i\in\mathbb{N}}\int_{A_i}\sum_{j\in\mathbb{N}} g_i h_i f d\sigma = \sum_{i,j\in\mathbb{N}}\int_{A_i} g_i h_i f d\sigma = \sum_{i,j\in\mathbb{N}}\int_{A_i\cap B_j} g_i h_i f d\sigma$$
$$= ... = \sum_{i\in\mathbb{N}}\int_{A_i} g_i f d\sigma$$

□

These "manifold" integrals have similar propertiess as usual ones, however changes of variables is a little different.

Lemma 1.10

$\forall a, b \in \mathbb{R}$ and $f, g \in C(A)$ holds

- $\int_A af + bg d\sigma = a\int_A f d\sigma + b\int_a g d\sigma$
- $f \leq g$ on A implies $\int_A f d\sigma \leq \int_A g d\sigma$
- $\big|\int_A f d\sigma\big| \leq \int_A |f| d\sigma$
- If $P : \mathbb{R}^n \to \mathbb{R}^n$ is euclidean motion (translation, reflection, rotation) and $s \in \mathbb{R}^+$ the scaling factor it holds
$$\int_A f d\sigma = s^k \int_{(sP)^{-1}(A)} f \circ (sP) d\sigma.$$

Remark 1.2

This last condition in the above means that we only need to correct for scaling when tranforming the submanifold integral.

We will now work with bounded and open sets $\Omega \subset \mathbb{R}$. Their boundaries can be very complicated, even fractal. We will require $\partial\Omega$ to be $(n-1)$ dimensional submanifolds. Compactness of $\partial\Omega$ yields boundedness of Ω. This is why we will require Ω to be bounded.
Vectors that lie perpendicular to the boundary can be found, as the boundary is a graph locally, which means we can find tangential vectos to the submanifold. The tangential vectors wrt. to the i^{th} component is are

$$v_i = \begin{pmatrix} 0 \\ \dots \\ 0 \\ 1 \\ 0 \\ \dots \\ 0 \\ \partial_i\lambda \end{pmatrix}.$$

Then we have

$$(0, 0, ..., 0, 1, 0, ..., 0, \partial_i\lambda)\begin{pmatrix} -\nabla\lambda \\ 1 \end{pmatrix} = -\partial\lambda + \partial_i\lambda = 0,$$

i.e. $\begin{pmatrix} -\nabla\lambda \\ 1 \end{pmatrix}$ is perpendicular to the boundary. The unit vector pointing outward from the boundary is then

$$N := \pm\frac{1}{\sqrt{1+|\nabla\lambda|^2}}\begin{pmatrix} -\nabla\lambda \\ 1 \end{pmatrix}.$$

Note that for $\Phi(x) = (x, \lambda(x))^T$ we have $\Phi' = \begin{pmatrix} Id_{n-1\times n-1} \\ \nabla\lambda^T \end{pmatrix}$ which implies that $\det((\Phi')^T\Phi') = 1 + |\nabla\lambda|^2$.

Theorem 1.11

Let $\Omega \subset \mathbb{R}^n$ be bounded and open such that $\partial\Omega$ is $(n-1)$ dimensional submanifold of $\mathbb{R}^n$. Let $F : \bar{\Omega} \to \mathbb{R}^n$ be continuous and differentiable on Ω such that F' continuosly extends to $\partial\Omega$. Then

$$\int_\Omega \nabla \cdot F d\mu = \int_{\partial\Omega} F \cdot N d\sigma$$

where N is the outward pointing normal.

Remark 1.3

Let f, g be two functions such that $\forall x \in \partial\Omega$: $f(x)g(x) = 0$, then

$$\int_\Omega f\partial_i g d\mu = -\int_\Omega \partial f g d\mu + \underbrace{\int_\Omega \partial_i(fg)d\mu}_{=\int_{\partial\Omega} fg\cdot N d\sigma = 0}$$

So for any multi-index $\gamma \in \mathbb{N}_0$ it holds

$$\int_\Omega f\partial^\gamma g d\mu = (-1)^{|\gamma|}\int_\Omega g\partial^\gamma f d\mu.$$

Remark 1.4

For any continousy differentiable function $F : \mathbb{R} \to \mathbb{R}^n$ we call

$$\dot{u}(x,t) + \nabla \cdot F(u(x,t)) = \dot{u}(x,t) + F'(u(x,t)) \cdot \nabla u(x,t) = 0$$

conservation law. This is because for any open and bounded $\Omega \subset \mathbb{R}^n$ with $(n-1)$-dimensional submanifold $\partial\Omega$ of $\mathbb{R}^n$ we obtain

$$\frac{d}{dt}\int_\Omega u(x,t)d^n x = \int_\Omega \dot{u}(x,t)d^n x = -\int_\Omega \nabla \cdot F(u(x,t))d^n x = -\int_{\partial\Omega} F(u(x,t)) \cdot N(x)d\sigma(x),$$

meaning the change over Ω only depends on the flux $-F(u(\cdot,t))$ on the boundary $\partial\Omega$.

Remark 1.5

The surface area of a ball in realtion to its volume is as follows (for later important). Let ω_n denote the volume of the n-dimensional unit ball. We scale it by r^n. Then n times that we have (using $\nabla \cdot x = n$)

$$n\omega_n r^n = \int_{B_r(0)} \nabla \cdot x d\mu = \int_{\partial B_r(x)} x \cdot N(x) d\sigma(x) = \int_{\partial B_r(0)} \underbrace{x \cdot \frac{x}{|x|}}_{=r\cdot x} d\sigma(x) = r\sigma_n(r),$$

although we used that $r = |x| = \sqrt{x \cdot x}$.

One can think of a ball $B_R(0)$ as spheres $\partial B_r(0)$, where $r \in (0, R)$. The origin can be ignored as it has measure zero. The spheres are level sets of a function.

Corollary 1.12

(Co-Area)
Let $l : \Omega \to \mathbb{R}$ be a C^1 non-negative function and suppose

$$\forall t \in [0, T] : \quad \Omega_t := \{x \in \Omega \mid l(x) < t\}$$

is a domain to which the divergence theorem applies, then

$$\forall f \in C^1(\Omega) : \quad \int_{\Omega_T} f|\nabla l| dx = \int_0^T \int_{\partial\Omega_t} f d\sigma dt.$$

Proof. Note that the gradient of a function is always normal to its level set and pointing in the direction of steepest increase, i.e. $N = |\nabla l|^{-1}\nabla l$ is the outward pointing unit normal of $\partial\Omega_t$. Define $F := fN$ a vector valued function. In particular $F \cdot N = fN \cdot N = f$. With the product rule it holds

$$\begin{aligned} \nabla \cdot ((T-l)F) &= \nabla(T-l) \cdot F + (T-l)\nabla \cdot F \\ &= -\nabla l \cdot F + (T-l)\nabla \cdot F \\ &= -\nabla l \cdot fN + (T-l)\nabla \cdot F \\ &= -l\frac{\nabla l \cdot \nabla l}{|\nabla l|} f + (T-l)\nabla \cdot F = -f|\nabla l| + (T-l)\nabla \cdot F. \end{aligned}$$

Using this it follows using the divergence theorem twice it follows

$$\begin{aligned}
\int_{\Omega_T} f|\nabla l|dx &= \int_{\Omega_T} (T-l)\nabla\cdot F dx - \int_{\Omega_T} \nabla\cdot((T-l)F)dx \\
&= \int_{\Omega_T} (T-l)\nabla\cdot F dx - \underbrace{\int_{\partial\Omega_T} (T-l)F\cdot N dx}_{=0, l=T \text{ on } \partial\Omega_T \text{ as } \Omega_T:=\{l<T\}} \\
&= \int_{\Omega_T}\int_{l(x)}^T 1 dt \nabla\cdot F dx = \int_{\Omega_T}\int_{l(x)}^T \nabla\cdot F dt dx \\
&= \int_{\Omega_T}\int_0^T \mathbf{1}_{\{t\geq l(x)\}}\nabla\cdot F dt dx = \int_0^T\int_{\Omega_T} \mathbf{1}_{\{t\geq l(x)\}}\nabla\cdot F dt dx \\
&= \int_0^T\int_{\mathbb{R}^n} \mathbf{1}_{\{t\geq l(x)\}\cap\Omega_T}\nabla\cdot F dx dt = \int_0^T\int_{\Omega_t} \nabla\cdot F dx dt \\
&= \int_0^T\int_{\partial\Omega_t} F\cdot N d\sigma dt = \int_0^T\int_{\partial\Omega_t} fN\cdot N d\sigma dt = \int_0^T\int_{\partial\Omega_t} f d\sigma dt
\end{aligned}$$

□

2 Distributions

We now introduce a generalization of the partial derivative and to the solution to a partial differential equation. A more general approach to a functions then satsfy these generalized partial derivatives and if they are differentiable the partial derivative as well. These functions are infinitely differentiable wrt. to this weak derivative, however they cannot be multiplied by each other.
In what way do non-linear PDEs result in a non linear equation on distribution???
We begin by defining a class of functions that are well behaved. For that define

$$supp\ f := \{x \mid f(x) \neq 0\}$$

On an open set $\Omega \subset \mathbb{R}^n$ we define the set of test functions $C_0^\infty(\Omega)$ as the set of all smooth functions whoose support is a compact subset of Ω. Within these test functions there is a subclass called mollifiers

Definition 2.1

The family of non-negative test functions $(\lambda_\epsilon)_{\epsilon\geq 0}$ with $supp\ \lambda_\epsilon = \overline{B_\epsilon(0)}$ and $\int \lambda_\epsilon d\mu = 1$ is called mollifiers.

Example 2.1

$$\lambda(x) = \begin{cases} Ce^{\frac{1}{|x|^2-1}} & , |x| < 1 \\ 0 & , |x| \geq 1 \end{cases}$$

is a smooth function on $\mathbb{R}^n$, its support is $\overline{B_1(0)}$ and its non-negative. We need to choose the constant C in such a way that the integral is one. Rescaling yields this property

$$\begin{aligned}\int_{\mathbb{R}^n} \lambda_\epsilon(x) d^n x &= \frac{1}{\epsilon^n} \int_{B_1(0)} \lambda(\frac{x}{\epsilon}) d^n x = \frac{1}{\epsilon^n} \int_{B_\epsilon(0)} \lambda(x) \epsilon^n d^n x \\ &= C \int_{B_\epsilon(0)} e^{\frac{1}{|x^2|-1}} d^n x = 1 \quad \iff \quad C = \frac{1}{\int_{B_\epsilon(0)} e^{\frac{1}{|x^2|-1}} d^n x}.\end{aligned}$$

We call this example standard mollifier.
For a continuous function f on Ω with $0 \in \Omega$ we call the following equation approximate identity

$$\int_\Omega f \lambda_\epsilon d\mu = \int_{B_\epsilon(0)} f \lambda_\epsilon d\mu \approx \int_{B_\epsilon(0)} f(0) \lambda_\epsilon d\mu = f(0),$$

where we used that

$$\forall x \in B_\epsilon(0): \quad f(x) \approx f(0).$$

In the next lemma we show that in the limit $\epsilon \searrow 0$ we get equality.

Lemma 2.2

Let $f \in C(\Omega)$ and $(\lambda_\epsilon)_{\epsilon \geq 0}$ a mollifier. The following family

$$f_\epsilon(x) := \int_\Omega f(y) \lambda_\epsilon(x-y) d^n y$$

is smooth and converges uniformly on any compact subset of Ω to f as $\epsilon \searrow 0$. As the derivative of smooth functions is again smooth, this result also holds for their derivatives.

Proof. 1.
Let $A \subset \Omega$ be compact then it holds that

$$\exists \epsilon > 0 \forall x \in A: \quad B_\epsilon(0) \subset \Omega.$$

Then we have that

$$\begin{aligned}|f_\epsilon(x) - f(x)| &= |\int_\Omega f(y) \lambda_\epsilon(x-y) d^n y - f(x)| \\ &= |\int_\Omega f(y) \lambda_\epsilon(x-y) d^n y - \underbrace{\int_\Omega \lambda_\epsilon(x-y) d^n y}_{=1} f(x)| \\ &= |\int_\Omega (f(y) - f(x)) \lambda_\epsilon(x-y) d^n y \\ &\leq \sup_{y \in B_\epsilon(0)} |f(y) - f(x)|.\end{aligned}$$

Due to the fact that continuous functions are uniformly continuous on a compact subset we get convergence $f_\epsilon \to f$ as $\epsilon \searrow 0$.
2.

As the partial derivaitve is the limit of the difference quotient, we need to show for all $x_0 \in \Omega$ on an open ball $B_\delta(x_0)$ that

$$\partial^\alpha f_\epsilon(x) = (\partial^\alpha f)_\epsilon(x).$$

Due to the fact that the mollifier is defined on $B_\epsilon(x)$ choose $\delta = 2\epsilon$. Then we have

$$\partial^\alpha f_\epsilon(x) = \partial^\alpha \int_{B_\epsilon(x)} \lambda_\epsilon(x-z) f(z) d^n z = \int_{B_\epsilon(0)} \lambda_\epsilon(z) \partial^\alpha f(x-z) d^n z = \int_{B_\epsilon(0)} \lambda_\epsilon(z)(\partial^\alpha f)(x-z) d^n z = (\partial^\alpha f)_\epsilon(x).$$

As the partial derivative of a smooth function is smooth again, the assertion follows with the first step. □

For $f, g \in C_0^\infty(\mathbb{R}^n)$ we define the convolution as

$$(g * f)(x) := \int_{\mathbb{R}^n} g(x-y) f(y) d^n y = \int_{\mathbb{R}^n} g(z) f(x-z) d^n z,$$

which can be shown to be linear, commutative and associative.

Remark 2.1

The advantage of convolution compared to pointwise multiplication is that it behaves nicely with differentiation. One can show that

$$\partial^\alpha (g * f) = (\partial^\alpha g) * f = g * (\partial^\alpha f).$$

Further one can show that convolution is volume preserving under the coordinate transform $z = y - x$ and $y = y$:

$$\begin{aligned}\int_{\mathbb{R}^n} (f * g)(x) dx &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x-y) g(y) dx dy = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} -(-1) f(z) g(y) dz dy \\ &= \left(\int_{\mathbb{R}^n} f(z) dz \right) \left(\int_{\mathbb{R}^n} g(y) dy \right).\end{aligned}$$

Lemma 2.3

Suppose f and g are rotationally symmetric around symm. around a and b respectively, i.e. for every O orthogonal transformation holds $f(a+x) = f(a+Ox)$. It then follows that $f * g$ is rotationally symmetric around $a + b$.

Proof.

$$\begin{aligned}(f*g)(a+b+Ox) &= \int_{\mathbb{R}^n} f(a+b+Ox-y)g(y)dy \\ &\overset{(*)}{=} \int_{\mathbb{R}^n} f(a+b+Ox-(Oz+b))g(Oz+b)dz \\ &= \int_{\mathbb{R}^n} f(a+O(x-z))g(Oz+b)dz \\ &\overset{VSS}{=} \int_{\mathbb{R}^n} f(a+x-z)g(b+z)dz \\ &= \int_{\mathbb{R}^n} f(a+x-(y'-b))g(b+y'-b)dy' \\ &= \int_{\mathbb{R}^n} f(a+x-y'+b)g(b+y'-b)dy' \\ &= (f*g)(a+b+x)\end{aligned}$$

Although we used in $(*)$ that $dy = dz$. This is due to $y(z) := Oz + b$ has the jacobian $\frac{\partial}{\partial z} y(z_0) = O$ and thus with the Jacobi-transformation theorem we have $det(O) = 1$. □

For every $f \in L^1_{loc}(\Omega)$ we define

$$F_f : C_0^\infty(\Omega) \to \mathbb{R}, \phi \mapsto \int_\Omega f\phi d\mu$$

and will see that information wrt. derivatives is retained in this linear form. The idea of distributions is to consider not just functions integrated against test functions, but all linear forms F defined on $C_0^\infty(\Omega)$??? To evade the difficulty of defining the topology on $C^\infty(\Omega)$, we use an equivalent criterion. $\mathcal{D}(\Omega)$ is the set of functions $f \in C_0^\infty(\Omega)$ that satisfy $f_n \to f \in C_0^\infty(\Omega) :\iff$

- $\exists K \subset \Omega$ compact $\forall n \in \mathbb{N}$: $supp f_n \subset K$
- $\forall \alpha \in \mathbb{N}_0^n$: $sup_K |\partial^\alpha f_n - \partial^\alpha f| \to 0$, $n \to \infty$.

Remark 2.2

We define the seminorm

$$\|\cdot\|_{K,\alpha} : C_0^\infty(\Omega) \to \mathbb{R}, \phi \mapsto \|\phi\|_{K,\alpha} := \sup_{x\in K} |\partial^\alpha \phi(x)|.$$

It is a seminorm, as constant functions $f \equiv const.$ have $\|f\|_{K,\alpha} = 0$, but they are not zero!
Note that convergence $\|f_n - f\|_{K,\alpha} \to 0$ is stronger than $\|f_n - f\|_\infty$. This can be seen by the following counterexample.

Example 2.2

Define $f_n(x) := \frac{1}{n}\sin(nx)\mathbf{1}_{[-1,1]}(x)$. Then we have that

$$\|f_n(x)\| \le \frac{1}{n}.$$

and because $\frac{1}{n}$ converges to zero, we get converge in $\|\cdot\|_\infty$. But on the other hand for $\alpha = 1$ and $n = 1$ we have

$$\begin{aligned}\sup_{x\in[-1,1]} |\partial_x f_n(x)| &= \sup_{x\in[-1,1]} |\cos(nx)\mathbf{1}_{[-1,1]}(x) + \frac{1}{n}\sin(nx)\frac{\partial}{\partial x}\mathbf{1}_{[-1,1]}(x)| \\ &\geq \sup_{x\in[-1,1]} |\cos(nx)| = 1,\end{aligned}$$

i.e. it is bounded by below by 1 for all $n \in \mathbb{N}$ and thus not convergent to zero.

Definition 2.4

On an open $\Omega \subseteq \mathbb{R}^n$ we define $\mathcal{D}'(\Omega)$ the space of all linear maps

$$F : \mathcal{D}(\Omega) \to \mathbb{R}$$

which are continiuous wrt. seminorms $\|\cdot\|_{K,\alpha}$ as the space of distributions, i.e.

$\forall K \subset \Omega$ compact $\exists \alpha_1, ..., \alpha_M, M < \infty, C_1, ..., C_M > 0 \forall \phi \in \mathcal{D}(\Omega) : |F(\phi)| \leq C_1\|\phi\|_{K,\alpha_1} + ... + C_M\|\phi\|_{K,\alpha_M}$.

Note that with continuity we get that

$$\phi_n \to \phi \text{ in } \mathcal{D}(\Omega) \quad \Rightarrow \quad F(\phi_n) \to F(\phi) \text{ in } \mathbb{R}.$$

and

$$\forall \phi \in D(\Omega): \quad F_n(\phi) \to F(\phi) \quad \Rightarrow \quad F_n \to F \text{ in } \mathcal{D}(\Omega).$$

Example 2.3

First, we show that $F : L^1_{loc}(\Omega) \to \mathbb{R}$ is a distribution. We show continuity:
$\forall K \subset \Omega$ compact and $\forall \phi \in \mathcal{D}(\Omega)$ with $supp\phi \subset K$:

$$|F_f(\phi)| \leq \int_K |f||\phi|dx \leq \sup_{x\in K} |\phi(x)| \int_K |f|dx = \sup_{x\in K} |\phi(x)| \underbrace{\|f\|_{L^1(K)}}_{\leq const.}$$

Next, one can show that

$$\delta_0 : \mathcal{D}(\mathbb{R}^n) \to \mathbb{R}, \quad \phi \mapsto \phi(0)$$

is a distribution, because $\forall \phi \in \mathcal{D}(\Omega)$ with $supp\ \phi \subset K$:

$$|\delta_0(\phi)| = |\phi(0)| \leq \sup_{x\in K} |\phi(x)|.$$

Distributions that can be written $\forall f \in L^1_{loc}(\Omega)$ as

$$F_f : C_0^\infty(\Omega) \to \mathbb{R}, \quad \phi \mapsto \int_\Omega f\phi d\mu$$

are called regular, i.e. they can be written in the integral form. The Dirac distribution cannot be written in this way, as no function in $L^1_{loc}(\Omega)$ exists that vanishes on $\mathbb{R}^n \setminus \{0\}$ and its total integral is one, i.e.

$$\int_{\mathbb{R}^n} f d\mu = 1, \quad \forall x \in \mathbb{R}^n \setminus \{0\} : \quad f(x) = 0.$$

One can also show that the Dirac distribution is the limit of mollifiers.

Lemma 2.5

If $f \in L^1_{loc}(\Omega)$ satisfies $F_f(\phi) \geq 0 \; \forall \phi \in C_0^\infty(\Omega)$ then it follows $f \geq 0$ a.e. and $L^1_{loc}(\Omega) \to \mathcal{D}'(\Omega)$, $f \mapsto F_f$ is injective.

Proof. As we only show the non negativity, which is a local statement, it suffices to show it for $f \in L^1(\Omega)$, i.e. any arbitary compact subset. We can then extend f to $\mathbb{R}^n$ by setting it on $\mathbb{R}^n \setminus \Omega$ to zero.
First, we want to show that $(f * \lambda_\epsilon)_{\epsilon>0}$ converges to $f \in L^1(\Omega)$ using the "standard machinery" (Gebetsmühle).
It holds that for a mollifier $(\lambda_\epsilon)_{\epsilon>0}$

$$\begin{aligned}
\|\lambda_\epsilon * f - f\|_1 &= \int_{\mathbb{R}^n} \left| \int_{B_\epsilon(0)} \lambda_\epsilon(y) f(x-y) d^n y - f(x) \right| d^n x \\
&\leq \int_{\mathbb{R}^n} \int_{B_\epsilon(0)} \lambda_\epsilon(y) \left| f(x-y) - f(x) \right| d^n y d^n x \\
&\leq \int_{B_\epsilon(0)} \int_{\mathbb{R}^n} \lambda_\epsilon(y) \left| f(x-y) - f(x) \right| d^n x d^n y \\
&\leq \sup_{y \in B_\epsilon(0)} \int_{\mathbb{R}^n} \lambda_\epsilon(y) \left| f(x-y) - f(x) \right| d^n x \\
&\leq \sup_{y \in B_\epsilon(0)} \| f(\cdot - y) - f(\cdot) \|_{L^1(\Omega)}
\end{aligned}$$

- For indicator function

$$\mathbf{1}_A(x) = \begin{cases} 1 & , x \in A \\ 0 & , x \notin A \end{cases}, \quad A = [a_1, b_1] \times ... \times [a_n, b_b]$$

  we have with the above

$$\|\lambda_\epsilon * \mathbf{1}_A - \mathbf{1}_A\|_1 \leq \sup_{y \in B_\epsilon(0)} \|\mathbf{1}_{A-y} - \mathbf{1}_A\| \to 0, \quad \epsilon \to 0.$$

- For step functions

$$f(x) := \sum_{i=1}^n a_i \mathbf{1}_{Q_i}(x), \quad \bigcup_{i=1}^n Q_i = \Omega$$

  Then using triangle inequality we get

$$\|\lambda_\epsilon * f - f\|_1 \leq \sum_{i=1}^n \|\lambda_\epsilon * (a_i \mathbf{1}_A) - a_i \mathbf{1}_A\|_{L^1(Q_i)} \leq \sum_{i=1}^n \sup_{y \in B_\epsilon(0)} |a_i| \|\mathbf{1}_{A-y} - \mathbf{1}_A\|_{L^1(Q_i)} \to 0, \quad \epsilon \to 0.$$

Because the step function lie dense in $L^1(\Omega)$ we get convegence.
Next, we need to show that the limit of $(\lambda_\epsilon * f)$, i.e. f is non negative, however, as $f \in L^1(\Omega)$ we have that f is only almost everywhere non negative. Both of these need to be shown.

- For $\epsilon > 0$ it holds

$$\forall x \in \Omega : \quad (\lambda_\epsilon * f)(x) = \int_{\mathbb{R}^n} \lambda_\epsilon(x-y) f(y) d^n y = F_f(\lambda_\epsilon(x - \cdot)) \overset{Assumption}{\geq} 0.$$

- In the above we have shown that there exists $(\epsilon_n)_{n\in\mathbb{N}}$ such that

$$\|\lambda_{\epsilon_n} * f - f\|_1 \leq 2^{-n}, \quad \forall n \in \mathbb{N},$$

which implies that $\sum_{n\in\mathbb{N}} |\lambda_{\epsilon_n} * f - f|$ converges in L^1. This means that almost everywhere the value of this series is finite. This then requires that almost everywhere we have

$$(f * \lambda_{\epsilon_n})(x) \to f(x).$$

□

Let $\Omega' \subset \Omega$ then every test function on Ω' is also a test function on Ω. If we restrict a distribution on Ω only to Ω' it is again a distribution. We call this restriction. Using restrictions we can define the support of a distribution. The compliment of the support of a distribution is the union over all stes on which the distribution/restriction vanishes:

$$(suppF)^C = \bigcup_{\Omega' \in \mathcal{P}(\Omega)} \{\Omega' \subset \Omega \mid F(\phi) = 0, \forall \phi \in \mathcal{D}(\Omega')\}$$

The support of a delta distribution is $\{0\}$ and for a $L^1_{loc}(\Omega)$ is the support in the classical sense.
We now want to define as many operations as possible on distributions and thus extend operations on functions. The idea is to compare F_f and F_{Af}, where A is some operation. If we can write the operation in a way that only depends on the distribution and not directly on the function it is a suitable definition to be generalized. First, we consider the multiplication by a smooth function $g \in C^\infty(\Omega)$ and $F : L^1_{loc}(\Omega) \to \mathbb{R}$

$$F_{fg}(\phi) = \int_\Omega (gf)\phi dx = \int_\Omega f(g\phi) dx = F_f(g\phi).$$

Thus we define the multiplication by a smooth function $g \in C^\infty(\Omega)$ in the following way.

Definition 2.6

For $g \in C^\infty(\Omega)$ define

$$gF : \mathcal{D}(\Omega) \to \mathbb{R}, \quad \phi \mapsto F(g\phi).$$

Note that the product of a distribution with a non-smooth function is not defined, because then $g\phi$ is not a test function. This is why we require g to be in $C^\infty(\Omega)$.

Remark 2.3

Remember that for two topoligical spaces X and Y the mapping $f : X \to Y$ is called an embedding of two topological spaces, if it is bijective, continuous and its inverse is continuous. ...???

The most important operation for distributions is its derivative. Integrating by parts yields

$$F_{\partial_i f}(\phi) = \int_\Omega \partial_i f \phi d^n x = -\int_\Omega f \partial_i \phi d^n x = -F_f(\partial_i \phi).$$

Definition 2.7

For all $F \in \mathcal{D}'(\Omega)$ we define
$$\partial_i F : \mathcal{D}' \to \mathbb{R}, \quad \phi \mapsto -F(\partial_i \phi).$$

Hence, because test functions are infinitely differentiable, it follows that distributions are infinitely differentiable.

Remark 2.4

These two new operations are distributions again and are clearly linear. One would need to show as well that they obey the continuity definition.

We now want to extend convolutions to distributions. Observe that
$$\begin{aligned} F_{g*f}(\phi) = \int_{\mathbb{R}^n} (g*f)\phi d^n x = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} g(x-y)f(y)\phi(x) d^n y d^n x \\ \stackrel{Fubini}{=} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \phi(x) g(x-y) d^n x f(y) d^n y = F_f(\phi * Pg), \end{aligned}$$
although $(Pg)(z) := g(-z)$ is the point reflection operator.

Definition 2.8

We define for $g \in C_0^\infty(\mathbb{R}^n)$ and $F \in \mathcal{D}'(\mathbb{R}^n)$
$$g * F : \mathcal{D}(\mathbb{R}^n) \to \mathbb{R}, \quad \phi \mapsto F(\phi * Pg).$$

One would need to check that this is again a distribution and it is even a regular distribution, i.e. $\phi * Pg \in (L^1_{loc}(\Omega))'$.

Lemma 2.9

Let $g \in C_0^\infty(\mathbb{R}^n)$ be a test function and a distribution $F \in \mathcal{D}'(\mathbb{R}^n)$, then their convolution satisfy
$$g * F \in C^\infty(\mathbb{R}^n)$$
and we can write it as
$$g * F : \mathbb{R}^n \to \mathbb{R}, \quad x \mapsto F(T_x Pg),$$
where $(T_x\phi)(y) := \phi(y-x)$ is the translation operator.
Finally, it holds that $supp g * F \subset supp g + supp F$.

Proof. We first show existence and continuity of $g * F$.
Existence:
$$\begin{aligned} suppT_x Pg &= \{y \in \mathbb{R}^n \mid y \in suppPg(\cdot - x)\} = \{y \in \mathbb{R}^n \mid y \in suppg(x - \cdot)\} \\ &= \{y \in \mathbb{R}^n \mid x - y \in suppg(\cdot)\} = x - supp\ g \end{aligned}$$
Thus, $\forall x \in \mathbb{R}^n$ and $F \in \mathcal{D}'(\mathbb{R}^n)$ we have that $F(T_x Pg)$ is well defined, because $supp\ g$ is well defined.
Continuity:

Fist, we see that $(T_xPg)(y) = Pg(y-x) = g(x-y)$, g is continuous and continuity implies uniform continuity on a compact subset, we have that $x \mapsto T_xPg$ is continuous wrt. $\|\cdot\|_{K,0}$. Remember, that $K \subset supp\, T_xPg$ is compact. We now need to show that

$$\frac{T_{x+\epsilon h} - T_x}{\epsilon} g \overset{uniformly}{\rightarrow} T_x \left(\sum_{i=1}^{n} -h_i \partial_i g \right), \quad \epsilon \to 0, \text{ on } \mathbb{R}^n.$$

This is due to the following: Remember that with Taylor and the mean value theorem

$$\frac{g(x-\epsilon h) - g(x)}{\epsilon} \overset{uniformly}{\rightarrow} -h\langle h, \nabla g(y)\rangle, \quad \epsilon \to 0$$

Using the fact that

$$\frac{T_{x+\epsilon h} - T_x}{\epsilon} g = T_x \frac{T_{\epsilon h} - Id}{\epsilon} g = T_x \frac{T_{\epsilon h} g - g}{\epsilon} = T_x \frac{g(\cdot - \epsilon h) - g(\cdot)}{\epsilon}$$

it follows that

$$\frac{T_{x+\epsilon h} - T_x}{\epsilon} g \overset{uniformly}{\rightarrow} T_x \langle h, \nabla g(x)\rangle = T_x \left(\sum_{i=1}^{n} -h_i \partial_i g \right).$$

Due to the fact that F is a distribution, i.e. continuous we get that

$$\forall x \in \mathbb{R}^n : \quad F\left(\frac{T_{x+\epsilon h} - T_x}{\epsilon} g\right) \to F\left(T_x \left(\sum_{i=1}^{n} -h_i \partial_i g \right)\right), \quad \epsilon \to 0.$$

which means that $x \mapsto F(\frac{T_{x+\epsilon h} - T_x}{\epsilon} g) \in C^\infty(\mathbb{R}^n; \mathbb{R})$, for $F \in \mathcal{D}'(\mathbb{R}^n)$.
We now show the equality. Above it was shown that $x \mapsto T_xPg$ is smooth on $\mathcal{D}(\mathbb{R}^n)$, i.e. wrt. the semi-norm $\|\cdot\|_{K,\alpha}$. We choose

$$K \subset supp\, T_xPg = x - supp\, g$$

as a compact subset. Thus a finite disjoint decomposition exists

$$K := \bigcup_{i=1}^{n} E_i.$$

Then the Riemann sum is given by

$$S_n := \sum_{i=1}^{n} (T_{x_j}Pg)(\cdot)\phi(x_i)|E_i|.$$

Due to $supp\, T_xPg = x - supp\, g$ we have that $T_xPg \in \mathcal{D}(\mathbb{R}^n)$. Multiplying with constants and a finite sum still leads to $S_n \in \mathcal{D}(\mathbb{R}^n)$. It then follows that for $x \in supp\phi$

$$\| \int_{\mathbb{R}^n} (T_xPg)(\cdot)\phi(x) d^n x - S_n \|_{K,\alpha} \leq \sum_{i=1}^{n} |E_i| \int_{\mathbb{R}^n} \|(T_xPg)(\cdot)\phi(x) - (T_{x_i}Pg)(\cdot)\phi(x_i)\|_{K,\alpha} d^n x$$

$\square$

Remark 2.5

This Lemma implies that the convolution of two distributions $F, G \in \mathcal{D}'(\mathbb{R}^n)$ where *supp* G is compact, is a well defined distribution

$$F * G : \mathcal{D}(\Omega) \to \mathbb{R}, \quad \phi \mapsto F(\phi * PG), \quad PG(\phi) := G(P\phi).$$

So the convolution with the δ-distribution is well defined. Remarkably, $F * \delta = F$, i.e. the δ-distribution is the neutral element of convolutions.

CHAPTER 3

LAPLACE EQUATION

We begin by solving the Laplace equation that is given by

$$\Delta u = \frac{\partial^2}{\partial x_1^2} u + ... + \frac{\partial^2}{\partial x_n^2} u = 0$$

If the right hand side $f \neq 0$ we call this equation Possion eqution, i.e. $-\Delta u = f$. This is a linear PDE with the solution $u : \mathbb{R}^n \to \mathbb{R}$.

1 Fundamendal Solution

This equation is invaraint wrt. all rotations and translations. This leads to the notion that our solution should also have these conditions. We defeine $r := |x| := \sqrt{x^T x}$ which defines the length of the position vector, i.e. the solution should only depend on this r. Further, define $u(x) := v(r)$. Then for $w(r) := v'(r)$

$$\nabla_x u(x) = v'(r) \nabla_x r = v'(r) \frac{2x}{2|x|}$$

which leads to

$$\Delta_x u(x) = \nabla_x \cdot \nabla_x u(x) = \nabla_x \cdot (v'(r) \frac{x}{|x|}) = v''(r) + \frac{n-1}{r} v'(r) = 0.$$

We see that this is only an ODE in r. Solving it is not hard: First it can be seen that

$$v''(r) + \frac{n-1}{r} v'(r) = 0 \quad \iff \quad (r^{n-1} v'(r))' = (n-1) v'(r) r^{n-2} + v'' r^{n-1} = r^n (\frac{n-1}{r} v'(r) + v'') = 0.$$

As the derivative stays constant we define $C_1 := r^{n-1} v'(r)$ which can also be written as $v'(r) = C_1 r^{1-n}$. This formulation now leads t0

$$v(r) = \begin{cases} C_2 + \frac{C_1}{2-n} r^{2-n} & , n \neq 2 \\ C_2 + C_1 \log r & , n = 2. \end{cases}$$

Note that for the solution the constant C_2 can be chosen arbitrarily as it will vanish whilst applying the Laplace operator. With this in mind we define

$$\Phi(x) := \begin{cases} \frac{C_1}{2-n} r^{2-n} & , n \neq 2 \\ C_1 \log r & , n = 2. \end{cases}$$

We now need to determine C_1.

Definition 1.1

The fundamental solution of the Laplace operator is

$$\Phi(x) := \begin{cases} -\frac{1}{2\pi}\log|x| & , n=2 \\ \frac{1}{n(n-2)\omega_n}\frac{1}{|x|^{n-2}} & , n\geq 3 \end{cases}$$

where ω_n denotes the volume of the n-dimensional unit ball.

In the following we will show that the solution of $-\Delta u = f$ is the convolution of f with the fundamental solution Φ. Note that due to

$$\|(f*g)(x)\|_1 \leq \|\int f dx\|_1 \|\int g dx\|_1,$$

it follows that the convolution is well defined for functions in $L^1(\mathbb{R}^n)$. However, as we need the solution to be twice differentiable, we need f to be twice differentiable.

Theorem 1.2

For $f\in C_0^2(\mathbb{R}^n)$ the solution of the Poisson equation $-\Delta u = f$ is given by the convolution

$$u(x) = (\Phi * f)(x) = \int_{\mathbb{R}^n}\Phi(x-y)f(y)d^n y = \int_{\mathbb{R}^n}\Phi(z)f(x-z)d^n z.$$

Proof. The first equality in the theorem comes from the substitution $z = x-y$. Note that the second integral is twice continuous differentiable, because we can pull in in the derivative in the Integral due to the fact that $f\in C_0^2(\mathbb{R}^n)$:

$$\frac{\partial^2}{\partial x_i \partial x_j}u(x) = \int_{\mathbb{R}^n}\Phi(z)\frac{\partial^2}{\partial x_i\partial x_j}f(x-z)d^n z.$$

We will now write $\Delta_x u(x) = \int_{\mathbb{R}^n}\Phi(y)\Delta_x f(x-y)d^n y$. The trick now will be to decompose the integral into a open area around zero (where there is a singularity) and everything else. Then we will let the radius of that open area go to zero. Then the integral evaluated at the singularity will be zero, as it will be a zero set. Define for $\epsilon > 0$:

$$\Delta_x u(x) = \underbrace{\int_{\mathbb{R}^n\setminus B_\epsilon(0)}\Phi(y)\Delta_x f(x-y)d^n y}_{=:J_\epsilon} + \underbrace{\int_{B_\epsilon(0)}\Phi(y)\Delta_x f(x-y)d^n y}_{=:I_\epsilon}$$

Step 1 I_ϵ:
We want to show that the integral

$$|I_\epsilon| := |\int_{B_\epsilon(0)}\Phi(x)\Delta_y f(y-x)d^n x| \leq \|\Delta_y f\|_{L^\infty(\mathbb{R}^n)}\int_{B_\epsilon(0)}|\Phi(x)|d^n x$$

is finite and converges to zero. Note that using the Co-Area theorem we have that

$$B_\epsilon(0) := \{x\in\mathbb{R}^n \mid l(x) := |x| < \epsilon\}$$

and due to $1 = |\nabla l(x)|$ we get that

$$\int_{B_\epsilon(0)} \phi(x) d^n x = \int_{B_\epsilon(0)} \phi(x)|\nabla l(x)| d^n x = \int_0^\epsilon \int_{\partial B_r(0)} \phi d\sigma dr.$$

We now want to do a Jacobain transformation to the space of S^{n-1}. For this define the map $T : S^{n-1} \to \partial B_r(0), s \mapsto T(s) := rs$. But, we have the problem, that the Jabobian is only defined for subsets for euclidean spaces! Thus we need to introduce two new maps. First, $\psi : V \subset \mathbb{R}^{n-1} \to \partial B_r(0)$ and $\phi : U \subset \mathbb{R}^{n-1} \to S^{n-1}$, where U, V are open subsets. Both are regular parametrizations. Then the write for the composition

$$F := \psi^{-1} \circ T \circ \phi : U \to V.$$

Its Jacobian we do can calculate. Fist, using the chain rule

$$DF(x) = D(\psi^{-1})(T(\phi(x)))DT(\phi(x))D\phi(x).$$

Then we use that for $y \in U$:

$$Id = D(\psi \circ \psi^{-1})(y) = D\psi(\psi^{-1}(y))D\psi^{-1}(y) \quad \Longleftrightarrow \quad D\psi^{-1}(y) = (D\psi(\psi^{-1}(y)))^{-1}.$$

Using this in the above we get that

$$\begin{aligned} DF(x) &= D(\psi^{-1})(T(\phi(x)))DT(\phi(x))D\phi(x) \\ &= (D\psi(\psi^{-1}(T(\phi(x)))))^{-1}DT(\phi(x))D\phi(x) \\ &= (D\psi(F(x)))^{-1}DT(\phi(x))D\phi(x) \end{aligned}$$

We then have that

$$DF(x) = (D\psi(F(x)))^{-1}DT(\phi(x))D\phi(x) \iff D\psi(F(x))DF(x) = DT(\phi(x))D\phi(x)$$

We now use the general fact that $AB = C$ implies that

$$(AB)^T AB = B^T A^T AB = C^T C$$

which leads to

$$\det(B^T B)\det(A^T A) = \det(C^T C) \iff \det(A^T A) = \frac{\det((C^T C))}{\det((B^T B))}.$$

We do this so complicated, as the images are manifolds, i.e. n-1 dimensional subspaces in $\mathbb{R}^n$ and thus we cannot calculate the determinant, as their Jacobians are not square matrices. Now using $DT(\phi(x)) = Id \cdot r$ we get that

$$\det(DF^T DF) = \frac{\overbrace{\det(Id \cdot r)^2}^{=r^{2(n-1)}} \det(D\phi^T D\phi)}{\det(D\psi(F)^T D\psi(F))}.$$

Due to the fact that F has a quadratic Jaconian we have

$$|\det(DF)| = r^{n-1}\sqrt{\frac{\det(D\phi^T D\phi)}{\det(D\psi(F)^T D\psi(F))}} \tag{3.1}$$

Let A_i be a cover of $\partial B_r(0)$ and O_i a cover of S^{n-1}, then due to $\psi \circ F = T \circ \phi$ the following calculation holds for any $f \in C^1(\mathbb{R}^n)$:

$$
\begin{aligned}
\int_{B_\epsilon(0)} f(x) d^n x &\overset{Co-Area}{=} \int_0^\epsilon \int_{\partial B_r(0)} f d\sigma dr \\
&= \int_0^\epsilon \sum_i \int_{A_i} \underbrace{h_i f}_{=: f_i} d\sigma dr \\
&= \int_0^\epsilon \sum_i \int_{V_i} f_i \circ \psi \sqrt{\det(D\psi^T D\psi)} d\mu_{\mathbb{R}^{n-1}} dr \\
&\overset{JacobiTransform}{=} \int_0^\epsilon \sum_i \int_{U_i} (f_i \circ \psi \circ F) \sqrt{\det((D\psi(F))^T D\psi(F))} |\det(DF)| d\mu_{\mathbb{R}^{n-1}} dr \\
&= \int_0^\epsilon \sum_i \int_{U_i} (f_i \circ T \circ \phi) \sqrt{\det((D\psi(F))^T D\psi(F))} |\det(DF)| d\mu_{\mathbb{R}^{n-1}} dr \\
&\overset{Eq. 3.1}{=} \int_0^\epsilon \sum_i \int_{U_i} (f_i \circ T \circ \phi) \sqrt{\det((D\psi(F))^T D\psi(F))} r^{n-1} \sqrt{\frac{\det(D\phi^T D\phi)}{\det(D\psi(F)^T D\psi(F))}} d\mu_{\mathbb{R}^{n-1}} dr \\
&= \int_0^\epsilon \sum_i \int_{U_i} (f_i \circ T \circ \phi) r^{n-1} \sqrt{\det(D\phi^T D\phi)} d\mu_{\mathbb{R}^{n-1}} dr \\
&= \int_0^\epsilon \sum_i \int_{O_i} (f_i \circ T) r^{n-1} d\sigma dr \\
&= \int_0^\epsilon \int_{S^{n-1}} (f \circ T) r^{n-1} d\sigma dr
\end{aligned}
$$

Then it holds that

$$
\int_{B_\epsilon(0)} |\Phi(y)| d^n y \overset{above}{=} \int_0^\epsilon \int_{S^{n-1}} |\Phi \circ T| d\sigma r^{n-1} dr = \int_0^\epsilon \int_{S^{n-1}} |\Phi(rx)| d\sigma(x) r^{n-1} dr.
$$

We see that for all $x \in S^{n-1}$ it follows $|x| = 1$ and thus

$$
\Phi(xr) = \Phi(re), \quad \text{where } e \text{ is any unit vector.}
$$

Due to the fact that

$$
\begin{aligned}
\int_{S^{n-1}} 1 d\sigma(x) &= \int_{S^{n-1}} |x|^2 d\sigma(x) = \int_{S^{n-1}} F\nu d\sigma = \int_{B_1(0)} divF d^n x \\
&= \int_{B_1(0)} \sum_{i=1}^n \frac{\partial}{\partial x_i} x_i d^n x = n\lambda^n(B_1(0)) =: n\omega_n
\end{aligned}
$$

it finally follows

$$\begin{aligned}
\int_{B_\epsilon(0)} |\Phi(y)| d^n y &= \int_0^\epsilon \int_{S^{n-1}} |\Phi(rx)| d\sigma(x) r^{n-1} dr = \int_0^\epsilon r^{n-1} |\Phi(re)| \int_{S^{n-1}} 1 d\sigma(x) dr \\
&= \int_0^\epsilon r^{n-1} |\Phi(re)| n\omega_n dr \\
&= \begin{cases} \int_0^\epsilon r |\log(r)| dr & , n = 2 \\ \int_0^\epsilon r^{n-1} \frac{1}{(n-2)r^{n-2}} dr & , n \geq 3 \end{cases} \\
&= \begin{cases} \frac{\epsilon^2}{2}(|\log(\epsilon)| + \frac{1}{2}) & , n = 2 \\ \int_0^\epsilon r \frac{1}{n-2} dr = \frac{\epsilon^2}{2(n-2)} & , n \geq 3 \end{cases}
\end{aligned}$$

Note that in the case $n = 2$ we have $\omega_n = \pi$. Both terms are finite! And because the upper bounds depend on ϵ it follows that $I_\epsilon \to 0$ as $\epsilon \to 0$.

Step 2 J_ϵ:

First notice that for $z := x - y$ it holds that $\sum_j \partial f(z) \frac{\partial}{\partial x_i} z = \partial_i f(x - z) = -\sum_j \partial f(z) \frac{\partial}{\partial y_i} z$ and thus $|\Delta_x f(x - y)| = |\Delta_y f(x - y)|$. It follows

$$\begin{aligned}
|J_\epsilon| &\leq \int_{\mathbb{R}^n \setminus B_\epsilon(0)} |\Phi(y)| |\Delta_y f(x - y)| d^n y \\
&\overset{\phi\Delta f = (\phi\nabla f)' - \nabla\phi\nabla f}{=} \int_{\mathbb{R}^n \setminus B_\epsilon(0)} |\nabla_y \Phi(y)| |\nabla f(x - y)| d^n y - \int_{\mathbb{R}^n \setminus B_\epsilon(0)} (|\Phi(y)| |\nabla f(x - y)|)' d^n y \\
&= \int_{\mathbb{R}^n \setminus B_\epsilon(0)} |\nabla_y \Phi(y)| |\nabla f(x - y)| d^n y - \int_{\partial B_\epsilon(0)} |\Phi(y)| |\nabla f(x - y)| N d\sigma(y) \\
&=: K_\epsilon - L_\epsilon.
\end{aligned}$$

First with the same arguments as above it follows

$$|L_\epsilon| \leq \|\nabla f\|_{L^\infty(\mathbb{R}^d)} \int_{\partial B_\epsilon(0)} |\Phi(y)| d\sigma(y) \leq \begin{cases} C\epsilon |\log(\epsilon)| & , n = 2 \\ C\epsilon & , n \geq 3 \end{cases}$$

This clearly converges to zero as $\epsilon \to 0$. For K_ϵ we have that

$$\begin{aligned}
K_\epsilon &= \int_{\mathbb{R}^n \setminus B_\epsilon(0)} \nabla_y \Phi(y) \nabla_y f(x - y) d^n y \\
&= \underbrace{\int_{\mathbb{R}^n \setminus B_\epsilon(0)} \Delta_y \Phi(y) f(x - y) d^n y}_{=0} - \int_{\mathbb{R}^n \setminus B_\epsilon(0)} (\nabla_y \Phi(y) f(x - y))' d^n y \\
&= -\int_{\partial B_\epsilon(0)} \nabla_y \Phi(y) f(x - y) N d\sigma(y).
\end{aligned}$$

Note that $\nabla \Phi(y) = -\frac{y}{n\omega_n |y|^n}$ and $N = -\frac{y}{|y|}$. Further, due to the fact that

$$\begin{aligned}
r^n n\omega_n &= n \int_{B_r(0)} 1 d^n x = \int_{B_r(0)} \nabla \cdot x d^n x = \int_{\partial B_r(0)} x N d\sigma(x) \\
&= \int_{\partial B_r(0)} x \frac{x}{|x|} d\sigma(x) = \int_{\partial B_r(0)} |x| d\sigma(x) = r \int_{\partial B_r(0)} d\sigma(x) = r\sigma_n(r)
\end{aligned}$$

The above is equivalent to $\sigma_n(r) = n\omega_n r^{n-1}$. This finally leads to

$$\begin{aligned}
K_\epsilon &= -\int_{\partial B_\epsilon(0)} \nabla_y \Phi(y) f(x-y) N d\sigma(y) \\
&= \int_{\partial B_\epsilon(0)} \frac{y}{n\omega_n |y|^n} f(x-y)(-1)\frac{y}{|y|} d\sigma(y) \\
&= -\int_{\partial B_\epsilon(0)} \frac{|y|}{n\omega_n \epsilon^n} f(x-y) d\sigma(y) \\
&= -\frac{\epsilon}{n\omega_n \epsilon^n} \int_{\partial B_\epsilon(0)} f(x-y) d\sigma(y) \\
&= -\underbrace{\frac{1}{\sigma_n(\epsilon)}}_{\to 1, \epsilon \to 0} \underbrace{\int_{\partial B_\epsilon(0)} f(x-y) d\sigma(y)}_{\to -f(x), \epsilon \to 0} \to -f(x), \quad \epsilon \to 0,
\end{aligned}$$

because f is continuous. Thus in total we have shown that

$$\Delta_x u(x) = J_\epsilon + I_\epsilon = J_\epsilon + K_\epsilon - L_\epsilon \le |J_\epsilon| + K_\epsilon + |L_\epsilon| \to -f(x), \quad \epsilon \to 0.$$

□

Thus due to the fact that we can write for every $\phi \in C_0^\infty(\mathbb{R}^n)$:

$$\begin{aligned}
(f * \delta_0)(\phi) = \delta_0(\phi * Pf) &= \int_{\mathbb{R}^n} (\phi * Pf)(x) d\delta_0(x) \\
&= (\phi * Pf)(0) = \int_{\mathbb{R}^n} f(x)\phi(x) d^n x \\
&= \int_{\mathbb{R}^n} (-\Delta_x u(x))\phi(x) d^n x = \int_{\mathbb{R}^n} -\Delta_x(\Phi * f)(x)\phi(x) d^n x \\
&= \int_{\mathbb{R}^n} ((-\Delta_x \Phi) * f)(x)\phi(x) d^n x \\
&= (f * (-\Delta_x \Phi))(\phi),
\end{aligned}$$

it follows that $-\Delta_x \Phi = \delta_0$ in the distributional sense.

2 Mean Value Property

The value of u at the center of any $B_r(0)$ with compact closure ($\bar{A} := \{x \in \Omega \mid \forall \epsilon > 0 : \mathcal{U}_\epsilon(x) \cap A \neq \emptyset\}$) in Ω is equal to the mean of u on the boundary of the ball. In the following we will show that this porperatry holds for any harmonic function. Remember, these were functions that satisfy $\Delta u = 0$. One can also show that if this mean value property holds for a function u then it is harmonic. We will also show that the mean on $B_r(x)$ is the mean over the means of u on $\partial B_{r'}(x)$ for all $r' \in [0, r]$. I.e. after having shown both directions, the mean value property is equivalent to functions beeing harmonic.

Lemma 2.1

- Let $u \in C^2(\Omega)$ be harmonic on an open domain $\Omega \subset \mathbb{R}^n$ containing $\overline{B_r(x)}$. The mean of u on $B_r(x)$ and on $\overline{B_r(x)}$ is equal to $u(x)$, i.e. the center.
- If for $u \in C^2(\Omega)$ the means on all Balls $B_r(x)$ with compact closure in Ω or all means on the boundaries of such balls is equal to $u(x)$ then u is harmonic.

Proof. Define for $x \in \Omega$ the mean of u on $\partial B_r(x) \subseteq \Omega$ as

$$\begin{aligned} S_x(r) &:= \frac{1}{r^{n-1} n\omega_n} \int_{\partial B_r(x)} u(y) d\sigma(y) = \frac{1}{r^{n-1} n\omega_n} \int_{\partial B_1(x)} (u \circ T)(y) r^{n-1} d\sigma(y) \\ &= \frac{1}{n\omega_n} \int_{\partial B_1(0)} (u \circ G)(ry) d\sigma(y) = \frac{1}{n\omega_n} \int_{\partial B_1(0)} u(x+ry) d\sigma(y) \end{aligned}$$

where $T : \partial B_1(x) \to \partial B_r(x), y \mapsto ry$ and $G : \partial B_r(0) \to \partial B_r(x), y \mapsto x + y$. The transformation with T has already been discussed. Transforming with G is only a translation and its Jabobian is trivially Id. Note that one would need to use parametrizations in order to calculate the Jacobian on euclidean spaces.
With the divergence theorem we get

$$\frac{d}{dr} S_x(r) = \frac{1}{n\omega_n} \int_{\partial B_1(0)} \frac{d}{dr} u(x+ry) d\sigma(y) = \frac{1}{n\omega_n} \int_{\partial B_1(0)} \nabla u(x+ry) y d\sigma(y)$$

Note that y is on $\partial B_1(0)$ a normal vector, i.e. $|y| = 1$ and if we consider $B_1(0)$ then it is an outward pointing normal vector. If we now transform back to $\partial B_r(x)$ then $\frac{y-x}{r} = \frac{y-x}{|y-x|}$ is an outward pointing normal vector! Thus we get

$$\begin{aligned} \frac{d}{dr} S_x(r) &= \frac{1}{n\omega_n} \int_{\partial B_1(0)} \nabla u(x+ry) y d\sigma(y) = \frac{1}{n\omega_n} \int_{\partial B_1(0)} \nabla u(x+ry) y d\sigma(y) \\ &= \frac{1}{r^{n-1} n\omega_n} \int_{\partial B_r(x)} \nabla u(y) \underbrace{\frac{y-x}{|y-x|}}_{=N} d\sigma(y) = \frac{1}{r^{n-1} n\omega_n} \int_{B_r(x)} \underbrace{\nabla \cdot \nabla u(y)}_{=\Delta u(y)} d^n y. \end{aligned}$$

The above only holds for $\overline{B_r(x)}$ compact in Ω, because ???. If u is harmonic, i.e. by definition $\Delta u \equiv 0$, then it follows that $\frac{d}{dr} S_x(r) = 0$ for all such r, i.e. $S_x(r)$ is constant in r. This just means that the average on $\partial B_r(x)$ does not depend on r. Further, because u is continuous it follows that $S_x(r) \to u(x), r \to 0$. THus means that the mean is the center value, because $S_x(0) = u(x)$ and $S_x(\cdot)$ is constant.
Now it holds that

$$\int_{B_r(x)} u(y) d^n y = \int_{B_r(0)} u(x+y) d^n y = \int_{B_r(0) \cap \{y_n > 0\}} u(x+y) d^n y + \int_{B_r(0) \cap \{y_n < 0\}} u(x+y) d^n y + \underbrace{\int_{B_r(0) \cap \{y_n = 0\}} u(x+y) d^n y}_{=0}.$$

In order to calculate these two integrals we need to do a Jacobi tranformation. We take the mapping

$$\phi_+ : \{(z,s) \in B_r^{n-1}(0) \times (0,r) \mid |z| < s\} \to B_r(0) \cap \{y_n > 0\}, (z,s) \mapsto \left(z, \sqrt{s^2 - |z|^2}\right)$$

and

$$\phi_- : \{(z,s) \in B_r^{n-1}(0) \times (0,r) \mid |z| < s\} \to B_r(0) \cap \{y_n > 0\}, (z,s) \mapsto \left(z, -\sqrt{s^2 - |z|^2}\right).$$

which gives us

$$\int_{B_r(0)\cap\{y_n>0\}} u(x+y)d^n y = \int_0^r \int_{B_s^{n-1}(0)} u\left(x+\left(z,\sqrt{s^2-|z|^2}\right)\right) \underbrace{|\det(D\phi_+(z,s))|}_{\overset{(*)}{=}\frac{s}{\sqrt{s^2-|z|^2}}} d^{n-1}z ds.$$

We now need to find a parametrization ψ that satisfies $det((\psi')^T\psi') = \frac{s}{\sqrt{s^2-|z|^2}}$ and then we can write it as an integral over a manifold. Fot that define

$$\psi_+ : B_s^{n-1}(0) \to \partial B_s^n(0) \cap \{y_n > 0\}, z \mapsto \left(z, \sqrt{s^2-|z|^2}\right).$$

This gives us

$$\begin{aligned}\int_{B_r(0)\cap\{y_n>0\}} u(x+y)d^n y &= \int_0^r \int_{B_s^{n-1}(0)\cap\{y_n>0\}} u\left(x+\underbrace{\left(z,\sqrt{s^2-|z|^2}\right)}_{=\psi_+(z)}\right)|\det((\psi_+')^T\psi_+')|d^n z ds \\ &= \int_0^r \int_{\partial B_s^{n-1}(0)\cap\{y_n>0\}} u(x+y)d\sigma(y)ds.\end{aligned}$$

Doing this analougously for ϕ_- and defining an respective ψ_- we get in total that

$$\begin{aligned}\int_{B_r(x)} u(y)d^n y &= \int_{B_r(0)\cap\{y_n>0\}} u(x+y)d^n y + \int_{B_r(0)\cap\{y_n<0\}} u(x+y)d^n y \\ &= \int_0^r \int_{\partial B_s^{n-1}(0)\cap\{y_n>0\}} u(x+y)d\sigma(y)ds + \int_0^r \int_{\partial B_s^{n-1}(0)\cap\{y_n<0\}} u(x+y)d\sigma(y)ds \\ &= \int_0^r \int_{\partial B_s^{n-1}(0)} u(x+y)d\sigma(y)ds \\ &= \int_0^r \int_{\partial B_s^{n-1}(x)} u(y)d\sigma(y)ds\end{aligned}$$

altough we used in the last step a translation on manifolds, which has Jacobain Id.
Now we can finally show that due to the fact that $S(\cdot) \equiv u(x)$ is constant we get

$$\begin{aligned}\frac{1}{r^n\omega_n}\int_{B_r(x)} u(y)d^n y &= \frac{n}{r^n n\omega_n}\int_0^r \frac{s^{n-1}}{s^{n-1}}\int_{\partial B_s^{n-1}(x)} u(y)d\sigma(y)ds \\ &= \frac{n}{r^n}\int_0^r s^{n-1}\frac{1}{s^{n-1}n\omega_n}\int_{\partial B_s^{n-1}(x)} u(y)d\sigma(y)ds \\ &= \frac{n}{r^n}\int_0^r s^{n-1}S(s)ds = u(x)\frac{n}{r^n}[\frac{1}{n}s^n]_0^r = u(x).\end{aligned}$$

Now we show the other direction:
Let $\overline{B_r(x)} \subset \Omega$ for every $r > 0$ be compact and assume the mean value property holds, i.e.

$$u(x) = \frac{1}{\omega_n r^n}\int_{B_r(x)} u(y)d^n y = \frac{1}{\omega_n r^n}\int_{B_r(x)} u(y)d^n y \overset{above}{=} \frac{n}{r^n}\int_0^r s^{n-1}S_x(s)ds.$$

Because u is constant in r it follows that

$$\begin{aligned}
0 &= \frac{d}{dr}\frac{n}{r^n}\int_0^r s^{n-1}S_x(s)ds \\
&= -\frac{n^2}{r^{n+1}}\int_0^r s^{n-1}S_x(s)ds + \frac{n}{r^n}\frac{d}{dr}\int_0^r s^{n-1}S_x(s)ds \\
&\overset{???}{=} -\frac{n^2}{r^{n+1}}\int_0^r s^{n-1}S_x(s)ds + \frac{n}{r^n}r^{n-1}S_x(r) \\
&= -\frac{n^2}{r^{n+1}}\frac{1}{n}r^nS_x(r) + \frac{n}{r}S_x(r)ds = -\frac{n}{r^n}S_x(r) + \frac{n}{r}S_x(r) \\
&\iff u(x) = S_x(r).
\end{aligned}$$

The above shows that the mean on $\partial B_r(x)$ is equalt to $u(x)$. We have shown that $S'(r)$ is the mean of Δu on $B_r(x)$ and because $S(\cdot)$ is constant it follows

$$S'(r) = 0 \qquad \iff \qquad \int_{B_r(x)} \Delta u d^n y = 0.$$

Finally, $S(\cdot)$ is twice differentiable, because $u \in C^2(\Omega)$.
We now want to show that $\Delta u \equiv 0$. For that, assume $\exists x \in \Omega$ such that $\Delta u(x) \neq 0$. Because Δu is continuous

$$\forall \xi > 0 \exists y \in B_\epsilon(x) : \quad |\Delta u(y) - \Delta u(x)| < \xi.$$

Choose $\xi := \frac{|\Delta u(x)|}{2}$. It follows

$$|\Delta u(y) - \Delta u(x)| < \frac{|\Delta u(x)|}{2}.$$

- Case 1: $\Delta u(x) > 0$

$$\Delta u(y) \geq \Delta(x) - |\Delta u(y) - \Delta u(x)| > \Delta u(x) - \frac{|\Delta u(x)|}{2} = \frac{\Delta u(x)}{2} > 0.$$

- Case 2: $\Delta u(x) < 0$

$$\Delta u(y) = \Delta u(X) - \Delta u(x) + \Delta u(y) < \Delta u(x) + |\Delta u(y) - \Delta u(x)| \leq \Delta u(x) + \frac{|\Delta u(x)|}{2} = \frac{\Delta u(x)}{2} < 0.$$

In total it was shown that $\frac{|\Delta u(x)|}{2} > 0$ and

$$\forall y \in B_\epsilon(x) : |\Delta u(y)| \geq \frac{|\Delta u(x)|}{2}.$$

Ultimiately this leads to

$$\left|\int_{B_r(x)} \Delta u(y) d^n y\right| \geq \frac{|\Delta u(x)}{2}\int_{B_r(x)} d^n y > 0,$$

which is a contradiction. Thus it follows that $\Delta u \equiv 0$.

(*)

$$D\phi_+(z,x) = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & \\ \vdots & \ddots & & & \vdots \\ \frac{2|z|}{\sqrt{s^2-|z|^2}} & \cdots & \frac{2|z|}{\sqrt{s^2-|z|^2}} & \frac{s}{\sqrt{s^2-|z|^2}} & \end{pmatrix} = \begin{pmatrix} Id & & & 0 \\ \frac{2|z|}{\sqrt{s^2-|z|^2}} & \cdots & \frac{2|z|}{\sqrt{s^2-|z|^2}} & \frac{s}{\sqrt{s^2-|z|^2}} \end{pmatrix}$$

This yields

$$\begin{aligned}\det(D\phi_+(z,x)) &= \frac{2|z|}{\sqrt{s^2-|z|^2}}\det(\begin{pmatrix} 0 & \cdots & 0 \\ & Id & \end{pmatrix}) + \cdots + \frac{2|z|}{\sqrt{s^2-|z|^2}}\det(\begin{pmatrix} & Id & \\ 0 & \cdots & 0 \end{pmatrix}) + \frac{s}{\sqrt{s^2-|z|^2}}\det(Id) \\ &= \frac{s}{\sqrt{s^2-|z|^2}}.\end{aligned}$$

□

The next result shows that smooth harmonic functions do not blow up in the interior of a domain, by showing that all their partial derivatives are bounded. On an open ball $B_r(x)$ the value of any partial derivative of a smooth harmonic function in the center of that ball is bounded by above by a finite constant depnding on the radius and the supremum on the closure of the ball. Liouvilles Theorem will then use this result and show that bounded harmonic functions are constant. This can be seen in the form of the upper bound of the following corollary, as this bound goes to zero as $r \to \infty$.

Corollary 2.2

Let u be a smooth harmonic function on an open domain $\Omega \subset \mathbb{R}^n$ and $B_r(x)$ with closure in Ω. Then $\forall \alpha \in \mathbb{N}_0^n$ it holds that

$$|\partial^\alpha u(x)| \leq C(n,|\alpha|)r^{-|\alpha|}\|u\|_{L^\infty(\overline{B_r(x)})}, \quad C(n,|\alpha|) = 2^{\frac{|\alpha|(1+|\alpha|)}{2}} n^{|\alpha|}.$$

Proof. First, its clear that all partial derivatives of a harmonic function are harmonic again. We show the assertion via induction but only consider the induction start:

$$\begin{aligned}|\partial_i\partial^\alpha u(x)| &\overset{meanvalue}{=} |\frac{2^n}{\omega_n r^n}\int_{B_{r/2}(x)} \partial_i\partial^\alpha u d^n x| \\ &\overset{Div.Thm.}{=} |\frac{2^n}{\omega_n r^n}\int_{\partial B_{r/2}(x)} \partial^\alpha u N d\sigma| \\ &\leq |\frac{2^n}{\omega_n r^n}| \|\partial^\alpha u\|_{L^\infty(\partial B_{r/2}(x))} \overbrace{|\int_{\partial B_{r/2}(x)} \underbrace{1_{n-1} N}_{=1} d\sigma|}^{=n\omega_n (r/2)^{n-1}} \\ &= \|\partial^\alpha u\|_{L^\infty(\partial B_{r/2}(x))}\frac{2n}{r}, \quad \forall i = 1,...,n.\end{aligned}$$

This shows the assertion for $|\alpha| = 1$, i.e. the induction start is $C(n,1) = 2n$. □

Theorem 2.3: Liouvilles

On $\mathbb{R}^n$ a bounded harmonic function is constant.

Proof. The previous corolarry has shown that a smooth (???) harmonic function is bounded in the following way

$$|\partial^\alpha u(x)| \leq C(n, |\alpha|) r^{-|\alpha|} \|u\|_{L^\infty(\overline{B_r(x)})}.$$

This goes to zero as $r \to \infty$, i.e. all partial derivatives of order α dissapear. Thus u is constant. □

The average of a function being constant on balls $B_r(x)$ is equivalent to taking a test function ψ that is rotationally symmetric and satisfies $\int \psi dx = 0$ and satisfies the condition $\int u\psi dx = 0$, as rotationally symmetric functions can be for example the difference of two averaging kernels.
We now want to transfer the mean value property to distributions. Let $u \in C^2(\Omega)$ be harmonic, then $\forall B_r(x) \subset \Omega$ and $\psi \in C_0^\infty(\Omega)$ it holds

$$\begin{aligned}\int_{B_r(x)} u(y) \frac{\psi(|y-x|)}{n|y-x|^{n-1}\omega_n} d^n y &= \int_0^r \int_{\partial B_s(x)} u(y) \frac{\psi(|y-x|)}{n|y-x|^{n-1}\omega_n} d\sigma(y) ds \\ &= \int_0^r \frac{\psi(s)}{ns^{n-1}\omega_n} \int_{\partial B_s(x)} u(y) d\sigma(y) ds = \int_0^r \psi(s) u(x) ds = u(x) \int_0^r \psi(s) ds.\end{aligned}$$

If we choose a smart assumption to get the right hand side to become zero, this equation will imply that the value of the mean of u does not depend on the radius and thus is equal to the center.

Theorem 2.4: Weak Mean Value Property

Let $U \in \mathcal{D}'(\Omega)$ be a harmonic distribution on an open domain $\Omega \subset \mathbb{R}^n$. For each ball $B_r(x) \subset \Omega$ and each $\psi \in C_0^\infty((0,r))$ with $\int_0^r \psi d\mu = 0$ the distribution U vanishes on the followsing test function $f \in C_0^\infty(\Omega)$ with

$$y \mapsto f(y) := \frac{\psi(|y-x|)}{n|y-x|^{n-1}\omega_n}, \quad \textit{supp}\ f \subset B_r(x) \subset \Omega,$$

i.e. $U(f) = 0$.

Proof. Due to the fact that U is harmonic it holds that

$$0 = (\Delta U)(g) = U(\Delta g) = U(f),$$

if we show that $\exists g \in C_0^\infty(\Omega)$ with $\Delta g = f$.
Because by assumption ψ is integrable ($\int_\Omega \psi d\mu = \int_0^r \psi d\mu = 0$) there exists a $\Psi \in C_0^\infty((0,r))$ such that $\Psi' = \psi$. Now define

$$g(y) := v(|y-x|), \quad v(t) := \int_r^t \frac{\Psi(s)}{ns^{n-1}\omega_n} ds.$$

This function has compact support in $B_r(x) \subset \Omega$, because of the following. As we defined $\Psi \in C_0^\infty((0,r))$ it holds that $\Psi(s) = 0$ for all $s \geq r$ and $\exists \epsilon > 0$ such that $\Psi(s) = 0$ for all $s \leq \epsilon$. Finally, we get that for $y \neq x$

$$\begin{aligned}
\Delta_y g(y) &= \nabla_y \cdot \left(\frac{\partial}{\partial y_i} |y-x| v'(y) \right) = \nabla_y \cdot \frac{y-x}{|y-x|} v'(y) \\
&= \sum_{i=1}^{n} \left(\frac{1}{|y-x|} v'(y) + v''(y) \frac{y_i - x_i}{|y-x|} \right) \\
&= \frac{n-1}{|y-x|} v'(y) + v''(y) \\
&= \frac{n-1}{|y-x|} v'(y) + \frac{\psi(s)}{n s^{n-1} \omega_n} + \frac{\Psi(s)}{n s^{n-1} \omega_n} \frac{1-n}{|y-x|} \\
&= \frac{n-1}{|y-x|} v'(y) + \frac{\psi(s)}{n s^{n-1} \omega_n} - \frac{\Psi(s)}{n s^{n-1} \omega_n} \frac{n-1}{|y-x|} \\
&= \frac{\psi(s)}{n s^{n-1} \omega_n} = f(y).
\end{aligned}$$

□

Lemma 2.5

On an open domain $\Omega \subset \mathbb{R}^n$ for each harmonic distribution $U \in \mathcal{D}'(\Omega)$ there exists $u \in C^\infty(\Omega)$ harmonic such that $U = F_u$.

Proof. • Idea: We construct $u \in C^\infty(\Omega)$ as

$$u(x) := U(g_x), \quad g_x(y) := \frac{\phi(|x-y|)}{n|y-x|^{n-1}\omega_n}, \quad \phi \in C_0^\infty((0,r)), \quad \int_0^r \phi(s) ds = 1$$

and show that $F_u =: \tilde{U}$ satisfies the weak mean value property, u is harmonic, smooth and $U = \tilde{U}$. This only works if u is constructed independently from ϕ and r.

First, we shall find a u as it was proposed in the lemma. For every $x \in \Omega$ choose a ball $B_r(x) \subset \Omega$ and a test function $\phi \in C_0^\infty((0,r))$ with $\int_0^r \phi(s) ds = 1$. Then we define

$$u(x) := U(g_x), \quad g_x(y) := \frac{\phi(|y-x|)}{n|y-x|^{n-1}\omega_n}.$$

Now we need to show that u is well defined, i.e. independent of the choice of r and ϕ. For that define

$$g_{x,1}(y) := \frac{\phi_1(|y-x|)}{n|y-x|^{n-1}\omega_n}, \quad g_{x,2}(y) := \frac{\phi_2(|y-x|)}{n|y-x|^{n-1}\omega_n},$$

for $\phi_1 \in C_0^\infty((0,r_1))$ and $\phi_2 \in C_0^\infty((0,r_2))$. We want to show that $U(g_{x,1}) = U(g_{x,2})$ which is equivalent by linearity to $U(g_{x,1} - g_{x,2}) = 0$. For ease of notation define

$$f(y) := \frac{\phi_1(|y-x|) - \phi_2(|x-y|)}{n|y-x|^{n-1}\omega_n}.$$

Thus we need to show that $U(f) = 0$, where $\phi := \phi_1 - \phi_2$. We do this by showing that the conditions from the mean value theorem are satisfied. It holds that

$$\int_0^{\max r_1, r_2} \phi ds = \int_0^{r_1} \phi ds - \int_0^{r_2} \phi_2 ds = 1 - 1 = 0.$$

Thus u is independent from ϕ and r. Additionally it holds that

$$g_x(y) = g_0(y-x) = (T_x g_0)(y)$$

which leads to

$$u(x) = U(g_x) = U(T_x g_0) = U(T_x P g_0) = (g_0 * U)(\cdot)$$

due to the fact that $g_0(\cdot)$ is symmetric, i.e. $g_0 = Pg_0$. Thus because we can write any distribution as a function

$$g * F : \mathbb{R}^n \to \mathbb{R}^n, \quad x \mapsto F(T_x P g)$$

that is in C^∞ it follows that u is smooth.
Next, we show that the distribution $\tilde{U} = F_u$ satisfies the weak Mean value property. With the above it holds

$$F_u(f) = \int_{\mathbb{R}^n} u(x) f(x) d^n x = \int_{\mathbb{R}^n} (g_0 * U)(x) f(x) d^n x = (g_0 * U)(f).$$

Note that the function that we characterized in the mean value property theorem satisfies the following three properties

1. It only depends on the distance $|y-x|$ to the center $x \in \Omega$
2. *supp* $f \subset B_r(x)$
3. $U(f) = 0$

Clearly the first property is equivalent to f is invariant under all rotains around the center. Remember that we have shown that for two functions f and g that are rotationally symmetric around a and b repsectively (i.e. for any orthonagonal transformation $O \in O(n, \mathbb{R})$ it holds $f(a+b) = f(a+Ox)$) follows

$$(f * g)(a+b+x) = (f * g)(a+b+Ox),$$

i.e. rotationally symmetric wrt. $a+b$. Thus, if we take $g_0 * f$ it is rotationally symmetric around the same center as f, because g_0 is rotaionally symmetric around 0.
Define as in the weak mean value thoerem $\psi \in C_0^\infty((0,r))$ the function

$$f(y) := \frac{\psi(|y-x|)}{n|y-x|^{n-1}\omega_n}$$

vanishes on for all $y \in B_\epsilon^n(x)$, if we choose $\epsilon > 0$ small enough.
Note that we used different test functions for f and g_0. For the first we have $\psi \in C_0^\infty((0,r))$ with $\int \psi d\mu = 0$ and for the second we have $\phi \in C_0^\infty((0,r))$ with $\int \phi d\mu = 1$. If g_0 has its support on $B_\xi(0)$ with $\xi < \epsilon$ then *supp* $g_0 * f \subset B_\xi(x)$. This and the fact that $\int_\Omega \phi(x)\psi(y-x)d\mu = 0$ gives us all the conditions we need to to use the weak mean value property in order for the following calculation

$$\tilde{U}(f) = (g_0 * U)(f) = U(f * Pg_0) = U(f * g_0) = 0.$$

Thus, $\tilde{U}$ satisfies the weak mean value property.
The next step is to show that u is harmonic.
Let $\phi_\epsilon \in C_0^\infty(\mathbb{R})$ be a mollifier. Because $B_r(x)$ has compact closure in Ω $\exists R > r$ such that $B_R(x) \subset \Omega$. Further, for any $0 < r_1 < r_2 < R$ and $\epsilon > 0$ small enough the function

$$\varphi(t) := \phi_\epsilon(t - r_1) - \phi_\epsilon(t - r_2),$$

has compact support in $(0,R)$ and a vanishing integral

$$\int \varphi(t)d\mu = \int \phi_\epsilon(t-r_1)d\mu - \int \phi_\epsilon(t-r_2)d\mu = 1-1=0.$$

We define

$$f_x^\epsilon(y) := \frac{\varphi(|y-x|)}{n|y-x|^{n-1}\omega_n}, \quad y \in B_\epsilon(x).$$

For sufficiently small $\epsilon > 0$ f_x^ϵ has compact support in Ω and is symmetric wrt. x. Since satisfies all the conditions from the weak mean value theorem, as $\epsilon \to 0$, it follows that with

$$\begin{aligned}
0 \leftarrow \tilde{U}(f_x^\epsilon) &= \int_\Omega u(y) f_x^\epsilon(y)dy = \int_\Omega u(y)\frac{\varphi(|y-x|)}{n|y-x|^{n-1}\omega_n}dy \\
&= \frac{1}{n\omega_n}\int_{B_\epsilon(x)} u(y)\frac{\varphi(|y-x|)}{|y-x|^{n-1}}dy = \frac{1}{n\omega_n}\int_0^\epsilon \int_{\partial B_r(x)} u(y)\frac{\varphi(|y-x|)}{|y-x|^{n-1}}d\sigma(y)dr \\
&= \int_0^\epsilon \varphi(r)\frac{1}{nr^{n-1}\omega_n}\int_{\partial B_r(x)} u(y)d\sigma(y)dr \\
&= \int_0^\epsilon \varphi(r)S_x(r)dr \\
&= \int_0^\epsilon \phi_\epsilon(r-r_1)S_x(r)dr - \int_0^\epsilon \phi_\epsilon(r-r_2)S_x(r)dr \\
&\to S_x(r_1) - S_x(r_2), \quad \epsilon \to 0.
\end{aligned}$$

we get $0 = S_x(r_1) - S_x(r_2)$. In the limit we have

$$S_x(r_1) = S_x(r_2) \to u(x), t \to 0.$$

Thus u satisfies the mean value property and is thus equivalently harmonic.
Finally, we show that $U = \tilde{U}$. For that define

$$\eta(t) := \phi_{\epsilon/3}(t - \frac{2}{3}\epsilon), \quad \text{has support } (\epsilon/3, \epsilon)$$

and $\int \eta d\mu = 1$. Define the mollifier

$$f_x^{\eta,\epsilon}(y) := \frac{\eta(|y-x|)}{n|y-x|^{n-1}\omega_n}.$$

Then $(f_0^{\eta,\epsilon}(\cdot))_{\epsilon \geq 0}$ is a sequence of smooth mollifiers. It then finally follows with $W := \tilde{U} - U$ that for any test function $\psi \in C_0^\infty(\mathbb{R}^n)$:

$$\begin{aligned}\tilde{U}(\psi) &= \lim_{\epsilon\to 0}\tilde{U}(\psi * f_0^{\eta,\epsilon}) = \lim_{\epsilon\to 0}\int_\Omega u(y)(\psi(y) * f_0^{\eta,\epsilon})(y)dy \\ &= \lim_{\epsilon\to 0}\int_\Omega u(y)\int_\Omega \psi(x) f_0^{\eta,\epsilon}(y-x)dxdy \\ &= \lim_{\epsilon\to 0}\int_\Omega \psi(x)\int_\Omega u(y) f_0^{\eta,\epsilon}(y-x)dydx \\ &= \lim_{\epsilon\to 0}\int_\Omega \psi(x)\tilde{U}(f_x^{\eta,\epsilon})dx \\ &= \int_\Omega \psi(x)u(x)dx \\ &= \int_\Omega \psi(x)U(g_x)dx \\ &= \lim_{\epsilon\to 0}\int_\Omega \psi(x)U(f_x^{\eta,\epsilon})dx \\ &= \lim_{\epsilon\to 0}\int_\Omega \psi(x)U(f_0^{\eta,\epsilon}(\cdot - x))dx \\ &= \lim_{\epsilon\to 0}\psi * U(f_0^{\eta,\epsilon}) \\ &= \lim_{\epsilon\to 0}U(\psi * f_0^{\eta,\epsilon}) = U(\psi)\end{aligned}$$

Although we used that for a mollifier $(\lambda_\epsilon)_{\epsilon\geq 0}$ it holds

$$\lim_{\epsilon\to 0}\psi * \lambda_\epsilon = \psi, \quad \psi \in C_0^\infty(\mathbb{R}^n).$$

□

Theorem 2.6: Harnack

Let Ω' be an open path-connected domaina, i.e.

$$\forall x,y \in \Omega' \exists \gamma : [0,1] \to \Omega' \text{ continuous: } \gamma(0) = x \text{ and } \gamma(1) = y,$$

with compact closure in the open domain $\Omega \subset \mathbb{R}^n$. Then there exists $C(\Omega,\Omega') > 0$ such that any non-negative harmonic function u on Ω satisfies

$$\sup_{x\in\Omega'} u(x) \leq C \inf_{x\in\Omega'} u(x)$$

and

$$\forall x,y \in \Omega' : \frac{1}{C}u(y) \leq u(x) \leq Cu(y).$$

Proof. Let Ω' be an open domain that is path connected and has compact slosure in the open domain $\Omega \subset \mathbb{R}^n$. Define

$$g(x) := \inf_{y\in\partial\Omega}\|y-x\|$$

Since $\bar{\Omega}'$ is compact in Ω it is disjoint from $\partial\Omega$ which implies that $g > 0$. Let

$$r := \frac{1}{2}\min_{x\in\bar{\Omega}'} g(x).$$

By construction it holds that for any $x \in \bar{\Omega}'$ we have $B_{2r}(x) \subset \Omega$. Next, let $x, y \in \Omega'$ be arbitrary with $\|x-y\| < r$. It the follows that for any $z \in B_r(y)$ it holds

$$\|z-x\| \leq \|z-y\| + \|y-x\| < 2r.$$

Thus $B_r(y) \subset B_{2r}(x)$. Since u is harmonic and non-negative it follows

$$u(x) = \frac{1}{2^n r^n \omega_n} \int_{B_{2r}(x)} u d\mu \geq \frac{2^{-n}}{r^n \omega_n} \int_{B_r(y)} u d\mu = 2^{-n} u(y)$$

holds for all $x, y \in \Omega'$ with $\|x-y\| < r$.
As we have required $\bar{\Omega}'$ to be compact, i.e. meaning that there exists a cover of finite many balls $B_1, ..., B_N$ with radii $\frac{r}{2}$ with

$$\forall k = 1, ..., N-1 : B_{k+1} \cap B_k \neq \emptyset.$$

Let $B_k := B_{r/2}(x_k)$, then $\forall z_k \in B_{k+1} \cap B_k$ holds

$$\|x_k - x_{k+1}\| \leq \|x_k - z_k\| + \|z_k - x_{k+1}\| < \frac{r}{2} + \frac{r}{2} = r,$$

which implies that $B_r(x_{k+1}) \subset B_{2r}(x_k)$. Thus, for all $k = 1, ..., N-1$:

$$u(x_k) = \frac{1}{2^n r^n \omega_n} \int_{B_{2r}(x_k)} u d\mu \geq \frac{2^{-n}}{r^n \omega_n} \int_{B_r(x_{k+1})} u d\mu = 2^{-n} u(x_{k+1})$$

which is equivalent to $2^n u(x_k) \geq u(x_{k+1})$. Doing this N times yields

$$u(x_N) \leq ... \leq 2^{n(N-1)} u(x_1).$$

Now choose $x, y \in \Omega'$ arbitrary and the cover in such a way that $x \in B_{r/2}(x_1)$ and $y \in B_{r/2}(x_N)$. Because Ω' is path connected there exist a sequence of open balls $B_{k_1}, ..., B_{k_n}$ that connect these points. For ease of notation let $k_1 = 1$ and $k_n := N$. Then it follows that

$$\|x - x_1\| < \frac{r}{2} < r \implies B_{r/2}(x_1) \subset B_r(x) \implies 2^n u(x) \geq u(x_1).$$

and

$$\|y - x_N\| < \frac{r}{2} < r \implies B_{r/2}(y) \subset B_r(x_N) \implies 2^n u(x_N) \geq u(y)$$

which finally leads to

$$u(y) \leq 2^n u(x_N) \leq ... \leq 2^{nN} u(x_1) \leq 2^{n(N+1)} u(x).$$

As the above holds for arbitrary $x, y \in \Omega'$ and its an open set the following immediatly follows

$$\sup_{x \in \Omega'} u(x) \leq 2^{n(N+1)} \inf_{x \in \Omega'} u(x).$$

□

Lemma 2.7: Harnack's Principle

On an open and path connected domain $\Omega \subset \mathbb{R}^n$ a monotone sequence of harmonic functions $(u_n)_{n \in \mathbb{N}}$ converges uniformly on all compact subsets $\Omega' \subset \Omega$, i.e.

$$\exists u \forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, x \in \Omega' : \quad |u_n(x) - u(x)| < \epsilon,$$

if and only if

$$\exists x \in \Omega : (|u_n(x)|)_{n \in \mathbb{N}} \text{ is bounded.}$$

Proof. "$\Rightarrow$":
Let $(u_n)_{n\in\mathbb{N}}$ be uniformly convergent on every compact subset $\Omega' \subset \Omega$. Then with analysis it follows $(u_n(x))_{n\in\mathbb{N}}$ converges for all $x \in \Omega$. Pointwise convergence implies boundedness.
"$\Leftarrow$":
Let $(|u_n(x)|)_{n\in\mathbb{N}}$ be bounded for a $x \in \Omega$. By assumption it is monotone. Boundedness an monotonicity imply pointwise convergence, i.e. $(u_n(x))_{n\in\mathbb{N}}$ converges. To show uniform convergence on Ω', we need to show that $(u_n)_{n\in\mathbb{N}}$ is uniformly a chauchy sequence on Ω'. Because Ω is a domain

$$\exists \Omega'' : \Omega' \subset \Omega'' \subset \Omega, \text{ with } x \in \Omega''.$$

Define $v_{n,m}(x) := u_n(x) - u_m(x)$, which is harmonic and non negative, because $(u_n)_{n\in\mathbb{N}}$ is monotone. Thus Harnacks inequality implies that

$$\sup_{y\in\Omega''} v_{n,m}(y) \le C \inf_{y\in\Omega''} v_{n,m}(y).$$

which implies for all $y \in \Omega''$:

$$v_{n,m}(y) \le C \inf_{y\in\Omega''} v_{n,m}(y).$$

Due to the fact that $x \in \Omega''$ it follows also that for all $y \in \Omega''$:

$$v_{n,m}(y) \le C v_{n,m}(x).$$

Finally, due to $\Omega' \subset \Omega''$ it follows

$$\sup_{y\in\Omega'} v_{n,m}(y) \le C v_{n,m}(x).$$

Now due to the fact that $(u_n(x))_{n\in\mathbb{N}}$ is a convergent sequence, it is a cauchy sequence, i.e.

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \ge m \ge N : \quad u_n(x) - u_m(x) = v_{n,m}(x) < \frac{\epsilon}{C}.$$

Thus

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \ge m \ge N : \quad \sup_{y\in\Omega'} v_{n,m}(y) \le C v_{n,m}(x) < \epsilon,$$

which is the uniform cauchy criterion on Ω'. □

3 Maximum Principle

The following theorem is a consequence of the mean value property.

Theorem 3.1: Strong Maximum Principle

Let u be harmonic on a path-connected open domain $\Omega \subset \mathbb{R}^n$ and taking its maximum. Then u is constant.

Proof. Let u be a harmonic function that takes its maximum at $x \in \Omega \subset \mathbb{R}^n$ on an open path connected domain. The MVP implies

$$\forall y \in B_r(x) \subset \Omega : \quad \frac{1}{r^n \omega_n} \int_{B_r(x)} |u(x) - u(y)| d^n y \overset{Max.}{=} \frac{1}{r^n \omega_n} \int_{B_r(x)} u(x) - u(y) d^n y \overset{MVP}{=} 0.$$

Hence u is constant on $B_r(x)$, i.e. $\forall y \in B_r(x) : \max_{z\in\Omega} u(z) = u(y) =: M$.
Define $A := \{z \in \Omega \mid u(z) = M\}$. We have shown in the above that $A \neq \emptyset$ because $B_r(x) \subset A$. We will show that A is closed and open and non-empty. This implies due to (*) that $A = \Omega$.
Closed: Because the preimage of closed sets for continuous functions is again closed it follows that $u^{-1}(\{M\}) = A$ is closed.
Open: Let $a \in A$ be arbitrary. Since Ω is open there exists $r > 0$ such that $B_r(a) \subset \Omega$. Due to the fact that $a \in A$ it follows that $u(a) = M$ is the global maximum. With the arguments above it follows that

$$\forall y \in B_r(a) \subset \Omega : \quad \frac{1}{r^n\omega_n}\int_{B_r(a)} |u(a) - u(y)| d^n y \overset{Max.}{=} \frac{1}{r^n\omega_n}\int_{B_r(a)} u(a) - u(y) d^n y \overset{MVP}{=} 0.$$

Hence u is constant on $B_r(a)$. As $a \in A$ was arbitrary it follows that A is open.
All the above arguments imply that u is constant on Ω where it takes its maximum.
(*): Ω is path connected implies Ω is connected, which is defined as

$$\nexists B, C \neq \emptyset \text{ open with } B \cap C = \emptyset \text{ and } \Omega = B \cup C.$$

Since A is closed it follows that A^C is open. Because $\Omega = A \cup A^C$ and Ω is connected one of the sets has to be empty. As $A \neq \emptyset$ it follows $A^C = \emptyset$. □

Theorem 3.2: Weak Maximum Principle

Let u be harmonic on a bounded open domain $\Omega \subset \mathbb{R}^n$ that extends continuously to $\partial\Omega$. Then the maximum of u is then taken on $\partial\Omega$.

Proof. Because Ω is bounded it follows that $\bar{\Omega} = \Omega \cup \partial\Omega$ is closed and bounded, which implies with Heine Borel that $\bar{\Omega}$ is compact. A continuous function on a compact set attains its maximum value M at some point $x \in \bar{\Omega}$. There are now two cases:

- Case 1: If $x \in \partial\Omega$ the assertion is shown.
- Case 2: If $x \in \Omega$ the strong Maximum Principle can be applied an yields that $\forall z \in \Omega : u(z) = M$. Because U extends continuously to $\partial\Omega$ it follows that also $\forall y \in \partial\Omega$: $u(y) = M$.

In both cases the maximum is attained on $\partial\Omega$. □

Since $-u$ is again harmonic if u is, the strong and weak minimum principle immediatly follows. We will now generalise the Maximum principle.

Theorem 3.3

Let L be an elliptic differntial operator on a bounded open domain $\Omega \subset \mathbb{R}^n$ with coefficients $a_{i,j}, b_i$ extending continuously and ellipticly to $\partial\Omega$. Then every twice differentiable solution $u \in C^2$ of $Lu \geq 0$ that extends continuously to $\partial\Omega$ takes its maximum on $\partial\Omega$.

Proof. Because the coefficients are continuos the function $g : (x, k) \mapsto \sum_{i,j=1}^n a_{ij}(x)k_i k_j$ is contious on the compact set $\bar{\Omega} \times S^{n-1} \subset \bar{\Omega} \times \mathbb{R}^n$. A continuous function attains its minimum on compact sets. Thus g attains its minimum $\lambda > 0$, as L is elliptic, this minimum is positve. This implies uniform ellipticity.
Define $v(x) := e^{x_1\alpha}, \alpha > 0, x = (x_1, ..., x_n)$, then with uniform ellipticity we get that

$$Lv = \alpha(\alpha a_{11}(x) + b_1(x))v(x) \geq \alpha(\alpha\lambda + b_1(x))v > 0,$$

for $\alpha > 0$ large enough, because b_i are bounded and continuous on $\bar{\Omega}$ and $\lambda > 0$. With linearity it follows for all $\epsilon > 0$ that

$$L(u + \epsilon v) = \underbrace{Lu}_{\geq 0} + \underbrace{\epsilon Lv}_{>0} > 0, \text{ on } \Omega \forall \epsilon > 0.$$

To show: $w_\epsilon := u + \epsilon v$ cannot have an interior maximum.
Assume w_ϵ attains its maximum at some $x \in \Omega$. Since $w_\epsilon \in C^2$ it clearly holds that

- $\nabla w_\epsilon(x_0) = 0$
- The Hessian $Hw_\epsilon(x_0)$ is negative semi-definit, i.e. all eigenvalues $\mu_1, ..., \mu_n$ are non-positive.

Then there exists $B \in O(n)$ orthogonal such that $HW_\epsilon(x_0) = B^T \begin{pmatrix} \mu_1 & & \\ & \ddots & \\ & & \mu_n \end{pmatrix} B$. Using this we can now reformulate the differential operator.

$$\begin{aligned} Lw_\epsilon(x_0) &= \sum_{i,j=1}^n a_{ij}(x_0) \frac{\partial^2}{\partial x_i \partial x_j} w_\epsilon(x_0) + \sum_{i=1}^n b_i(x_0) \cdot 0 \\ &= trace(A(x_0) Hw_\epsilon(x_0)) = trace(A(x_0) B^T \begin{pmatrix} \mu_1 & & \\ & \ddots & \\ & & \mu_n \end{pmatrix} B) \\ &= \sum_{i,j=1}^n a_{ij}(x_0) \sum_{k=1}^n \mu_k b_{ki} b_{kj} \\ &= \sum_{k=1}^n \underbrace{\mu_k}_{\leq 0} \underbrace{\sum_{i,j=1}^n a_{ij}(x_0) b_{ki} b_{kj}}_{\geq \lambda > 0} \leq 0 \end{aligned}$$

which is a contradiction. Thus Lw_ϵ attains its maximum on $\partial\Omega$. Then

$$\begin{aligned} &\sup_{x\in\bar{\Omega}} u(x) + \epsilon \inf_{x\in\bar{\Omega}} v(x) \leq \sup_{x\in\bar{\Omega}} (u(x) + \epsilon v(x)) \overset{above}{=} \max_{x\in\partial\Omega} (u(x) + \epsilon v(x)) \leq \max_{x\in\partial\Omega} u(x) + \epsilon \max_{x\in\partial\Omega} v(x) \\ \iff & \sup_{x\in\bar{\Omega}} u(x) \leq \max_{x\in\partial\Omega} u(x) + \epsilon \underbrace{(\max_{x\in\partial\Omega} v(x) - \inf_{x\in\bar{\Omega}} v(x))}_{=const.}. \end{aligned}$$

Due to the fact that $\epsilon > 0$ was chosen arbitrary it follows that

$$\sup_{x\in\bar{\Omega}} u(x) = \max_{x\in\partial\Omega} u(x).$$

□

With the same proof idea one can show that $Lu \leq 0$ implies that u takes its minima on the boundary. Note that if $Lu = 0$ holds then both minima and maxima are on the boundary.

4 Green's Function

Now we want to find conditions that ensure the existence and uniqueness of harmonic functions on path-connected, open and bounded domains $\Omega \subset \mathbb{R}^n$. We will introduce boundary value problems, that means the harmonic function or some of its derivatives extend continuously to the boundary and coincide with some function on $\partial\Omega$.

Definition 4.1

We call u m-times differentiable on $\bar{\Omega}$ if it is m-times continuously differentiable on Ω and all partial derivates of order of at most m extend continuously to $\partial\Omega$.

Lemma 4.2: Greens First Formula

Let the divergence theorem hold true on an open and bounded domain $\Omega \subset \mathbb{R}^n$. Then for two $u, v \in C^2(\bar{\Omega})$ it holds

$$\int_\Omega v(y)\Delta u(y) d^n y + \int_\Omega \nabla v(y) \nabla u(y) d^n y = \int_\Omega (v(y)\nabla u(y))' d^n y = \int_{\partial\Omega} v(z)\nabla u(z) \cdot N d\sigma(z).$$

If we subtract the above by itself, while exchanging the role of u and v we get the second Formula.

Lemma 4.3: Greens Second Formula

Let the divergence theorem hold true on an open and bounded domain $\Omega \subset \mathbb{R}^n$. Then for two $u, v \in C^2(\bar{\Omega})$ it holds

$$\begin{aligned}
&\int_\Omega v(y)\Delta u(y) d^n y + \int_\Omega \nabla v(y)\nabla u(y) d^n y - \int_\Omega \Delta v(y) u(y) d^n y - \int_\Omega \nabla v(y) \nabla u(y) d^n y \\
&= \int_{\partial\Omega} v(z)\nabla(z) N d\sigma(z) - \int_{\partial\Omega} u(z)\nabla v(z) N d\sigma(z) \\
&\iff \int_\Omega v(y)\Delta u(y) - \Delta v(y) u(y) d^n y = \int_{\partial\Omega} (v(z)\nabla u(z) - u(z)\nabla v(z)) N d\sigma(z)
\end{aligned}$$

Theorem 4.4: Greens Representation Theorem

Let the divergence theorem hold true on an open and bounded domain $\Omega \subset \mathbb{R}^n$. Then for two $u, v \in C^2(\bar{\Omega})$ it holds

$$u(x) = -\int_\Omega \Phi(x-y)\Delta u(y) d^n y + \int_{\partial\Omega} (\Phi(x-y)\nabla u(y) - u(y)\nabla_y \Phi(x-y)) N d\sigma(y)$$

Proof. With the second Greens Identity it holds

$$\int_{\Omega\setminus B_\epsilon(x)} u(y)\Delta\Phi(x-y) - \Phi(x-y)\Delta u(y) d^n y = \int_{\partial(\Omega\setminus B_\epsilon(x))} (u(y)\nabla_y\Phi(x-y) - \Phi(x-y)\nabla u(y)) N d\sigma(y).$$

Because $\Phi(x-\cdot)$ was constructed to be the solution of $-\Delta u = 0$ for $x \neq y$ it follows that the above is equivalent to

$$
\begin{aligned}
&-\int_{\Omega\setminus B_\epsilon(x)} \Phi(x-y)\Delta u(y) d^n y = \int_{\partial(\Omega\setminus B_\epsilon(x))} (u(y)\nabla_y \Phi(x-y) - \Phi(x-y)\nabla u(y)) N d\sigma(y)\\
&= \int_{\partial\Omega} (u(y)\nabla_y \Phi(x-y) - \Phi(x-y)\nabla u(y)) N d\sigma(y) + \underbrace{\int_{\partial B_\epsilon(x)} (u(y)\nabla_y \Phi(x-y) - \Phi(x-y)\nabla u(y)) N d\sigma(y)}_{=:K_\epsilon}.
\end{aligned}
$$

Due to construction it follows $\Phi(x-y) \in \mathcal{O}(|x-y|^{2-n}), y \to x$, $\nabla u \cdot N$ is bounded and $\int_{\partial B_\epsilon(x)} d\sigma(y) \in \mathcal{O}(\epsilon^{n-1})$ which yields

$$
\lim_{\epsilon\to 0} \int_{\partial B_\epsilon(x)} \Phi(x-y) \underbrace{\nabla(y) N}_{\leq const.} d\sigma(y) \leq \lim_{\epsilon \to 0} \epsilon^{2-n} const. \int_{\partial B_\epsilon(x)} d\sigma(y) = \lim_{\epsilon\to 0} \epsilon = 0.
$$

So the Sphere area $\partial B_\epsilon(x)$ decays faster than the Fundamental solution explodes. Using this and the fact that $\nabla \Phi(y) = -\frac{1}{n\omega_n}\frac{y}{|y|^n}$ and $N = -\frac{x-y}{|x-y|}$ we get that

$$
\begin{aligned}
\lim_{\epsilon\to 0} K_\epsilon &= \lim_{\epsilon\to 0} \int_{\partial B_\epsilon(x)} u(y)\nabla_y \Phi(x-y) N d\sigma(y)\\
&= -\lim_{\epsilon\to 0} \int_{\partial B_\epsilon(x)} u(y) \frac{1}{n\omega_n}\frac{x-y}{|x-y|^n} N d\sigma(y)\\
&= \lim_{\epsilon\to 0} \frac{1}{n\omega_n}\frac{\epsilon}{|\epsilon|^n} \int_{\partial B_\epsilon(x)} u(y)\frac{x-y}{|x-y|} d\sigma(y)\\
&= \lim_{\epsilon\to 0} \frac{1}{n\omega_n}\frac{\epsilon^2}{|\epsilon|^{n+1}} \int_{\partial B_\epsilon(x)} u(y) d\sigma(y)\\
&= \lim_{\epsilon\to 0} \frac{1}{n\omega_n}\frac{1}{|\epsilon|^{n-1}} \int_{\partial B_\epsilon(x)} u(y) d\sigma(y) = \lim_{\epsilon\to 0} S_x(\epsilon) = u(x).
\end{aligned}
$$

Thus in total we have

$$
\begin{aligned}
&u(x) = -\int_\Omega \Phi(x-y)\Delta u(y) d^n y - \int_{\partial\Omega} (u(y)\nabla_y \Phi(x-y) - \Phi(x-y)\nabla u(y)) N d\sigma(y)\\
\iff\ & u(x) = -\int_\Omega \Phi(x-y)\Delta u(y) d^n y + \int_{\partial\Omega} (\Phi(x-y)\nabla u(y) - u(y)\nabla_y \Phi(x-y)) N d\sigma(y)
\end{aligned}
$$

□

Definition 4.5: Greens Function

A function

$$G_\Omega : \{(x,y) \in \Omega\times\Omega \mid x \neq y\} \to \mathbb{R},$$

is called Greens function on the open domain $\Omega \subset \mathbb{R}^n$ if the following are satisfied

- $\forall x \in \Omega$ the function $y \mapsto G_\Omega(x,y) - \Phi(x-y)$ is harmonic on Ω
- $\forall x \in \Omega$ the function $y \mapsto G_\Omega(x,y)$ extends continuously to $\partial\Omega$ and vanishes on $\partial\Omega$.

Note that if we use the Greens Representation theorem, the second Greens formula and define $v(y) := G_\Omega(x,y) - \Phi(x-y)$ (which is by definition a harmonic function) we get

$$
\begin{aligned}
u(x) &= -\int_\Omega \Phi(x-y)\Delta u(y)d^n y + \int_{\partial\Omega} (\Phi(x-z)\nabla u(z) - u(z)\nabla_z \Phi(x-z)) \cdot N d\sigma(z) \\
&= -\int_\Omega (G_\Omega(x,y) - v(y))\Delta u(y)d^n y + \int_{\partial\Omega} ((G_\Omega(x,z) - v(z))\nabla u(z) - u(z)\nabla_z (G_\Omega(x,z) - v(z))) \cdot N d\sigma(z) \\
&= -\int_\Omega \underbrace{\Delta v(y)u(y)}_{=0, v \text{ harmonic}} - v(y)\Delta u(y)d^n y - \int_{\partial\Omega} (u(z)\nabla v(z) - \nabla u(z)v(z))N d\sigma(z) \\
&\quad - \int_\Omega G_\Omega(x,y)\Delta u(y)d^n y + \int_{\partial\Omega} (G_\Omega(x,z)\nabla u(z) - u(z)\nabla_z G_\Omega(x,z)) \cdot N d\sigma(z) \\
&\overset{2.Greens}{=} 0 - \int_\Omega G_\Omega(x,y)\Delta u(y)d^n y + \int_{\partial\Omega} (G_\Omega(x,z)\nabla u(z) - u(z)\nabla_z G_\Omega(x,z)) \cdot N d\sigma(z) \\
&\overset{Def.}{=} -\int_\Omega G_\Omega(x,y)\Delta u(y)d^n y - \int_{\partial\Omega} u(z)\nabla_z G_\Omega(x,z) \cdot N d\sigma(z) \\
&\overset{Dirichlet}{=} \int_\Omega G_\Omega(x,y)f(y)d^n y - \int_{\partial\Omega} g(z)\nabla_z G_\Omega(x,z) \cdot N d\sigma(z).
\end{aligned}
$$

Thus the solution of the Dirichlet problem satisfies for $f : \bar{\Omega} \to \mathbb{R}$ and $g : \partial\Omega \to \mathbb{R}$ sufficiently regular

$$u(x) = \int_\Omega G_\Omega(x,y)f(y)d^n y - \int_{\partial\Omega} g(z)\nabla_z G_\Omega(x,z) \cdot N d\sigma(z).$$

Remark 4.1

Because $u(x)$ only depends on f, g, G_Ω and f, g are known, the Dirichlet problem reduces to a search for the Greens function, that satisfies this integral eqaution.
We can split up the Dirichlet probelm

$$
\begin{aligned}
-\Delta u &= f, \quad \text{on } \Omega \\
u &= g, \quad \text{on } \partial\Omega,
\end{aligned}
$$

into two subporblems. We can do this by homogenizing the boundary and homogenizing the equation:

$$
\begin{cases} -\Delta u = f & \text{on } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad \text{and} \quad \begin{cases} -\Delta v = 0 & \text{on } \Omega \\ v = g & \text{on } \partial\Omega \end{cases}.
$$

Then $u + v$ solves the original Dirichlet problem.

For $x \in \Omega$ the function $y \mapsto G_\Omega(x,y) - \Phi(x-y)$ is harmonic on Ω and on $\partial\Omega$ is equal to $-\Phi(x-y)$. This function is the solution of the Dirichlet Problem $f = 0$ on Ω and $g(x) = -\Phi(x-y)$ on $\partial\Omega$.

Theorem 4.6: Symmetry

If there exists a Greens function G_Ω for the domain Ω then

$$\forall x \neq y : \quad G_\Omega(x,y) = G_\Omega(y,x).$$

Proof. For $x \neq y \in \Omega$ and $\epsilon > 0$ sufficiently small such that the balls $B_\epsilon(x), B_\epsilon(y) \subset \Omega$ are disjoint, the Greens function G_Ω is by definition harmonic on $\Omega_\epsilon := \Omega \setminus (B_\epsilon(x) \cup B_\epsilon(y))$. Note that in the definition of Ω_ϵ the normal vector for the balls $\partial B_\epsilon(x)$ and $\partial B_\epsilon(y)$ respectively, is inward facing. To apply the Greens second Identiy, we need an outward facing vector. This is why we always need to multiply by (-1) when we apply it in the following! Using Greends second Identity we get

$$
\begin{aligned}
&\int_{\Omega_\epsilon} G_\Omega(x,z)\Delta G_\Omega(y,z) - G_\Omega(y,z)\Delta G_\Omega(x,z) d^n z = \int_{\partial\Omega_\epsilon} (G_\Omega(x,z)\nabla G_\Omega(y,z) - G_\Omega(y,z)\nabla G_\Omega(x,z)) \cdot N d\sigma(z) \\
&\iff 0 = \int_{\partial\Omega_\epsilon} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y,z)) \cdot N d\sigma(z) \\
&\iff 0 = \int_{\partial\Omega \cup \partial B_\epsilon(x) \cup B_\epsilon(y)} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y,z)) \cdot N d\sigma(z) \\
&\overset{(*)}{\iff} 0 = \int_{\partial B_\epsilon(x) \cup B_\epsilon(y)} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y,z)) \cdot N d\sigma(z) \\
&\iff -\int_{\partial B_\epsilon(y)} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y,z)) \cdot N d\sigma(z) = \int_{\partial B_\epsilon(x)} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y
\end{aligned}
$$

Although we used in (*) that by definition the Greens function vanishes on $\partial\Omega$. Applying Greens second formula again yields

$$
\begin{aligned}
&\lim_{\epsilon \to 0} \int_{\partial B_\epsilon(x)} (G_\Omega(y,z)\nabla_z G_\Omega(x,z) - G_\Omega(x,z)\nabla_z G_\Omega(y,z)) N d\sigma(z) \\
&= -\lim_{\epsilon \to 0} \int_{B_\epsilon(x)} G_\Omega(y,z)\Delta_z G_\Omega(x,z) - G_\Omega(x,z)\Delta_z G_\Omega(y,z) d^n z \\
&= -\lim_{\epsilon \to 0} \int_{B_\epsilon(x)} G_\Omega(y,z)\Delta_z G_\Omega(x,z) d^n z = -\lim_{\epsilon \to 0} \int_{B_\epsilon(x)} G_\Omega(y,z)(-\delta(x-z)) d^n z = G_\Omega(y,x)
\end{aligned}
$$

and in a similiar fashion

$$
\begin{aligned}
&-\lim_{\epsilon \to 0} \int_{\partial B_\epsilon(y)} (G_\Omega(y,z)\nabla G_\Omega(x,z) - G_\Omega(x,z)\nabla G_\Omega(y,z)) \cdot N d\sigma(z) \\
&= \lim_{\epsilon \to 0} \int_{B_\epsilon(y)} G_\Omega(y,z)\Delta G_\Omega(x,z) - G_\Omega(x,z)\Delta G_\Omega(y,z) d^n z \\
&= -\lim_{\epsilon \to 0} \int_{B_\epsilon(y)} G_\Omega(x,z)\Delta G_\Omega(y,z) d^n z \\
&= -\lim_{\epsilon \to 0} \int_{B_\epsilon(y)} G_\Omega(x,z)(-\delta(y-z)) d^n z = G_\Omega(x,y)
\end{aligned}
$$

Thus the assertion is shown. □

We will now calculate Greens functions for $B_\epsilon(x) = \Omega \subset \mathbb{R}^n$. For now only consider the unit ball. Define the inversion in the unit spehre as

$$\tilde{x} : B_1(0) \to \mathbb{R}^n \setminus \overline{B_1(0)}, x \mapsto \tilde{x}(x) := \frac{x}{|x|^2}.$$

Points close to the origin are mapped to close to infinity and points near the boundary are mapped to points near the boundary on the outside.
Later it will help us to solve the Dirichlet problem for $f = 0$ and $g(x) = \Phi(x - y)$???.

Lemma 4.7: Greens function on the unit Ball

The Greens function on the unit Ball is given by

$$G_{B_1(0)}(x,y) = \Phi(x-y) - \Phi(|x|(\tilde{x}(x)-y)) = \begin{cases} \Phi(x-y) - |x|^{2-n}\Phi(\tilde{x}(x)-y), & n>2 \\ \Phi(x-y) - \Phi(\tilde{x}(x)-y) - \Phi(x), & n=2 \end{cases}$$

Proof. We want to show that $v(y) := G_\Omega(x,y) - \Phi(x-y)$ is harmonic. Define $v(y) := -\Phi(|x|(\tilde{x}(x)-y))$. Now the conditions of the Green function need to be shown under that definition. We first show that $y \mapsto G_{B_1(0)}(x,y)$ vanishes on the boundary, i.e. for $y \in \partial B_1(0)$:

$$\begin{aligned} |x|^2|\tilde{x}(x)-y|^2 &= |x|^2 \sum_{i=1}^n \left(\frac{x_i}{|x|^2} - y_i\right)^2 = \sum_{i=1}^n \left(\frac{x_i}{|x|^2} - 2x_iy_i + y_i^2|x|^2\right) \\ &= \frac{|x|^2}{|x|^2} - 2\sum_{i=1}^n x_iy_i + \underbrace{|y|}_{=1}|x|^2 = 1 - 2\sum_{i=1}^n x_iy_i + |x|^2 \\ &= |y|^2 - 2\sum_{i=1}^n x_iy_i + |x|^2 = |x-y|^2 \\ &\iff |x||\tilde{x}(x)-y| = |x-y| \end{aligned}$$

Which implies that

$$\forall y \in \partial B_1(0): \quad G_{B_1(0)}(x,y) = \Phi(x-y) - \Phi(|x|(\tilde{x}(x)-y)) = 0.$$

Next, in distribution it needs to be shown that $\Delta_y \Phi(|x|(\tilde{x}(x)-y)) = 0$ on $x,y \in B_1(0)$. Note that

$$\begin{aligned} \Phi(|x|(\tilde{x}(x)-y)) &= \begin{cases} -\frac{1}{2\pi}(\log(|\tilde{x}(x)-y|) + \log(|x|)) & ,n=2 \\ \frac{1}{n(n-2)\omega_n|\tilde{x}(x)-y|^{n-2}}\frac{1}{|x|^{n-2}} & ,n>2 \end{cases} \\ &= \begin{cases} \Phi(\tilde{x}(x)-y) + \Phi(|x|) & ,n=2 \\ \Phi(\tilde{x}(x)-y)\frac{1}{|x|^{n-2}} & ,n>2 \end{cases}. \end{aligned}$$

We already have shown that $\Delta_y\Phi(x-y) = -\delta(x-y)$ and $\Delta_z\Phi(z) = -\delta(z)$. We will now show that $\Phi(|\cdot|(\tilde{x}(\cdot)-y))$ has its singularity outside of $B_1(0)$. This is the case because first for $x \in B_1(0)$ it holds that $|x|<1$ which implies that

$$|\tilde{x}(x)| = |\frac{x}{|x|^2}| = \frac{1}{|x|} > 1$$

and second because

$$\Phi(|x|(\tilde{x}(x)-y)) \text{ singular } \iff y = \tilde{x}(x).$$

Because by construction Φ has only on singularity and because $\tilde{x}(x) \notin B_1(0)$ for $x \in B_1(0)$ it follows that $-\Phi(|x|(\tilde{x}(x)-y)) = G_\Omega(x,y) - \Phi(x-y)$ is harmonic on $B_1(0)$. $\square$

Theorem 4.8: Poisson Represation Formula

For $f \in C^2(\overline{B_r(z)})$ and $g \in C(\partial B_r(z))$ the unique solution of the Dirichlet Problem on $\Omega = B_r(z)$ is given by

$$u(x) = \frac{1}{r^{n-2}} \int_{B_r(z)} G_{B_1(0)}\left(\frac{x-z}{r}, \frac{y-z}{r}\right) f(y) d^n y + \frac{1 - \frac{|x-z|^2}{r^2}}{n\omega_n} \int_{\partial B_1(0)} \frac{g(z+ry)}{|\frac{x-z}{r} - y|^n} d\sigma(y).$$

Proof. Remember that we can decompose the Dirichlet probelm into two subproblem.

1. $-\Delta u = f$ on Ω and $u = 0$ on $\partial\Omega$ (Poisson equation with homogeneuous boundary data)
2. $\Delta u = 0$ on Ω and $u = g$ on $\partial\Omega$ (Laplace equation with inhomogenous boundary data).

The affine map $\psi : \overline{B_r(z)} \to \overline{B_1(0)}, x \mapsto \frac{x-z}{r}$ is a homomorphism from $B_r(z)$ to $B_1(0)$ and also from $\partial B_r(z)$ onto $\partial B_1(0)$. Further note that

$$\begin{aligned}
&r^{2-n}\Phi\left(\frac{x-z}{r} - \frac{y-z}{r}\right) - \Phi(x-y) \\
&= r^{2-n}\frac{1}{n(n-2)\omega_n|\frac{x-z}{r} - \frac{y-z}{r}|^{n-2}} - \frac{1}{n(n-2)\omega_n|x-y|^{n-2}} = 0
\end{aligned}$$

vanishes for $n > 2$ and for $n = 2$ it is constant:

$$\begin{aligned}
&r^{2-n}\Phi\left(\frac{x-z}{r} - \frac{y-z}{r}\right) - \Phi(x-y) \\
&= \frac{1}{2\pi}\log(|\frac{x-z}{r} - \frac{y-z}{r}|) + \frac{1}{2\pi}\log(|x-y|) \\
&= \frac{1}{2\pi}\left(\log(|x-y|) - \log(|r|)\right) + \frac{1}{2\pi}\log(|x-y|) = \frac{1}{2\pi}\log(|r|).
\end{aligned}$$

Due to the fact $\Phi(rx) = r^{2-n}\Phi(x)$ we can write for $x, y \in B_r(z)$ the Greens function as

$$\begin{aligned}
G_{B_1(0)}(\psi(x), \psi(y)) &= \Phi(\psi(x) - \psi(y)) - \Phi(|\psi(x)|(\tilde{x}(\psi(x)) - \psi(y))) \\
&= \frac{1}{r^{2-n}}\Phi(x - z + z - y) - \frac{1}{r^{2-n}}\Phi(|x-z|(\frac{x-z}{r|\psi(x)|^2} - y - z)) \\
&= \frac{1}{r^{2-n}}\Phi(x-y) - \Phi(|\frac{x-z}{r}|(\tilde{x}(\frac{x-z}{r}) - \frac{y-z}{r})) \\
&= \frac{1}{r^{2-n}}\Phi(x-y) - \Phi(|\frac{x-z}{r}|(\frac{x-z}{r|\frac{x-z}{r}|^2} - \frac{y-z}{r})) \\
&= \frac{1}{r^{2-n}}\Phi(x-y) - \Phi(|\frac{x-z}{r}|(r\frac{x-z}{|x-z|^2} - \frac{y-z}{r})) \\
&= \frac{1}{r^{2-n}}\Phi(x-y) - \Phi(\frac{1}{r}|\frac{x-z}{r}|(\underbrace{r^2\frac{x-z}{|x-z|^2} + z}_{=:\tilde{x}_z^r(x) \in \partial B_r(z)} - y)) \\
&= \frac{1}{r^{2-n}}\left(\Phi(x-y) - \Phi(|x-z|(\tilde{x}_z^r(x-z) - y))\right) = \frac{1}{r^{2-n}}G_{B_r(z)}(x,y).
\end{aligned}$$

The above is equivalent to

$$G_{B_r(z)}(x,y) = r^{2-n} G_{B_1(0)}(\frac{x-z}{r}, \frac{y-z}{r}).$$

We now do the first subproblem. Here the solution of the Poisson equation $-\Delta u = f$ is given by

$$u(x) = \int_{B_r(z)} G_{B_r(z)}(x,y) f(y) d^n y - \underbrace{\int_{\partial B_r(z)} g(z) \nabla_z G_{B_r(z)}(x,z) N d\sigma(z)}_{=0},$$

i.e. the solution of the homognenous possion equation differs by a constant term from the solution of the solution of the inhmomogenous Poission equation (???). By definition of the Greens function the limit to the boundary of the second term vanishes. By symmetry we can thus show that for all $x_0 \in \partial B_r(z)$:

$$\lim_{x\to x_0} u(x) = \lim_{x\to x_0} \int_{B_r(z)} G_{B_r(z)}(x,y) f(y) d^n y = \int_{B_r(z)} \lim_{x\to x_0} G_{B_r(z)}(y,x) f(y) d^n y = 0.$$

Therefore, the solution on $B_r(z)$ extends continuosly to $\partial B_r(z)$ and is equal to $g = 0$. Now we focus on the second problem. Note that for $x \in B_r(z)$ the solution then is given by

$$\begin{aligned}
u(x) &= \underbrace{\int_{B_r(z)} G_{B_r(z)}(x,y) f(\bar{y}) d^n y}_{=0} - \int_{\partial B_r(z)} g(y) \nabla_y G_{B_r(z)}(x,y) \cdot N d\sigma(y) \\
&= -r^{2-n} \int_{\partial B_r(z)} g(y) \nabla_{\frac{y-z}{r}} G_{B_1(0)}(\frac{x-z}{r}, \frac{y-z}{r}) \cdot N d\sigma(y) \\
&= -r^{2-n} \int_{\partial B_r(z)} g(y) \nabla_{\psi(y)} G_{B_1(0)}(\psi(x), \psi(y)) \cdot N d\sigma(y) \\
&= -r^{2-n} \int_{\partial B_r(z)} g(y) \nabla_y (G_{B_1(0)} \circ \psi)(x,y) \cdot N d\sigma(y) \\
&= -r^{2-n} \int_{\partial B_r(z)} g(y) \nabla_y G_{B_1(0)}(\psi(x), \psi(y)) r^{-1} \cdot N d\sigma(y) \\
&= -r^{2-n} \int_{\partial B_1(0)} g(yr+z) \nabla_y G_{B_1(0)}(\psi(x), \psi(yr+z)) r^{-1} r^{n-1} \cdot N d\sigma(y) \\
&= - \int_{\partial B_1(0)} g(yr+z) \nabla_y G_{B_1(0)}(\psi(x), y) \cdot N d\sigma(y)
\end{aligned}$$

In the following we will write $x := \psi(x) \in B_1(0)$ (a little crude definition) and analyze for $y \in \partial B_1(0)$ the term $K(x,y) := \nabla_y G_{B_r(z)}(x,y) \cdot N$. We will show that this term is harmonic. Clearly it holds

$$\begin{aligned}
&\nabla_y |x-y|^{2-n} = (2-n)|x-y|^{1-n} \nabla_y |x-y| = (2-n)|x-y|^{1-n} \nabla_y \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2} \\
&= (2-n)|x-y|^{1-n} \frac{1}{2} \frac{1}{|x-y|} \nabla (x-y)^2 \\
&= (2-n)|x-y|^{1-n} \frac{(-1)(x-y)}{|x-y|}.
\end{aligned}$$

Using the above and the fact that $|y| = 1$ we get for $n > 2$ (not done for $n = 2$)

$$
\begin{aligned}
K(x,y) &:= -\nabla_y \left(\Phi(x-y) - |x|^{2-n}\Phi(\tilde{x}(x)-y)\right)\frac{y}{|y|} \\
&= -\frac{1}{n(n-2)\omega_n}\nabla_y\left(\frac{1}{|x-y|^{n-2}} - |x|^{2-n}\frac{1}{|\tilde{x}(x)-y|^{n-2}}\right) y \\
&= -\frac{y}{n(n-2)\omega_n}\nabla_y\left(|x-y|^{2-n} - |x|^{2-n}|\tilde{x}(x)-y|^{2-n}\right) \\
&= -\frac{y}{n(n-2)\omega_n}\left((2-n)|x-y|^{1-n}\frac{(-1)(x-y)}{|x-y|} - |x|^{2-n}(2-n)|\tilde{x}(x)-y|^{1-n}\frac{(-1)(\tilde{x}(x)-y)}{|\tilde{x}(x)-y|}\right) \\
&= \frac{y}{n\omega_n}\left(|x-y|^{1-n}\frac{(-1)(x-y)}{|x-y|} - |x|^{2-n}|\tilde{x}(x)-y|^{1-n}\frac{(-1)(\tilde{x}(x)-y)}{|\tilde{x}(x)-y|}\right) \\
&= \frac{y}{n\omega_n}\left(\frac{y-x}{|x-y|^n} - |x|^{2-n}\frac{y-\tilde{x}(x)}{|\tilde{x}(x)-y|^n}\right) \\
&= \frac{y(y-x)}{n\omega_n|x-y|^n} - \frac{y(y-\tilde{x}(x))}{n\omega_n|\tilde{x}(x)-y|^n}|x|^{2-n} \\
&= \frac{y(y-x)}{n\omega_n|x-y|^n} - \frac{y(y-\tilde{x}(x))}{n\omega_n|\tilde{x}(x)-y|^n}|x|^{2-n}
\end{aligned}
$$

Due to the fact that

$$
\begin{aligned}
&|\tilde{x}(x)-y|^2|x|^2 = \sum_{i=1}^n |x|^2\left(\frac{x_i}{|x|^2} - y_i\right)^2 \\
&= \sum_{i=1}^n |x|^2\left(\frac{x_i^2}{|x|^4} - 2\frac{x_i}{|x|^2}y_i + y_i^2\right) \\
&= \sum_{i=1}^n \left(\frac{x_i^2}{|x|^2} - 2x_iy_i + |x|^2y_i^2\right) \\
&= \frac{\sum_{i=1}^n x_i^2}{|x|^2} - \sum_{i=1}^n 2x_iy_i + |x|^2\sum_{i=1}^n y_i^2 \\
&= 1 - \sum_{i=1}^n 2x_iy_i + |x|^2 = |y|^2 - \sum_{i=1}^n 2x_iy_i + |x|^2 = |x-y|^2 \\
&\iff |\tilde{x}(x)-y||x| = |x-y|
\end{aligned}
$$

we get that the above is

$$\begin{aligned}
K(x,y) &= \frac{y(y-x)}{n\omega_n|x-y|^n} - \frac{y(y-\tilde{x}(x))}{n\omega_n|\tilde{x}(x)-y|^n}|x|^{2-n} \\
&= \frac{y(y-x)}{n\omega_n|x-y|^n} - \frac{y(y-\tilde{x}(x))}{n\omega_n|x-y|^n}|x|^2 \\
&= \frac{yy-yx)}{n\omega_n|x-y|^n} - \frac{yy|x|^2-yx}{n\omega_n|x-y|^n} \\
&= \frac{\overbrace{yy}^{=|y|=1}}{n\omega_n|x-y|^n} - \frac{yy|x|^2}{n\omega_n|x-y|^n} = \frac{1-|x|^2}{n\omega_n|x-y|^n}
\end{aligned}$$

Because $y \in \partial B_1(0)$ it follows for all $x \in B_1(0)$ that

$$\Delta_x K(x,y) = -\Delta_x \nabla_y G_{B_1(0)}(x,y)\frac{y}{|y|} = -\nabla_y \underbrace{\Delta_x G_{B_1(0)}(x,y)}_{=\Delta_x G_{B_1(0)}(x,y)=-\delta_0(x-y)=0} \frac{y}{|y|} = 0.$$

Thus due to symmetry $x \mapsto K(x,y)$ is harmonic.
We now want to show that u extends continuously to $\partial B_r(z)$ and is there equal to 1. For $x \in B_1(0)$ and $x_0 \in \partial B_1(0)$ it holds with symmetry

$$\lim_{x\to x_0} u(z+rx) = \int_{\partial B_1(0)} g(z+ry) \lim_{x\to x_0} K(x,y)d\sigma(y) = \int_{\partial B_1(0)} g(z+ry) \lim_{x\to x_0} K(y,x)d\sigma(y) \text{ exists.}$$

Next, observe that

- $\forall (x,y) \in B_1(0) \times \partial B_1(0) : K(x,y) = \frac{1-|x|^2}{n\omega_n|x-y|^n} > 0.$
- The mapping $y \mapsto K(\lambda x, y)$ is uniform convergent as $\lambda \to 1$ for $x \in \partial B_1(0)$. This is due to the following: Choose $x \in \partial B_1(0)$ and $y \in \partial B_1(0) \setminus \partial B_\epsilon(x)$ for $\epsilon > 0$ arbitary. Then

  $$K(x\lambda, y) = \frac{1-|\lambda x|^2}{n\omega_n|\lambda x-y|^n} = \frac{1-\lambda^2}{n\omega_n|\lambda x-y|^n} \le \frac{1-\lambda^2}{n\omega_n(\epsilon-(1-\lambda))^n} \to 0, \lambda \to 1,$$

  although we used that $|\lambda x - y| \ge |x-y| - |x-\lambda x| \ge \epsilon - |x-\lambda x| = \epsilon - (1-\lambda)$.
- Now choose $x \in \partial B_1(0)$ and $y = x$, then

  $$K(\lambda x, y) = \frac{1-|\lambda x|^2}{n\omega_n|\lambda x - x|^n} = \frac{1-\lambda^2}{n\omega_n(1-\lambda)^n} = \frac{(1-\lambda)(1+\lambda)}{n\omega_n(1-\lambda)^n} = \frac{1+\lambda}{n\omega_n(1-\lambda)^{n-1}} \to \infty, \lambda \to 1.$$

We now want to show that for $x_0 \in \partial B_r(z)$ and $x \in B_r(z)$ that

$$\lim_{x\to x_0} u(x) = g(x_0).$$

For that define $w \equiv 1$ on $\overline{B_1(0)}$, then it holds

$$\begin{aligned}
&\int_{\partial B_1(0)} K(x,y)d\sigma(y) = \int_{\partial B_1(0)} w(y)\nabla_y G_{B_1(0)}(x,y)\cdot N d\sigma(y)\\
&= \int_{\partial B_1(0)} (w(y)\nabla_y G_{B_1(0)}(x,y) - G_{B_1(0)}(x,y)\nabla_y w(y))\cdot N d\sigma(y)\\
&\overset{Greens2}{=} \int_{B_1(0)} w(y)\Delta_y G_{B_1(0)}(x,y) - G_{B_1(0)}(x,y)\Delta_y w(y) d^n y\\
&= \int_{B_1(0)} \Delta_y G_{B_1(0)}(x,y)d^n y = \int_{B_1(0)} \delta_x(y)d^n y = 1.
\end{aligned}$$

Using that it follows for all $\epsilon > 0$ and $\delta > 0$ such that $|x-y| < \delta$ it follows with continuity that $g(y)-g(x_0) < \frac{\epsilon}{2}$. Due to the fact that $\int_{\partial B_1(0)\setminus\partial B_\delta(x_0)} K(x,y)d\sigma(y) \to 0$ as $x \to x_0$ for $|y-x_0| < \delta$ such that

$$\int_{\partial B_1(0)\setminus\partial B_\delta(x_0)} K(x,y)d\sigma(y) \leq \frac{\epsilon}{4\|g\|_\infty}$$

it follows

$$\begin{aligned}
|u(x)-g(x_0)| &\leq \int_{\partial B_1(0)} K(x,y)|g(y)-g(x_0)|d\sigma(y)\\
&\leq \int_{\partial B_1(0)\setminus\partial B_\delta(x_0)} K(x,y)|g(y)-g(x_0)|d\sigma(y) + \int_{\partial B_\delta(x_0)} K(x,y)|g(y)-g(x_0)|d\sigma(y)\\
&\leq 2\|g\|_\infty \int_{\partial B_1(0)\setminus\partial B_\delta(x_0)} K(x,y)d\sigma(y) + \int_{\partial B_\delta(x_0)} K(x,y)\frac{\epsilon}{2}d\sigma(y)\\
&\leq 2\|g\|_\infty \underbrace{\int_{\partial B_1(0)\setminus\partial B_\delta(x_0)} K(x,y)d\sigma(y)}_{\to 0, x\to x_0} + \frac{\epsilon}{2}\underbrace{\int_{\partial B_1(0)} K(x,y)d\sigma(y)}_{=1}\\
&\leq 2\|g\|_\infty \frac{\epsilon}{4\|g\|_\infty} + \frac{\epsilon}{2} = \epsilon.
\end{aligned}$$

Because $\epsilon > 0$ was chosen arbitrary the assertion follows.
Finally, we have shown in this proof that the solution of 1. is

$$u_1(x) = \int_{B_r(z)} G_{B_r(z)}(x,y)f(y)d^n y = \frac{1}{r^{n-2}}\int_{B_r(z)} G_{B_1(0)}(\psi(x),\psi(y))f(y)d^n y$$

and for 2. is

$$u_2(x) = \int_{\partial B_r(z)} g(yr+z)\nabla_y G_{B_1(0)}(\psi(x),y)\cdot N d\sigma(y) = \frac{1-|\psi(x)|^2}{n\omega_n}\int_{\partial B_r(z)} \frac{g(yr+z)}{|\psi(x)-y|^n}d\sigma(y).$$

Thus in total the solution of the Dirichlet problem is

$$u(x) = \frac{1}{r^{n-2}}\int_{B_r(z)} G_{B_1(0)}(\frac{x-z}{r},\frac{y-z}{r})f(y)d^n y + \frac{1-|\frac{x-z}{r}|^2}{n\omega_n}\int_{\partial B_r(z)} \frac{g(yr+z)}{|\frac{x-z}{r}-y|^n}d\sigma(y).$$

$\square$

Corollary 4.9

Harmonic functions on an open domain $\Omega \subset \mathbb{R}^n$ are analytic.

Proof. Let u be harmonic on $B_r(z)$ and extend continuously to $\partial B_r(z)$ where $u \equiv g$, for some $g \in C^2(\partial B_r(z))$. This can be represented as a Dirichlet problem. Its solution is the given by the result of the last theorem:

$$\begin{aligned} u(x) &= \frac{1}{r^{n-2}} \int_{B_r(z)} G_{B_1(0)}\left(\frac{x-z}{r}, \frac{y-z}{r}\right) \underbrace{f(y)}_{=-\Delta u = 0} \, d^n y + \frac{1 - \frac{|x-z|^2}{r^2}}{n\omega_n} \int_{\partial B_1(0)} \frac{g(z+ry)}{|\frac{x-z}{r} - y|^n} d\sigma(y) \\ &= \frac{1 - \frac{|x-z|^2}{r^2}}{n\omega_n} \int_{\partial B_1(0)} \frac{u(z+ry)}{|\frac{x-z}{r} - y|^n} d\sigma(y) \\ &= \frac{1 - \frac{|x-z|^2}{r^2}}{n\omega_n} \int_{\partial B_r(z)} \frac{u(y)}{|\frac{x-z}{r} - \frac{y-z}{r}|^n} r^{1-n} d\sigma(y) \\ &= \frac{r^2 - |x-z|^2}{nr^2\omega_n} \int_{\partial B_r(z)} \frac{u(y)}{|x-y|^n} r^n r^{1-n} d\sigma(y) \\ &= \frac{r^2 - |x-z|^2}{nr\omega_n} \int_{\partial B_r(z)} \frac{u(y)}{|x-y|^n} d\sigma(y) \end{aligned}$$

Note that for $x = z$ it follows that $|x - y| = r$ and

$$u(x) = \frac{r^2}{nr\omega_n} \int_{\partial B_r(z)} \frac{u(y)}{r^n} d\sigma(y) = \frac{1}{nr^{n-1}\omega_n} \int_{\partial B_r(z)} u(y) d\sigma(y),$$

which is exactly the mean value property.
Remember for a sequence of the form $(\sum_{i=1}^n a_i x^i)_{n\in\mathbb{N}}$ the radius of convergence is defined as $R := \frac{1}{\lim_{i\to\infty} \sqrt[n]{|a_i|}}$.
A function is called analytic in x_0 if there exists a Taylor series $\sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ that converges to $f(x_0)$ as $n \to \infty$ and its convergence radius R is positive. ...? □

Lemma 4.10

Let $\Omega \subset \mathbb{R}^n$ be an open neighborhood of 0 and u a bounded harmonic function on $\Omega \setminus \{0\}$. Then u extends as a harmonic function to Ω.

Proof. Let $B_r(0) \subset \Omega$ have compact closure in Ω. With the theorem before the solution for the Dirichlet Problem where $f = 0$ on $B_r(0)$ and $g = u$ on $\partial B_r(0)$ there exists a unique $\tilde{u} \in C^2(B_r(0)) \cap C^0(\overline{B_r(0)})$. Note that u is as defined in the lemma. We define

$$u_\epsilon(x) := \tilde{u}(x) - u(x) + \epsilon G_{B_r(0)}(x, 0), \quad x \in B_r(0) \setminus \{0\}, \quad \epsilon \in \mathbb{R}.$$

By definition of the Greens function it holds in distribution that $-\Delta G_{B_r(0)}(x, 0) = \delta_0$, $G_{B_r(0)}(x, 0)$ vanishes on $\partial B_r(0)$ and it has a singularity in $x = 0$, due to the fact that $G_{B_r(x,0)}$ is singular iff $\tilde{x}(x) = 0$, which is equivalent to $x = 0$.
Due to the fact that $G_{B_r(0)}(x, 0)$ vanishes on $\partial B_r(0)$ and that for $x_0 \in \partial B_r(0)$ $\tilde{u}(x_0) = u(x_0)$, it follows $u_\epsilon(x_0) = 0$. Further, as stated before, $G_{B_r(0)}(x, 0)$ is singular in $x = 0$. Due to the fact that $\tilde{u}$ and u are

bounded it follows for every $\epsilon > 0$ that $u_\epsilon(x) \to \infty, x \to 0$. Note that u_ϵ is a harmonic function on $B_r(0) \setminus \{0\}$ (linear combination of harmonic functions).
We now want to show that for $\epsilon > 0$ the function u_ϵ is non-negative. For that assume that u_ϵ takes a negative value on $B_r(0) \setminus \{0\}$. Due to the fact that there is a pole in $x = 0$ with $u_\epsilon(0) = \infty$ and on the boundary we have $u_\epsilon(x) = 0$ it follows that there exists a negative minimum value in the interior. With the strong minimum principle it follows that u_ϵ is constant. This a contradiction to the fact that the pole in $x = 0$ is equal to $+\infty$.
The argument is analougous for $\epsilon < 0$ and u_ϵ non-positive.
In total it holds

$$0 \leftarrow \lim_{\epsilon \nearrow 0} u_\epsilon(x) = \tilde{u}(x) - u(x) = \lim_{\epsilon \searrow 0} u_\epsilon(x) \to 0, \quad x \in B_r(0) \setminus \{0\}.$$

Thus $\tilde{u}$ is a harmonic extension of u on Ω. □

In the above lemma we used the condition that the harmonic function u is bounded. This condition could be weakend and we still would get the same result: For every u harmonic on $\Omega \setminus \{0\}$ with

$$\forall \epsilon > 0 \exists \delta_\epsilon > 0 \forall x \in B_{\delta_\epsilon}(0) \setminus \{0\} : \quad |u(x)| \leq \epsilon G_{B_{\delta_\epsilon}}(x, 0),$$

has a harmonic extension to Ω.

5 Dirichlet Principle

Theorem 5.1

Let $\Omega \subset \mathbb{R}^n$ be bounded, open domain and $\partial\Omega$ a $n-1$ dimensional submanifold. Let $f : \bar{\Omega} \to \mathbb{R}$ and $g : \partial\Omega \to \mathbb{R}$ be continuous and u the solution of the respective Dirichlet Problem, that is given by the minimizer of the following functional

$$I : \{w \in C^2(\bar{\Omega}) \mid w\big|_{\partial\Omega} = g\} \to \mathbb{R}, \quad w \to I(w) := \int_\Omega \frac{1}{2} \nabla w \cdot \nabla w - wf d^n x.$$

Proof. We show the equivalence of u being the solution of the Dirichlet Problem and u being a minimizer.
"$\Rightarrow$"
Let $u, w \in \{w \in C^2(\bar{\Omega}) \mid w\big|_{\partial\Omega} = g\}$ where u is the solution of the Dirichlet problem and w just any function in this set. Due to the fact that u is the solution of the Dirichlet problem we have on Ω that $-\Delta u = f$, i.e.

$$\begin{aligned}
0 &= \int_\Omega (-\Delta u - f) d^n x = \int_\Omega (-\Delta u - f)(u - w) d^n x \\
&= \int_\Omega -\Delta u u + \Delta u w - fu + fw d^n x \\
&= \int_{\partial\Omega} \underbrace{(w - u)}_{=0} \nabla u \cdot N d\sigma(z) - \int_\Omega \nabla u \nabla (w - u) - f(w - u) d^n x \\
&= \int_\Omega \nabla u \nabla (u - w) - f(u - w) d^n x.
\end{aligned}$$

This is now equivalent to

$$\int_\Omega \nabla u \nabla u - fu d^n x = \int_\Omega \nabla u \nabla w - fw d^n x.$$

Using this if follows

$$\begin{aligned}\int_\Omega \nabla u \nabla u - fud^n x &= \int_\Omega \nabla u \nabla w - fwd^n x\\ &= \int_\Omega \nabla u \nabla w d^n x - \int_\Omega fwd^n x\\ &\overset{CS}{\leq} \int_\Omega \nabla u \nabla w + \frac{1}{2}(\nabla u - \nabla w)(\nabla u - \nabla w)d^n x - \int_\Omega fwd^n x\\ &= \int_\Omega \frac{1}{2}\nabla u \nabla u + \frac{1}{2}\nabla w \nabla w - fwd^n x = \int_\Omega \frac{1}{2}\nabla u \nabla u d^n x + I(w),\end{aligned}$$

which is equivalent to

$$I(u) \leq I(w).$$

"$\Leftarrow$"
Let u be a minum of that function $I(\cdot)$ then for every $v \in C^2(\bar{\Omega})$ that vanish on $\partial\Omega$ satisfy

$$\begin{aligned}0 &= \frac{d}{dt} I(u+tv)\big|_{t=0} = \frac{d}{dt}\left(\int_\Omega \frac{1}{2}\nabla(u+tv)\nabla(u+tv) - (u+tv)fd^n x\right)\big|_{t=0}\\ &= \frac{d}{dt}\left(I(u) + \int_\Omega \nabla u t \nabla v + \frac{1}{2}t^2 \nabla u \nabla v - tvfd^n x\right)\big|_{t=0}\\ &= \int_\Omega \nabla u \nabla v d^n x + \int_\Omega t \nabla u \nabla v d^n x\big|_{t=0} - \frac{d}{dt}\left(\int_\Omega tvfd^n x\right)\big|_{t=0}\\ &= \int_\Omega \nabla u \nabla v - vfd^n x = \int_\Omega (-\Delta u - f)vd^n x\\ &\iff -\int_\Omega \Delta u v d^n x = \int_\Omega fvd^n x.\end{aligned}$$

Because v was chosen arbitrary, it follows that $-\Delta u = f$ on Ω. □

Remember from numerics of PDEs that this last equation in the proof was called variational equation of the Poission equation, that was formulted to find a weak solution.

Remark 5.1

Let u, v be two solutions of the Dirchlet problem, then

$$\Delta(u-v) = \text{on } \Omega \text{ and } u - v = g - g = 0 \text{ on } \partial\Omega.$$

This implies $I(u-v) = 0$??? and larger set of functions???

6 A PDE with no solutions

Skipped for now ???

7 Summary

We began by exploiting the rotational and translational invariance of the Laplacian to derive the fundamental solution Φ. With that we get the solution of the Poisson equation as the convolution $u = \Phi * f$. On bounded domains we then showed that a function is harmonic, i.e. $\Delta u = 0 \iff$ it satisfies the mean value property. This results from the fact that the mean value function $\mathcal{S}(r)$ is constant which is equivalent to $\Delta u = 0$. This then implies regularity for smooth harmonic functions in the interior:

$$|\partial^\alpha u(x)| \leq C(n, |\alpha|) r^{-|\alpha|} \|u\|_{L^\infty(\overline{B_r(x)})}.$$

With Liouvilles we got that harmonic functions on $\mathbb{R}^n$ which are bounded are constant.
The next step was to generalize these results to distributions. For that we introduced rotationally symmertric kernels f such that $U(f) = 0$ results in the mean of u being constant regardless of the radius of the average. Weyl's Lemma then shows that harmonic distributions are regular, i.e. they are of the form F_u, where $u \in C^\infty(\Omega)$, i.e. smooth. Using the fundamental Lemma of the calculus of variations, i.e $L^1_{loc}(\Omega) \to \mathcal{D}'(\Omega), f \mapsto F_f$ being injective, we get the result that every distribution U that satisfies the weak mean value property corresponds to a smooth harmonic function, meaning that every weak soltion to the Laplace equation coincides with a smooth strong solution.
Harnacks inequality then yields that the solution on open path-connected Domains with compact closure at individual points are comparable by a finite multiplicative factor from other points. This tells us, that the solution deos not osccilate pointwise too much. Harnacks princpile tells us that if a sequence of monotone harmonic functions is bounded (which implies converge on $\mathbb{R}^n$) at a specific point, this is equivalent to convergence of that sequence on entire compact subsets of open path connected domains. Later the Dirichlet principle gives us that the limit of energy functionals solve Dirichlet problem. A sequence of energy functionals that converge to that minimizer does then with the Dirichlet principle exist and is indeed harmonic.
The strong maximum principle states that harmonic functions cannot attain their extremum in the interior of an open path connected domain unless its constant. The weak maximum principle ensures that the maximum is attained on the boundary of bounded domains. This result is then applied to arbitrary linear elliptic differential operators.
So far we have not considered boundary conditions. To solve the Dirichlet problem, we introced the Greens functions in order to construct solutions for the Poisson equation with boundary condition $u = g$. Using Greens identities and the Greens representation theorem we can write down the solution to the Poisson equation in terms of the Greens function. Using the symmetry of the Greens function we can write it on $B_1(0)$ in terms of the fundamental solution Φ. Then Poissions representation theorem gives us a specific form of the solution of the Dirichlet Problem on $B_r(z)$, which can be written such that it only depends on the boundary $\partial B_r(z)$. This can be then used to imply that all harmonic functions are analytic. Finally, we can conclude that every bounded harmonic function on $\Omega \setminus \{0\}$ has to extend to Ω harmonically, which means that it is smooth at the origin and it does not explode there.
The Dirichlet principle gives us an equivalent formulation of the Dirichlet problem to a minimization problem of a variational equation with unqiue solution.

CHAPTER 4

HEAT EQUATION

We now focus on $\frac{\partial}{\partial t}u - \Delta u = 0$ the heat eqaution and is inhomogenous form $\dot{u} - \Delta u = f$, where u is defined on the open domain $\Omega \subset \mathbb{R}^n \times \mathbb{R}$ and f on Ω. We will extend statements about harmonic function to solution of the heat equation. This equation models the diffusion of some quantity. (The heat equation follows from the scalar conservation law???)

1 Fundamental Solution of the Heat Equation

Since in the heat equation there are only first order derivatives in time and second order in space it holds that

$$u(x,t) \text{ is solution } \Rightarrow \forall \lambda \in \mathbb{R}: \quad u(\lambda x, \lambda^2 t) \text{ is solution.}$$

This is because

$$\frac{\partial}{\partial t}u(\lambda x, \lambda^2 t) - \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j} u(\lambda x, \lambda^2 t) = \lambda^2 \frac{\partial}{\partial(\lambda^2 t)} u(\lambda x, \lambda^2 t) - \lambda^2 \sum_{i,j=1}^{n} \frac{\partial^2}{\partial(\lambda x_i)\partial(\lambda x_j)} u(\lambda x, \lambda^2 t) = 0.$$

This suggests that the ratio $\frac{x^2}{t}$ is of interest. We invoke

$$u(x,t) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right), x \in \mathbb{R}^n, t \in \mathbb{R}^+, \alpha, \beta \in \mathbb{R}$$

and $v : \mathbb{R}^n \to \mathbb{R}$ an unknown function. Using this it now holds

$$\lambda^\alpha u(\lambda^\beta x, \lambda t) = \lambda^\alpha \frac{1}{(\lambda t)^\alpha} v\left(\frac{\lambda^\beta x}{(\lambda t)^\beta}\right) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right) = u(x,t).$$

With the choice $\lambda = \frac{1}{t}$ we have

$$u(\frac{x}{t^\beta}, 1) = t^\alpha u(x,t) = v\left(\frac{x}{t^\alpha}\right)$$

Then it follows with $y := \frac{x}{t^\beta}$ that

$$\begin{aligned}
0 &= \frac{\partial}{\partial t}u(x,t) - \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j} u(x,t) = \frac{\partial}{\partial t}\left(\frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right)\right) - \frac{1}{t^\alpha} \sum_{i,j=1}^{n} \frac{\partial^2}{\partial x_i \partial x_j}\left(v\left(\frac{x}{t^\beta}\right)\right) \\
&= -\frac{\alpha}{t^{\alpha+1}} v\left(\frac{x}{t^\beta}\right) - \frac{1}{t^\alpha} \sum_{i=1}^{n} \beta \frac{x_i}{t^{\beta+1}} \frac{\partial}{\partial y_i} v(y) - \frac{1}{t^{\alpha+2\beta}} \sum_{i,j=1}^{n} \frac{\partial^2}{\partial y_i \partial y_j} v(y) \\
&= -\alpha t^{-(\alpha+1)} v(y) - \beta t^{-(\alpha+1)} y^T \nabla v(y) - t^{-(\alpha+2\beta)} \Delta v(y).
\end{aligned}$$

If we set $\beta = \frac{1}{2}$ then

$$\begin{aligned} & -\alpha t^{-(\alpha+1)}v(y) - \frac{1}{2}t^{-(\alpha+1)}y^T\nabla v(y) - t^{-(\alpha+1)}\Delta v(y) = 0 \\ \iff & \alpha v(y) + \frac{1}{2}y^T\nabla v(y) + \Delta v(y) = 0. \end{aligned}$$

Define $r := |y| = \sqrt{y^T y}$ and $v(y) := w(r)$ then

$$\Delta v(y) = \nabla_y \cdot \nabla_y w(|y|) = \nabla_y \cdot \left(w'(r)\frac{y}{|y|}\right) = w''(r) + \frac{n-1}{r}w'(r)$$

that gives us

$$0 = \alpha w(r) + \frac{1}{2}y^T w'(r)\frac{y}{|y|} + w''(r) + \frac{n-1}{r}w'(r) = \alpha w(r) + \frac{1}{2}rw'(r) + w''(r) + \frac{n-1}{r}w'(r).$$

This with $\alpha = \frac{n}{2}$ it follows

$$\begin{aligned} & 0 = r^{n-1}\left(\frac{n}{2}w(r) + \frac{1}{2}rw'(r) + w''(r) + \frac{n-1}{r}w'(r)\right) \\ \iff & 0 = r^{n-1}\frac{n}{2}w(r) + r^{n-1}\frac{1}{2}rw'(r) + r^{n-1}w''(r) + r^{n-1}\frac{n-1}{r}w'(r) \\ \iff & 0 = \frac{1}{2}\left(r^n w(r)\right)' + \left(r^{n-1}w'(r)\right)' \quad \text{the second summand is the same for the Laplace eq.} \\ \iff & 0 = \frac{d}{dr}\left(\frac{1}{2}r^n w(r) + r^{n-1}w'(r)\right) \\ \iff & \int_{\mathbb{R}} 0 dr = \int_{\mathbb{R}} \frac{d}{dr}\left(\frac{1}{2}r^n w(r) + r^{n-1}w'(r)\right) dr = \frac{1}{2}r^n w(r) + r^{n-1}w'(r) \\ \iff & a = \frac{1}{2}r^n w(r) + r^{n-1}w'(r), \end{aligned}$$

where $a \in \mathbb{R}$ is an arbitrary constant. Due to the fact that w and w' for $r \to \infty$ it follows $a = 0$. This gives us the ODE

$$w'(r) = -\frac{1}{2}rw(r),$$

whose solution is given by $w(r) = be^{-\frac{r^2}{4}}$ and $b \in \mathbb{R}$ is arbitrary. For a special choice of b we get the fundamental solution

Definition 1.1

The fundamental solution of the heat equation is giveny by

$$\Phi(x,t) := \begin{cases} \frac{1}{(4\pi t)^{n/2}}e^{-\frac{|x|^2}{4t}} & , t > 0 \\ 0 & , t < 0 \end{cases}, x \in \mathbb{R}^n.$$

For $t = 0$ the Fundamental solution is not defined. It is defined as the limit from above, i.e. $\lim_{t\searrow 0}\Phi(x,t)$.

Lemma 1.2

For all $t > 0$ the fundamental solution satisfies

$$\int_{\mathbb{R}^n} \Phi(x,t) d^n x = 1.$$

Proof. For $y := \frac{x}{\sqrt{4t}}$ it holds

$$\frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} e^{-\frac{|x|^2}{4t}} d^n x = \frac{1}{\pi^{n_2}} \int_{\mathbb{R}^n} e^{-y^2} d^n y = \frac{1}{\pi^{n_2}} \left(\int_{\mathbb{R}^n} e^{-y^2} dy \right)^n = 1.$$

□

Theorem 1.3

For $h \in C_b(\mathbb{R}^n, \mathbb{R})$ the function

$$u(x,t) := \int_{\mathbb{R}^n} \Phi(x-y,t) h(y) d^n y$$

satisfies

- $u \in C^\infty(\mathbb{R}^n, \mathbb{R})$
- $\dot{u} - \Delta u = 0$ on $\mathbb{R}^n \times \mathbb{R}^+$
- u extends continuously and bounded to $\mathbb{R}^n \times [0, \infty)$ with $\lim_{t \searrow 0} u(x,t) = h(x)$ for all $x \in \mathbb{R}^n$.

Proof. i)

Since $\Phi(\cdot,\cdot)$ is smooth on $\mathbb{R}^n \times \mathbb{R}^+$ and h is bounded it follows $u(x,t)$ is well defined, bounded and continuous on $\mathbb{R}^n \times \mathbb{R}^+$. Define

$$F : \mathbb{R}^n \times \mathbb{R}^+ \to L^1(\mathbb{R}^n), (x,t) \mapsto F(x,t) := \big(y \mapsto \Phi(x-y,t) h(y) \big)$$

is infinitely integrable and all derivatives stay in $L^1(\mathbb{R}^n)$. Further, define

$$I : L^1(\mathbb{R}^n) \to \mathbb{R}, \quad f \mapsto I(f) := \int_{\mathbb{R}^n} f(y) dy,$$

is linear and continuous. Then

$$u(x,t) := I(F(x,t))$$

is smooth, because as I is constinuous and linear and F is smooth.
ii)
Because Φ solves the heat equation it follows

$$\dot{u}(x,t) - \Delta u(x,t) = \int_{\mathbb{R}^n} \left(\dot{\Phi}(x-y,t) - \Delta_x \Phi(x-y,t) \right) h(y) d^n y = 0.$$

iii)
The continuity of h implies that it is uniformly continuos on compact subsets of $\mathbb{R}^n$, i.e.

$$\forall \epsilon \forall x \in K \subset \mathbb{R}^n \text{ compact } \exists \delta > 0 \forall y \in B_\delta(x) : \quad |h(y) - h(x)| < \epsilon.$$

Furthermore, by construction $\forall \delta > 0 \; \exists T > 0$ such that forall $t < T$ it holds

$$\int_{\mathbb{R}^n \setminus B_\delta(0)} \Phi(y,t) d^n y = \int_{\mathbb{R}^n \setminus B_\delta(0)} \frac{1}{t^{n/2}} \Phi(\frac{y}{\sqrt{t}}, 1) d^n y = \frac{t^{n/2}}{t^{n/2}} \int_{\mathbb{R}^n \setminus B_{\delta/t}(0)} \Phi(z,1) d^n z < \epsilon,$$

where we used that

$$\Phi(y,t) = \frac{1}{(4\pi t)^{n_2}} e^{-\frac{|x|^2}{4t}} = \frac{1}{t^{n/2}} \frac{1}{(4\pi)^{n_2}} e^{-\frac{|x|^2}{4t}} = \frac{1}{t^{n/2}} \Phi(\frac{y}{\sqrt{t}}, 1).$$

It then follows that for all $t < T$:

$$\begin{aligned}
|u(x,t) - h(x)| &= \left| \int_{\mathbb{R}^n} \Phi(x-y,t)(h(y) - h(x)) d^n y \right| \\
&\leq \int_{B_\delta(x)} \Phi(x-y,t) \left| h(y) - h(x) \right| d^n y + \int_{\mathbb{R}^n \setminus B_\delta(x)} \Phi(x-y,t) \left| h(y) - h(x) \right| d^n y \\
&\leq \epsilon \int_{\mathbb{R}^n} \Phi(x-y,t) d^n y + 2\|h\|_\infty \int_{\mathbb{R}^n \setminus B_\delta(x)} \Phi(x-y,t) d^n y \\
&\leq \epsilon + 2\|h\|_\infty \epsilon = (1 + 2\|h\|_\infty)\epsilon.
\end{aligned}$$

Because $\epsilon > 0$ was chosen arbitrary the assertion follows. $\square$

2 Inhomogeneous Initial value problem

The homogenuous heat equation can be written as $\dot{u} = \Delta u$. Because the Laplace operator is a linear mapping from $C^2(\mathbb{R}^n) \to C^0(\mathbb{R}^n)$ the heat equation becomes an ODE on an infinite dimensional Banach space.

There is the homogeneous initial value problem

$$\begin{aligned}
\dot{u}(x,t) - \Delta u(x,t) = 0, &\quad (x,t) \in \mathbb{R}^n \times \mathbb{R}^+, \\
u(x,0) = h(x), &\quad x \in \mathbb{R}^n
\end{aligned}$$

and the inhomogenous initial value problem

$$\begin{aligned}
\dot{u}(x,t) - \Delta u(x,t) = f(x,t), &\quad (x,t) \in \mathbb{R}^n \times \mathbb{R}^+, \\
u(x,0) = 0, &\quad x \in \mathbb{R}^n.
\end{aligned}$$

Assume that u is the solution of the inhomegenous initial value problem and v_t the solution of the homogenous inital value problem with $v_t(x,0) = h_t(x)$. Further, assume that $f(x,t) = h_t(x)$, then define

$$u(x,t) := \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s) f(y,s) d^n y ds = \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s) h_s(y) d^n y ds.$$

We now want to show that u solves the inhomogeneous PDE $\dot{u} - \Delta u = f$. Remember the Leipniz rule: For any differentiable function $G(x,t)$ it holds

$$\frac{d}{dt} \int_0^t G(t,s) ds = \int_0^t \frac{\partial}{\partial t} G(t,s) ds + G(t,t).$$

Using this we get

$$
\begin{aligned}
\dot{u}(x,t) - \Delta u(x,t) &= \frac{\partial}{\partial t}\int_0^t v_s(x,t-s)ds + \int_0^t \Delta v_s(x,t-s)ds \\
&= \int_0^t \frac{\partial}{\partial t} v_s(x,t-s)ds + v_t(x,0) + \int_0^t \Delta v_s(x,t-s)ds \\
&= \int_0^t \int_{\mathbb{R}^n} \Big(\dot{\Phi}(x-y,t-s) - \Delta\Phi(x-y,t-s)\Big) f(y,s)d^n y ds + \lim_{s\to t}\int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,t)d^n y \\
&= \int_{\mathbb{R}^n} \delta(x-y)f(y,t)d^n y = f(x,t).
\end{aligned}
$$

It solves the inhomogeneous heat equation. Because

$$u(x,t) = \int_0^t v_s(x,t-s)ds$$

it suffices to study the homogeneous initial value problem as we can just integrate over time and thus get the solution to the non-homogenous problem.

Theorem 2.1

If f is twice continuously and bounded differentiable on $\mathbb{R}^n \times \mathbb{R}^+$ then the solution of the inhomogenous initial value problem

$$\dot{u} - \Delta u = f \text{ on } \mathbb{R}^n \times \mathbb{R}^+, \quad \lim_{t\to 0} u(x,t) = 0$$

is given by

$$u(x,t) = \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,s)d^n ds = \int_0^t \int_{\mathbb{R}^n} \Phi(y,s)f(x-y,t-s)d^n y ds.$$

Proof. We have already shown that

$$v_s(x,t) := \int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,s)d^n y$$

solves the homoegenous initial value problem

$$\dot{u} - \Delta u = 0 \text{ on } \mathbb{R}^n \times \mathbb{R}^+, \quad \lim_{t\to 0} u(x,t) = f(x,t)$$

and $v_s \in C^\infty(x,t)$.
We can now show that for arbitary $\epsilon > 0$ it holds

$$u_\epsilon(x,t) := \int_0^{t-\epsilon} v_s(x,t)ds = \int_0^{t-\epsilon}\int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,s)d^n y ds$$

satisfies

$$\dot{u}_\epsilon(x,t) - \Delta u_\epsilon(x,t) = \int_{\mathbb{R}^n} \Phi(x-y,t-t+\epsilon)f(y,t-\epsilon)d^n y = \int_{\mathbb{R}^n} \Phi(x-y,\epsilon)f(y,t-\epsilon)d^n y$$

which satisfies all the conditions from the previous theorem for homogenous initial value problems, giving us

$$\dot{u}_\epsilon(x,t) - \Delta u_\epsilon(x,t) \to f(x,t), \quad \epsilon \searrow 0.$$

Finally, combining the above yields the assertion:

$$f(x,t)=\lim_{\epsilon\to 0}\left(\dot{u}_\epsilon(x,t)-\Delta u_\epsilon(x,t)\right)=\lim_{\epsilon\to 0}\left(\frac{\partial}{\partial t}-\Delta\right)u_\epsilon(x,t)=\left(\frac{\partial}{\partial t}-\Delta\right)u(x,t)=\dot{u}(x,t)-\Delta u(x,t)$$

and the continuity of v_s gives

$$u(x,0)=\lim_{\epsilon\to 0}u_\epsilon(x,0)=\lim_{\epsilon\to 0}\int_0^{-\epsilon}\int_{\mathbb{R}^n}\Phi(x-y,-s)f(y,s)d^nyds=\int_{\mathbb{R}^n}\delta_0(x-y)f(y,0)d^ny=0.$$

□

In the following Corallary we call this initial value problem also Cauchy problem. It can be easily solved by splitting the solution $u=v+w$ into the initial value problems of the homogeneous and inhomogeneous heat equation.

Corollary 2.2

The inhomogenous initial value problem

$$\dot{u}-\Delta u=f\ \text{on}\ \Omega,\quad u(x,0)=h(x),\ \text{on}\ \partial\Omega,$$

has the solution of the form

$$u(x,t)=\int_{\mathbb{R}^n}\Phi(x-y,t)h(y)d^ny+\int_0^t\int_{\mathbb{R}^n}\Phi(x-y,t-s)f(y,s)d^nyds.$$

The first sum is the solution of the homogeneous initial value problem and the second is the solution to the inhomogeneous initial value problem.

3 Mean Value Property

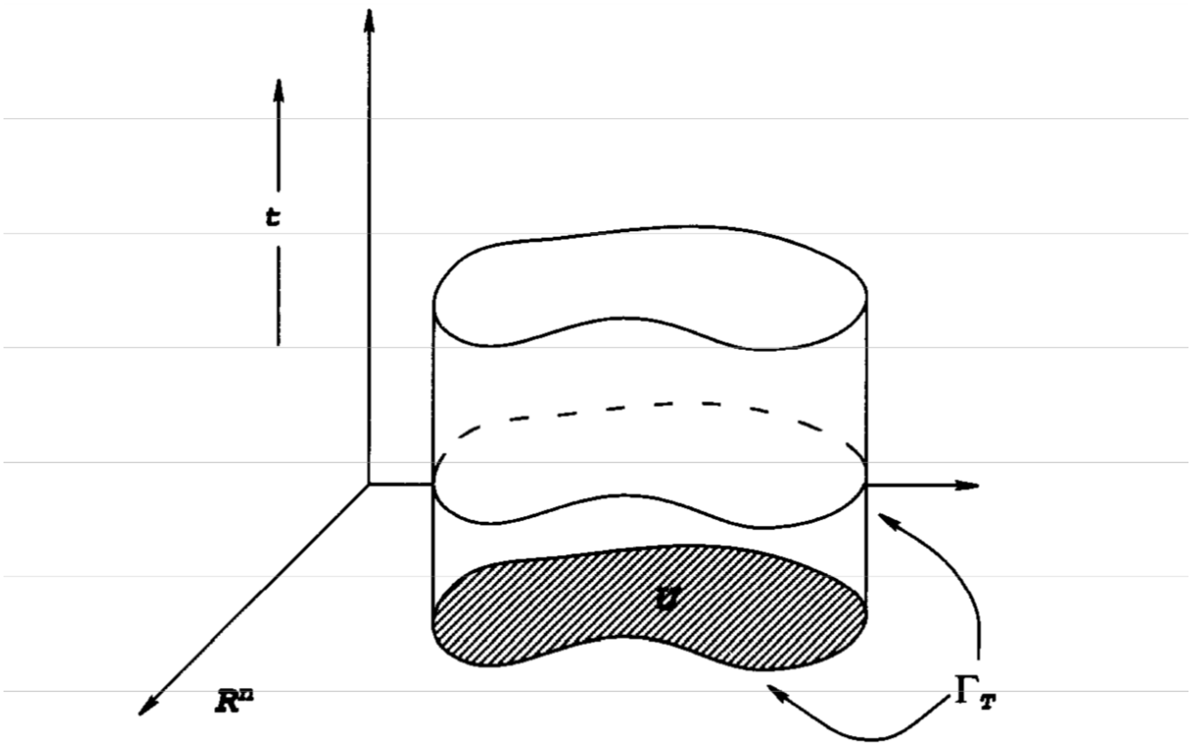


Figure 4.1: From Evans, *Partial Differential Equations*

We want to use the fundamental solution Φ to describe the solution $u(x,t)$ as some mean on a smart chosen ball.

Definition 3.1

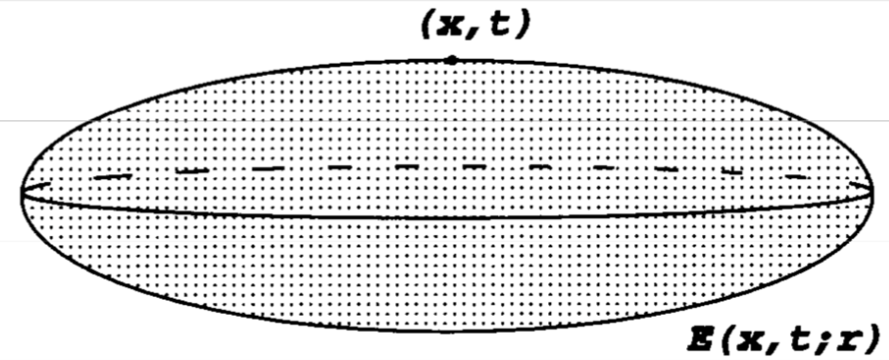
For every $(x,t) \in \mathbb{R}^n \times \mathbb{R}$ and $r > 0$ define

$$E(x,t,r) := \{(y,s) \in \mathbb{R}^{n+1} \mid s \leq t, \Phi(x-y,t-s) \geq \frac{1}{r^n}\}.$$

It then holds that

$$\begin{aligned}
&\Phi(x-y,t-s) \geq \frac{1}{r^n} \iff e^{-\frac{|x-y|^2}{4(t-s)}} \geq \frac{(4\pi)^{n/2}(t-s)^{n/2}}{r^n} \\
&\iff \left(\frac{r^2}{4(t-s)}\right)^{n/2} \frac{1}{\pi^{n/2}} \geq e^{\frac{|x-y|^2}{4(t-s)}} \\
&\iff -\frac{n}{2}\log(\pi) + \frac{n}{2}\left(2\log(r) - \log(4(t-s))\right) \iff \geq \frac{|x-y|^2}{4(t-s)} \\
&\iff 2(t-s)n\big(2\log(r) - \log(t-s) - \log(4\pi)\big) \geq |x-y|^2
\end{aligned}$$

We see that if $s \to t$ then the right hand side goes to 0. The heat ball has to be at that time a singular point. Further, if we let $s \to t - \frac{r}{4\pi}$ then the terms inside the logarithm go to 1 and thus the left hand side goes to 0. Thus the heat ball is some kind of paraboloid in space-time.



A "heat ball"

Figure 4.2: From Evans, *Partial Differential Equations*

Theorem 3.2: Mean Value Property for the Heat Equation

Let u be the solution of the heat equation on an open domain $\Omega \subset \mathbb{R}^n \times \mathbb{R}$. Then $\forall (x,t) \in \Omega$ and $r > 0$ with $E(x,t,r) \subset \Omega$:

$$u(x,t) = \frac{1}{C_n r^n} \int_{E(x,t,r)} u(y,s) \frac{|x-y|^2}{(t-s)^2} d^n y ds, \quad C_n := \int_{E(0,0,r)} \frac{|y|^2}{s^2} d^n y ds.$$

Proof. Due to translation invariance (???) $(x,t) = (0,0)$, define

$$\begin{aligned} S_{0,0}(r) &:= \frac{1}{r^n} \int_{E(0,0,r)} u(z,q) \frac{|z|^2}{q^2} d^n z dq = \frac{1}{r^n} \int_{E(0,0,r)} u(ry, r^2 s) \frac{|ry|^2}{(r^2 s)^2} d^n (ry) d(r^2 s) \\ &= \frac{1}{r^n} \int_{E(0,0,1)} u(ry, r^2 s) \frac{r^2}{r^4} \frac{|y|^2}{s^2} r^n r^2 d^n y ds = \int_{E(0,0,1)} u(ry, r^2 s) \frac{|y|^2}{s^2} d^n y ds. \end{aligned}$$

The map $T : E(x,t,r) \to E(rx, r^2 t, r), \quad (y,s) \mapsto (ry, r^2 s)$ is bijective, because it is first a isomorphism, due to its linearity and

$$\det \begin{pmatrix} \frac{\partial}{\partial y_1} T_1(y,s) & & & \\ & \ddots & & \\ & & \frac{\partial}{\partial y_n} T_n(y,s) & \\ & & & r^2 \end{pmatrix} = \det \begin{pmatrix} r \, Id_{n \times n} & 0 \\ 0 & r^2 \end{pmatrix} = r^n r^2 > 0.$$

Further, because

$$\Phi(r(x-z), r^2 t) = \frac{1}{(r^2)^{n/2}} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|r(x-z)|^2}{4r^2 t}} = \frac{1}{r^n} \Phi(x-z, t)$$

it follows that

- $\forall (x,t) \in E(x,t,1) \Rightarrow T(x,t) \in E(rx, r^2 t, r)$
- $\forall (x,t) \in E(rx,r^2t,r) \Rightarrow T^{-1}(x,t) \in E(x,t,1)$

It then follows that

$$\begin{aligned}
S'_{0,0}(r) &= \int_{E(0,0,1)} \frac{|y|^2}{s^2}\Big(y^T \nabla u(ry, r^2 s) + 2rs\dot{u}(ry,r^2 s)\Big)\, d^n y ds \\
&= \frac{1}{r^{n+1}} \int_{E(0,0,r)} r^2 \frac{|\frac{y}{r}|^2}{s^2}\left(\frac{y^T}{r}\nabla u(y,s) + 2r\frac{s}{r^2}\dot{u}(y,s)\right) d^n y ds \\
&= \frac{1}{r^{n+1}} \int_{E(0,0,r)} \frac{|y|^2}{s^2}\Big(y^T \nabla u(y,s) + 2s\dot{u}(y,s)\Big)\, d^n y ds.
\end{aligned}$$

Then for $x, t = 0$ it holds

$$\begin{aligned}
&|x-y|^2 \le 2(t-s)n\Big(2\log(r) - \log(t-s) - \log(4\pi)\Big) \\
\iff\ & |y|^2 \le -2sn\Big(2\log(r) - \log(-s) - \log(4\pi)\Big) \\
\iff\ & |y|^2 \le -4sn\Big(\log(r) - \frac{1}{2}\log(-4\pi s)\Big) \\
\iff\ & 0 \le -|y|^2 - 4s\Big(n\log(r) - \frac{n}{2}\log(-4\pi s)\Big) \\
\iff\ & 0 \le \frac{|y|^2}{4s} + n\log(r) - \frac{n}{2}\log(-4\pi s)
\end{aligned}$$

Define $\psi(y,s) := -\frac{n}{2}\log(-\pi s) + \frac{|y|^2}{4s} + n\log(r)$, then

$$E(0,0,r) = \{(y,s) \in \mathbb{R}^{n+1} \mid \psi(y,s) \ge 0, \underbrace{s \le t}_{???}\}$$

can be written which yields that by definition ψ vanishes on $\partial E(0,0,r)$. Using that

$$\frac{\partial}{\partial y_i}\psi(y,s) = \frac{1}{4s}2y_i = \frac{y_i}{2s} \quad \Rightarrow \quad 2y^T\nabla\psi(y,s) = \frac{|y|^2}{s}$$

it follows

$$\begin{aligned}
&\frac{1}{r^{n+1}}\int_{E(0,0,r)} 2\dot{u}\frac{|y|^2}{s} d^n y ds = \frac{1}{r^{n+1}}\int_{E(0,0,r)} 4\dot{u} y^T \nabla\psi d^n y ds \\
&= \frac{1}{r^{n+1}}\left(\int_{E(0,0,r)} (-4)\nabla\cdot(\dot{u}y)\psi d^n y ds + \int_{\partial E(0,0,r)} 4\dot{u}y^T\psi N d\sigma(y,s)\right) \\
&= -\frac{1}{r^{n+1}}\int_{E(0,0,r)} 4y^T\nabla\dot{u}\psi + 4\dot{u}n\psi d^n y ds \\
&= -\frac{1}{r^{n+1}}\int_{E(0,0,r)} 4\frac{\partial}{\partial t}\left(y^T\nabla u\psi\right) - 4y^T\nabla u\dot{\psi} + 4\dot{u}n\psi d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,r)} 4y^T\nabla u\dot{\psi} - 4\dot{u}n\psi d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,r)} 4y^T\nabla u\left(-\frac{n}{2}\frac{-4\pi}{-4\pi s} - \frac{|y|^2}{4}s^{-2}\right) - 4\dot{u}n\psi d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,r)} y^T\nabla u\left(-\frac{2n}{s} - \frac{|y|^2}{s^2}\right) - 4\dot{u}n\psi d^n y ds
\end{aligned}$$

Then it follows

$$\begin{aligned}
S_{0,0}'(r) &= \frac{1}{r^{n+1}}\int_{E(0,0,r)} \frac{|y|^2}{s^2}\left(y^T\nabla u + 2s\dot{u}(y,s)\right) d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,r)} \frac{|y|^2}{s^2} y^T \nabla u d^n y ds + \frac{1}{r^{n+1}}\int_{E(0,0,r)} y^T\nabla u\left(-\frac{2n}{s}-\frac{|y|^2}{s^2}\right) - 4\dot{u}n\psi d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,r)} (-1)y^T\nabla u\frac{2n}{s} - 4\dot{u}n\psi d^n y ds \\
&\overset{HeatEq}{=} -\frac{1}{r^{n+1}}\int_{E(0,0,r)} y^T\nabla u\frac{2n}{s} + 4\Delta u n\psi d^n y ds \\
&= -\frac{1}{r^{n+1}}\int_{E(0,0,r)} y^T\nabla u\frac{2n}{s} - 4n\nabla u^T\nabla\psi d^n y ds \\
&= -\frac{1}{r^{n+1}}\int_{E(0,0,r)} y^T\nabla u\frac{2n}{s} - 4n\nabla u^T\frac{y}{2s} d^n y ds \\
&= -\frac{1}{r^{n+1}}\int_{E(0,0,r)} \nabla u^T\left(y\frac{2n}{s} - 2\frac{y}{s}\right) d^n y ds = 0.
\end{aligned}$$

This shows that $S_{0,0}$ is constant. Further, it holds

$$\begin{aligned}
C_n :&= \int_{E(0,0,1)} \frac{|y|^2}{s^2} d^n y ds = \frac{1}{r^{n+1}}\int_{E(0,0,1)} \frac{1}{r}\frac{|y|^2}{s^2} r^n r^2 d^n y ds \\
&= \frac{1}{r^{n+1}}\int_{E(0,0,1)} \frac{|ry|^2}{(rs)^2} r^n r^2 d^n y ds = \frac{1}{r^{n+1}}\int_{E(0,0,1)} \frac{|z|^2}{q^2} d^n z dq.
\end{aligned}$$

Finally,

$$\begin{aligned}
&\lim_{r\to 0}\frac{1}{C_n}\int_{E(0,0,1)} u(ry, r^2 s)\frac{|y|^2}{s^2} d^n y ds = \frac{1}{C_n}\int_{E(0,0,1)} \lim_{r\to 0} u(ry, r^2 s)\frac{|y|^2}{s^2} d^n y ds \\
&= u(0,0)\frac{1}{C_n}\int_{E(0,0,1)} \frac{|y|^2}{s^2} d^n y ds = u(0,0).
\end{aligned}$$

Because $S_{0,0}(r)$ is constant and equalt to $u(0,0)$ for $r \to 0$ it follows it is equal to that point for every choice of $r > 0$.

□

4 Maximumsprinciple

The following theorem shows that diffusion is a consequence of curvature in space and time. If there is is no curvature, i.e. no change in space or time, the solution is constant.
For any open domain $\Omega \subset \mathbb{R}^n$ we define the parabolic cyclinder $\Omega_T := \Omega \times (0, T]$ and $\partial\Omega_T(\partial\Omega \times (0, T]) \cup (\bar{\Omega} \times \{0\})$.

Theorem 4.1

Let Ω be a path connected and u twice continuous differentiable solution of the heat equation on Ω_T with continuous extension to $\bar{\Omega}_T$. If u takes the maximum in Ω_T then u is constant Ω_T.

Proof. Let $(x_0, t_0) \in \Omega_T$ be the point where u takes its maximum. Define

$$(x_0, t_0) \in A := \{(x,t) \in \Omega_T \mid u(x,t) = \max_{(z,q)\in\bar{\Omega}_T} u(z,q)\}.$$

Because Ω_T is an open domain

$$\exists r_0 > 0 : \quad E(x_0, t_0, r_0) \subset \Omega_T.$$

Due to the path connectedness

$$\forall (x,t) \in \Omega \times (0, t_0) \exists n \in \mathbb{N} \text{ finite and } E(x_1, t_1, r_1), ..., E(x_n, t_n, r_n) \subset \Omega \times (0, t_0)$$

where $t_0 > ... > t_n$, $(x_i, t_i) \in E(x_{i-1}, t_{i-1}, r_{i-1})$ and $(x,t) \in E(x_n, t_n, r_n)$. This implies $u(x,t)$ is also the maximum. Thus $A = \mathring{\Omega}_{t_0}$. Because u is continuous in the interior and extends contiusly to the closure, it follows that u is also equal to the maximum on $\overline{\Omega_{t_0}}$.
In order to show that the solution is constant on Ω_T we need analiticity.??? Not done here. $\square$

Theorem 4.2: Weak Maximum Principle

Let $\Omega \subset \mathbb{R}^n$ be open and bounded and u a twice differntiable solution of the heat equation on Ω_T that continuously extends to $\bar{\Omega}_T$, then the maximum of u is taken on $\partial\Omega_T$.

Theorem 4.3

On an open and bounded domain $\Omega \subset \mathbb{R}^n$ there exists at most one solution u of the inhomogenous heat equation that extends continuously to $\bar{\Omega}_T$ and coincides on $\partial\Omega_T$ with a given function.

Proof. Assume u_1, u_2 are tow solution of this problem and define $w := u_1 - u_2$ then $(\partial_t - \Delta)w = 0$ and $w\big|_{\partial_P \Omega_T} = 0$. With the weak maximum priciple it follows that $\max_{(x,t)\in\bar{\Omega}_T} w(x,t) = 0$ implies

$$\forall (x,t) \in \Omega_T : \quad w(x,t) \leq 0.$$

Conversely, using the minimum principle it follows that

$$\forall (x,t) \in \Omega_T : \quad w(x,t) \geq 0.$$

Thus in total $\forall (x,t) \in \Omega_T : \quad w(x,t) = 0.$ $\square$

In order to prove uniqueness of the initial value problem on $\mathbb{R}^n \times \mathbb{R}^+$ we need a bound on the growth of the solution at infinity.

Theorem 4.4: Maximum Principle for the Cauchy Problem

For a bounded and continuous initial value h on $\mathbb{R}^n$ and u a solution on $\mathbb{R}^n \times (0, T]$ of

$$\dot{u} - \Delta u = 0 \text{ on } \mathbb{R}^n \times (0,T) \text{ and } u(x,0) = h(x) \text{ on } \mathbb{R}^n \times \{0\},$$

which is bounded by
$$u(x,t) \leq Ae^{a|x|^2}, \quad (x,t) \in \mathbb{R}^n \times [0,T] \text{ for } A, a > 0,$$
follows that u is bounded by
$$\sup_{(x,t)\in\mathbb{R}^n\times[0,T]} u(x,t) = \sup_{x\in\mathbb{R}^n} h(x).$$

Proof. The weak maximum principle cannot be applied here directly because thge domain is bounded. However, we can do it locally and then take the limit.
Case $4aT < 1$:
There exists $\epsilon > 0$ such that $4a(T+\epsilon) < 1$. For all $y \in \mathbb{R}^n$ and $\mu > 0$
$$v(x,t) := u(x,t) - \mu \cdot (T+\epsilon-t)^{-\frac{n}{2}} e^{\frac{|x-y|^2}{4(T+\epsilon-t)}}$$
is a sultion to the heat equation. This can be seen when plugging this into the differential operator $(\partial_t - \Delta_x)(\cdot)$. Because the second summand is of the form of the fundamental solution, this works. If we look at the problem only on $B_r(y)$, then the weak maximum principle applies. Thus, consider the parabolic boundary $\partial_P \Omega_T := B_r(y) \times \{0\} \cup \partial B_r(y) \times (0,T]$, where to solution takes its maximum. Then due to $\frac{1}{4(t+\epsilon)} > a$

- For $x \in B_r(y)$ it holds
$$v(x,0) = u(x,0) - \underbrace{\mu(T+\epsilon)^{-n/2} e^{\frac{|x-y|^2}{4(T+\epsilon)}}}_{\geq 0} \leq u(x,0) = h(x) \leq \sup_{x\in\mathbb{R}^n} h(x).$$
Now we need to consider the limit $r \to \infty$. Because u is bounded by $Ae^{a|x|^2}$ it follows that
$$v(x,0) \leq Ae^{a|x|^2} - \mu(T+\epsilon)^{-n/2} e^{\frac{|x-y|^2}{4(T+\epsilon)}}.$$
Through the assumption $4a(T+\epsilon) < 1$ it follows that the second summand grows faster than the first one. Thus, in the limit $r \to \infty$ the second summand dominates and we get
$$v(x,0) \to -\infty < \sup_{x\in\mathbb{R}^n} h(x).$$
This means $\exists R > 0 \forall r > R: \quad v(x,0) \leq \sup_{x\in\mathbb{R}^n} h(x)$. If we finally let $\mu \searrow 0$ we get the desired bound for u at $t=0$.

- For $(x,t) \in \partial B_r(y) \times (0,T]$ we want to find an upper bound that eventually drops beneath $\sup_{x\in\mathbb{R}^n} h(x)$. First, notice that
$$|x| = |x+y-y| \leq |x-y| + |y| = r + |y|.$$
Then we can show:
$$v(x,t) \leq Ae^{a|x|^2} - \mu(T+\epsilon-t)^{-n/2} e^{\frac{r^2}{4(t+\epsilon-t)}} \leq Ae^{a(|y|+r)^2} - \mu(T+\epsilon-t)^{-n/2} e^{\frac{r^2}{4(t+\epsilon-t)}}$$
the second summand will drop underneath the first due to the fact that $\frac{1}{4(t+\epsilon-t)} > a$. Thus, on that intervall $v(x,t) \to -\infty$ as $r \to \infty$, as constants are neglegible in the limit.

Both cases yield that on $\partial_P\Omega_T$ $v(x,t)$ is bounded by $\sup_{x\in\mathbb{R}^n} h(x)$ and $\mu > 0$ arbitary. Because the second summand can be written as $\mu w(x,t)$ we can take the limit $\mu \searrow 0$ wich leads to u being bounded by $\sup_{x\in\mathbb{R}^n} h(x)$. This implies

$$\sup_{(x,t)\in\mathbb{R}^n\times[0,T)} u(x,t) \leq \sup_{x\in\mathbb{R}^n} h(x).$$

The verse trivially holds because

$$\sup_{x\in\mathbb{R}^n} h(x) = \sup_{x\in\mathbb{R}^n} u(x,0) \leq \sup_{(x,t)\in\mathbb{R}^n\times[0,T)} u(x,t).$$

Case $4aT > 1$:
Decompose the intervall $[0,T] = [0,T_1] \cup .. \cup [T_M, T]$, although $4a(T_{m+1} - T_m) < 1$. Then on the inteverval $[0,T_1]$ we can use the first proof part. The second intervall is treated as a new initial value property. On this intervall we can again use the first proof. This is then condinued M times. □

Theorem 4.5: Existence and uniqueness of the Initial Value Problem

For $h \in C(\mathbb{R}^n)$ and $f \in C(\mathbb{R}^n \times [0,T])$ there exists at most one solution to the initial value problem

$$\dot{u} - \Delta u = f \text{ on } \mathbb{R}^n \times (0,T), \quad u = h \text{ on } \mathbb{R}^n \times \{0\},$$

on $\mathbb{R}^n \times [0,T_0]$ which is bounded by $|u(x,t)| \leq Ae^{a|x|^2}$, $a, A > 0$ and $0 < T_0 \leq T$.
Additionally, if h and f are also bounded by $Ae^{a|x|^2}$ on $\mathbb{R}^n \times [0,T]$, $a, A, T > 0$, then the unique solution is given by

$$u(x,t) = \int_{\mathbb{R}^n} \Phi(x-y,t)h(y)d^ny + \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,s)d^nyds.$$

This solution (might???) explode at some finite time $t \nearrow T_0 \geq \frac{1}{16a}$.

Proof. Using the maximum principle for the Cauchy problem we can again show that the vanishing of the difference of two solutions imply uniqueness.
To show existence we use that the solution can be written as

$$u(x,t) = \int_{\mathbb{R}^n} \Phi(x-y,t)h(y)d^ny + \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)f(y,s)d^nyds.$$

and need to show boundedness. Choose $0 \leq t-s \leq \frac{1}{16a}$, then

$$-\frac{|x-y|^2}{4(t-s)} = -\frac{2+2}{4}\frac{|x-y|^2}{4(t-s)} \leq -2a|x-y|^2 - \frac{|x-y|^2}{8(t-s)},$$

implies

$$\begin{aligned}
e^{-2a|x|^2}e^{a|y|^2}\Phi(x-y,t-s) &= e^{-2a|x|^2}e^{a|y|^2}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{4(t-s)}} \\
&\leq e^{-2a|x|^2}e^{a|y|^2}\frac{1}{(4\pi(t-s))^{n/2}}e^{-2a|x-y|^2-\frac{|x-y|^2}{8(t-s)}} \\
&= e^{-2a|x|^2-2a|x-y|^2+a|y|^2}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}
\end{aligned}$$

Now due to

$$-2a|x-y|^2 = -2a\left(|x|^2 - 2x^Ty + |y|^2\right) \iff 2x^Ty = |x|^2 - |x-y|^2 + |y|^2$$

it follows that the above is equal to

$$\begin{aligned}
e^{-2a|x|^2-2a|x-y|^2+a|y|^2}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}} &= e^{-a\left(2|x|^2+2|x-y|^2-|y|^2\right)}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}\\
&= e^{-a\left(2|x|^2+(2|x|^2-4x^Ty+2|y|^2)-|y|^2\right)}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}\\
&= e^{-a\left(2|x|^2+(2|x|^2-(|y|^2+4|x|^2-|2x-y|^2)+2|y|^2)-|y|^2\right)}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}\\
&= e^{-a|2x-y|^2}\frac{1}{(4\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}\\
&= 2^{n/2}e^{-a|2x-y|^2}\frac{1}{(8\pi(t-s))^{n/2}}e^{-\frac{|x-y|^2}{8(t-s)}}\\
&= 2^{n/2}e^{-a|2x-y|^2}\frac{1}{(4\pi2(t-s))^{n/2}}e^{-\frac{|x-y|^2}{4\cdot2(t-s)}}.
\end{aligned}$$

Due to the fact that $e^{-a|2x-y|^2} \leq e^0 = 1$ it follows

$$\Phi(x-y,t-s) \leq 2^{n/2}e^{2a|x|^2}e^{-a|y|^2}\Phi(x-y,2(t-s)).$$

Then it follows

$$\begin{aligned}
u(x,t) &= \int_{\mathbb{R}^n}\Phi(x-y,t)h(y)d^ny + \int_0^t\int_{\mathbb{R}^n}\Phi(x-y,t-s)f(y,s)d^nyds\\
&\leq \int_{\mathbb{R}^n}2^{n/2}\Phi(x-y,2t)e^{2a|x|^2}e^{-a|y|^2}Ae^{a|y|^2}d^ny\\
&+ \int_0^t\int_{\mathbb{R}^n}2^{n/2}\Phi(x-y,2(t-s))e^{2a|x|^2}e^{-a|y|^2}Ae^{a|y|^2}d^nyds\\
&\leq 2^{n/2}e^{2a|x|^2}A\int_{\mathbb{R}^n}\Phi(x-y,2t)d^ny + e^{2a|x|^2}A2^{n/2}\int_0^t\int_{\mathbb{R}^n}\Phi(x-y,2(t-s))d^nyds\\
&\leq 2^{n/2}e^{2a|x|^2}A + e^{2a|x|^2}A2^{n/2}t\\
&\leq 2^{n/2}e^{2a|x|^2}A(1+t).
\end{aligned}$$

Thus u is bounded if and only if $t-s \leq \frac{1}{16a}$. To incorporate the case that this constant may be larger than T we require that $t-s \in [0, \min\{T, \frac{1}{16a}\}]$.
??? Something with

$$\tilde{u}(x,t) = \frac{1}{(T_0-t)^{n/2}}e^{\frac{|x|^2}{4(T_0-t)}}u\left(\frac{x}{T_0-t}, \frac{1}{T_0-t}\right)$$

is again a solution??? THis implIES the existence up to time T_0. □

The following example shows the heat equation on $\mathbb{R}^n$ is not well posed, unless you impose a constraint on how fast the solution is allowed to grow. Specifically, it is shown that the heat equation

$$\partial_t u - \Delta u = 0, \quad u(x,0) = 0$$

has two solutions. The first is $u \equiv 0$ and the second will be constructed as $u(x,t) := \sum_{l\in\mathbb{N}} g_l(t)x^l$ a power series. It will be shown that

$$(\partial_t - \partial_{xx})\sum_{l\in\mathbb{N}} g_l(t)x^l = 0 \qquad \Longleftrightarrow \qquad g_{2l}(t) = \frac{g_0^{(l)}(t)}{(2l)!},$$

i.e. the solution only depends on only $g_0 \in C^\infty$. If we choose g_0 the power series exists. The bound

$$|u(x,t)| \leq e^{\frac{7|x|^2}{2t} - \frac{1}{2t^2}}$$

will be derived, which shows that at the smalles $t > 0$ and $x \to \pm\infty$ we have infinite temperature, which will propagate into the center with infinite speed.

Example 4.1

We want to show here that if we have no bound for the initial value problem for the heat equation, then the solution is not unique. For $n = 1$ define

$$u(x,t) = \sum_{l=0}^\infty g_l(t)x^l$$

then

$$\begin{aligned}
0 = (\frac{\partial}{\partial t} - \frac{\partial^2}{\partial x^2})u(x,t) &= \sum_{l=0}^\infty \left(\dot{g}_l(t)x^l - l(l-1)g_l(t)x^{l-2}\right) \\
&= \sum_{l=0}^\infty \dot{g}_l(t)x^l - \sum_{l=2}^\infty l(l-1)g_l(t)x^{l-2} \\
&= \sum_{l=0}^\infty (\dot{g}_l(t) - (l+2)(l+1)g_{l+2}(t))x^l
\end{aligned}$$

Then we have for all even $l \in \mathbb{N}$ that

$$\dot{g}_l(t) - (l+2)(l+1)g_{l+2}(t) = 0 \qquad \Longleftrightarrow \qquad g_{l+2}(t) = \frac{\dot{g}_l(t)}{(l+2)(l+1)} = \frac{\ddot{g}_{l-2}(t)}{(l+2)(l+1)l(l-1)} = ... = \frac{g_0^{((l+2)/2)}(t)}{(l+2)!}$$

Thus for all $l \in \mathbb{N}$ the relation holds

$$g_{2l}(t) = \frac{g_0^{(l)}(t)}{(2l)!}$$

We choose now all g_l with odd $l \in \mathbb{N}$ as zero and then get

$$u(x,t) = \sum_{l=0}^\infty g_l(t)x^l = \sum_{l=0}^\infty g_{2l}(t)x^{2l} = \sum_{l=0}^\infty \frac{g^{(l)}(t)}{(2l)!}x^{2l}.$$

Thus, we have shown that the solution can be written as a series of an arbitrary function $g_0 \in C^\infty$. If this series exists and converges to 0 as $t \searrow 0$, then we have found another solutition. A specific example is:

$$g_0(t) =: g(t) =: e^{-t^{-2}}\mathbf{1}_{t>0}, \quad u(x,t) = \sum_{l=0}^\infty \frac{g^{(l)}(t)}{(2l)!}x^{2l} =: \sum_{l=0}^\infty g_l(t)x^l.$$

For $t > 0$ it then holds

$$g'(t) = \frac{2}{t^3}e^{-t^{-2}} = t^{-1}2t^{-2}e^{-t^{-2}} =: t^{-1}p_1(t^{-2})e^{-t^{-2}},$$
$$g''(t) = (-6)t^{-4}e^{-t^{-2}} + 2t^{-3}2t^{-3}e^{-t^{-2}} = \left(-6t^{-4} + 4t^{-6}\right)e^{-t^{-2}} = t^{-2}\underbrace{\left(-6t^{-2} + 4t^{-4}\right)}_{=:p_2(t^{-2})}e^{-t^{-2}}$$
$$\vdots$$
$$g^{(l)}(t) = t^{-l}p_l(t^{-2})e^{-t^{-2}}$$

where p_l is of polynomial Degree l. Using the chain rule it then holds

$$g^{(l+1)}(t) = \frac{d}{dt}g^{(l)}(t) = -lt^{-(l+1)}p_l(t^{-2})e^{-t^{-2}} - 2t^{-(l+3)}p_l'(t^{-2})e^{-t^{-2}} + 2t^{-(l+3)}p_l(t^{-2})e^{-t^{-2}}$$
$$= t^{-(l+1)}e^{-t^{-2}}\left((-l)p_l(t^{-2}) - 2t^{-2}p_l'(t^{-2}) + 2t^{-2}p_l(t^{-2})\right)$$
$$\iff t^{-l}p_l(t^{-2})e^{-t^{-2}} = t^{-(l+1)}e^{-t^{-2}}\left((-l)p_l(t^{-2}) - 2t^{-2}p_l'(t^{-2}) + 2t^{-2}p_l(t^{-2})\right)$$
$$\iff p_{l+1}(t^{-2}) = 2t^{-2}p_l(t^{-2}) - lp_l(t^{-2}) - 2t^{-2}p_l'(t^{-2})$$

For $z := t^{-2}$ we have

$$p_{l+1}(z) = 2zp_l(z) - lp_l(z) - 2zp_l'(z).$$

We set $p_0(z) = 1$ then with the above $p_1(z) = 2z$. Because p_l is a polynomial we can write it as

$$p_l(z) = \sum_{k=0}^{l} a_{l,k}z^k$$

We now want to show that $|a_{l,k}| \leq \frac{l!z^l}{2^k k!}$ for all $l, k \in \mathbb{N}$.
Induction start: $l = 0, k = 0$ is trivial
Induction step: Note that

$$p_{l+1}(z) = 2zp_l(z) - lp_l(z) - 2z\sum_{k=0}^{l} a_{l,k}kz^{k-1}$$
$$= 2\sum_{k=0}^{l} a_{l,k}z^{k+1} - l\sum_{k=0}^{l} a_{l,k}z^k - 2\sum_{k=0}^{l} a_{l,k}kz^k$$
$$= 2\sum_{k=0}^{l} a_{l,k}z^{k+1} - l\sum_{k=0}^{l} a_{l,k}z^k - 2\sum_{k=0}^{l} a_{l,k}kz^k$$
$$= 2\sum_{k=1}^{l+1} a_{l,k-1}z^k - l\sum_{k=0}^{l} a_{l,k}z^k - 2\sum_{k=0}^{l} a_{l,k}kz^k$$
$$= \sum_{k=1}^{l} z^k(2a_{l,k-1} - la_{l,k} - 2ka_{l,k}) + 2a_{l,l}z^{l+1} - la_{l,0} - 2a_{l,0}\cdot 0$$
$$= \sum_{k=1}^{l} z^k(2a_{l,k-1} - la_{l,k} - 2ka_{l,k}) + 2a_{l,l}z^{l+1} - la_{l,0}$$
$$= \sum_{k=0}^{l+1} z^k(2a_{l,k-1} - la_{l,k} - 2ka_{l,k})$$

This the yields with the induction hypothethis that

$$
\begin{aligned}
|a_{l+1,k}| &\le 2|a_{l,k-1}| + l|a_{l,k}| + 2k|a_{l,k}| \\
&\le 2\frac{l!7^l}{2^{k-1}(k-1)!} + l\frac{l!7^l}{2^k k!} + 2k\frac{l!7^l}{2^k k!} \\
&\le 2\cdot 2k\frac{l!7^l}{2^k k!} + l\frac{l!7^l}{2^k k!} + 2k\frac{l!7^l}{2^k k!} \\
&= \frac{(l+1)!7^{l+1}}{2^k k!}\frac{2\cdot 2k + l + 2k}{(l+1)7} = \frac{(l+1)!7^{l+1}}{2^k k!}\frac{l+6k}{(l+1)7}
\end{aligned}
$$

Since $k \le k+1$ it follows

$$
|a_{l+1,k}| \le \frac{(l+1)!7^{l+1}}{2^k k!}\frac{l+6(l+1)}{(l+1)7} \le \frac{(l+1)!7^{l+1}}{2^k k!}\frac{7l+7}{(l+1)7} = \frac{(l+1)!7^{l+1}}{2^k k!}.
$$

We can use this bound and the fact that $\frac{l!}{(2l)!} \le \frac{1}{2^l l!}$ to find a bound for u:

$$
\begin{aligned}
|u(x,t)| &\le \sum_{l=0}^{\infty}\left|\frac{t^{-l}p_l(t^{-2})e^{-t^{-2}}}{(2l)!}x^{2l}\right| \le \sum_{l=0}^{\infty}\frac{t^{-l}e^{-t^{-2}}}{(2l)!}|x^{2l}|\sum_{k=0}^{l}|a_{l,k}|t^{-2k} \\
&\le \sum_{l=0}^{\infty}\frac{t^{-l}e^{-t^{-2}}}{(2l)!}|x^{2l}|\sum_{k=0}^{l}\frac{l!7^l}{2^k k!}t^{-2k} \\
&\le \sum_{l=0}^{\infty}\frac{t^{-l}e^{-t^{-2}}}{2^l l!}|x^{2l}|\sum_{k=0}^{l}\frac{7^l}{2^k k!}t^{-2k} \\
&\le \sum_{l=0}^{\infty}\frac{1}{l!}\left(\frac{7|x|^2}{2t}\right)^l e^{-t^{-2}}\sum_{k=0}^{\infty}\frac{1}{k!}\left(\frac{1}{2t^2}\right)^k \\
&= e^{-t^{-2}}e^{\frac{7|x|^2}{2t}+\frac{1}{2t^2}} = e^{\frac{7|x|^2}{2t}-\frac{1}{2t^2}}
\end{aligned}
$$

For every $t > 0$ this series is finitely bounded and thus absolutely convergent. Further, as $t \searrow 0$ it follows that on every compact subset $K \subset \mathbb{R}$ this series converges to zero, because

$$
0 \le |u(x,t)| \le e^{\frac{7|x|^2}{2t}-\frac{1}{2t^2}} \le e^{\frac{7\sup_{x\in K}|x|^2}{2t}-\frac{1}{2t^2}} \to 0.
$$

In the theorem of the uniqueness of the solution we required that $|u(x,t)|$ be bounded independent of t on $\mathbb{R}\times[0,T_0]$. However, our bound depends on time. Assume that there exists $a, C \in \mathbb{R}$ such that

$$
e^{\frac{7|x|^2}{2t}-\frac{1}{2t^2}} \le e^{\frac{7|x|^2}{2t}} \le e^{ax^2+C}.
$$

This would imply that $\frac{7}{2t} \le a$. But as $t \searrow 0$ $\frac{7}{2t} \to \infty$, which is a contradiction. Thus, we have found another solution of the heat equation that does not satify these upper bound conditions.
This example show that we found a solution that is zero on $\mathbb{R}$ at $t = 0$ and then as soon as $t > 0$ there is infinite heat at $+\infty$ and $-\infty$ that propagate into the center at infinite speed and result in a value of the solution at time $t > 0$ and point $x \in \mathbb{R}$ of $u(x,t) = \sum_{l=0}^{\infty}\frac{g^{(l)}(t)}{(2l)!}x^{2l}$.

An alternative to using the maximum principle is using the yet to be defined Energy functional. There we cannot use an unbounded domain $\mathbb{R}$ but instead have to focus on a bounded domain $\Omega \subset \mathbb{R}$. The energy function is given by

$$e(t) := \int_\Omega u^2(x,t) d^n x.$$

Now using $\dot{u} = \Delta u$ it follows

$$\begin{aligned}\dot{e}(t) &= \int_\Omega \frac{d}{dt} u^2(x,t) d^n x = 2 \int_\Omega u(x,t)\dot{u}(x,t) d^n x = 2 \int_\Omega u(x,t) \Delta u(x,t) d^n x \\ &= -2 \underbrace{\int_\Omega (\nabla u(x,t))^2 d^n x}_{\geq 0} \leq 0.\end{aligned}$$

This implies that $e(\cdot)$ is monotonically decreasing. To show uniqueness of the solution, take two solutions u_1, u_2 from the heat eqaution and define their difference $w := u_1 - u_2$. Then w is zero at $t = 0$. Further,

$$\int_\Omega w^2(x,0) d^n x = 0$$

and the fact that $e(\cdot)$ is monotonically decreasing it follows that

$$\forall t > 0 : \int_\Omega w^2(x,t) d^n x = 0$$

which itself implies that $w(x,t) = 0$ for all $(x,t) \in \Omega \times [0,T]$.

5 Heat Kernel

In analogy to Greens functions for the Laplace equation we define on open subsets $\Omega \subset \mathbb{R}^n$ the heat kernel H_Ω:

Definition 5.1

For all $\Omega \subset \mathbb{R}^n$ open domain the heat kernel $H_\Omega : \Omega \times \Omega \times \mathbb{R}^+ \to \mathbb{R}$ of Ω is characterized by the two properties

- For $(x,t) \in \Omega \times \mathbb{R}^+$ the function $y \mapsto H_\Omega(x,y,t)$ extends continuously to $\bar{\Omega}$ with value 0 on $\partial\Omega$.
- For $x \in \Omega$ the function $(y,t) \mapsto H_\Omega(x,y,t) - \Phi(x-y,t)$ solves the homogeneous heat equation and extends continuously to $\bar{\Omega} \times \mathbb{R}_0^+$ with value 0 on $\bar{\Omega} \times \{0\}$.

Lemma 5.2

If u, v are two functions on $\Omega \times \mathbb{R}^+$ with an open domain $\Omega \subset \mathbb{R}^n$, then under sufficient regularity we have

$$\begin{aligned}&\int_0^T \int_\Omega u(x,t) \left(\dot{v}(x,T-t) + \Delta v(x,T-t)\right) d^n x dt + \int_0^T \int_\Omega \left(\dot{u}(x,t) - \Delta u(x,t)\right) v(x,T-t) d^n x dt \\ &= \int_0^T \int_{\partial\Omega} \left(u(y,t)\nabla v(y,T-t) - \nabla u(y,t) v(y,T-t)\right) \cdot N d\sigma(y) dt + \int_\Omega u(x,T)v(x,0) - u(x,0)v(x,T) d^n x\end{aligned}$$

Proof.

$$\begin{aligned}
&\int_0^T\int_\Omega u(x,t)\left(\dot v(x,T-t)+\Delta v(x,T-t)\right)d^nxdt+\int_0^T\int_\Omega\left(\dot u(x,t)-\Delta u(x,t)\right)v(x,T-t)d^nxdt\\
&=\int_0^T\int_\Omega u(x,t)\dot v(x,T-t)+\dot u(x,t)v(x,T-t)d^nxdt+\int_0^T\int_\Omega u(x,t)\Delta v(x,T-t)-\Delta u(x,t)v(x,T-t)d^nxdt\\
&=\int_0^T\int_\Omega \frac{d}{dt}(v(x,T-t)u(x,t))d^nxdt+\int_0^T\int_\Omega \nabla(u(x,t)\nabla v(x,T-t))-\nabla(\nabla u(x,t)v(x,T-t))d^nxdt\\
&=\int_\Omega v(x,0)u(x,T)-v(x,T)u(x,0)d^nx+\int_0^T\int_\Omega\big(u(y,t)\nabla v(y,T-t)-\nabla u(y,t)v(y,T-t)\big)\cdot N\,d\sigma(y)dt
\end{aligned}$$

$\square$

Because Φ has a singularity at $(0,0)$ and $(y,t)\mapsto H_\Omega(x,y,t)-\Phi(x-y,t)$ is zero on $\bar\Omega\times\{0\}$ it must follow that $H_\Omega(x,y,t)$ has a singularity at $(x,x,0)$.
This is why we only integrate on $[0,T-\epsilon]$ and then take the limit $\epsilon\to 0$. We then get with the above

$$\begin{aligned}
&\int_0^T\int_\Omega\left(\dot u(y,t)-\Delta u(y,t)\right)H_\Omega(x,y,T-t)d^nydt\\
&=\int_0^T\int_{\partial\Omega}\left(u(y,t)\nabla_yH_\Omega(x,y,T-t)-\nabla_yu(y,t)H_\Omega(x,y,T-t)\right)\cdot Nd\sigma(y)dt\\
&+\int_\Omega u(y,T)H_\Omega(x,y,0)-u(y,0)H_\Omega(x,y,T)d^ny\\
&-\int_0^T\int_\Omega u(y,t)\left(\partial H_\Omega(x,y,T-t)+\Delta H_\Omega(x,y,T-t)\right)d^nydt\\
&=\int_0^T\int_{\partial\Omega}u(y,t)\nabla_yH_\Omega(x,y,T-t)\cdot Nd\sigma(y)dt\\
&+\int_\Omega u(y,T)H_\Omega(x,y,0)d^ny-u(y,0)H_\Omega(x,y,T)d^ny\\
&=\int_0^T\int_{\partial\Omega}u(y,t)\nabla_yH_\Omega(x,y,T-t)\cdot Nd\sigma(y)dt+u(x,T)-\int_\Omega u(y,0)H_\Omega(x,y,T)d^ny
\end{aligned}$$

Although we used that

$$\begin{aligned}
&\int_\Omega u(y,T)H_\Omega(x,y,0)d^ny=\int_\Omega u(y,T)\lim_{t\searrow 0}H_\Omega(x,y,t)d^ny\\
&=\int_\Omega u(y,T)\lim_{t\searrow 0}\Phi(x-y,t)d^ny=\lim_{t\searrow 0}\int_\Omega u(y,T)\Phi(x-y,t)d^ny\\
&=\lim_{t\searrow 0}(u(\cdot,T)*\Phi(\cdot,t))(x)=u(x,T).
\end{aligned}$$

Thus, it holds

$$\begin{aligned}
u(x,T)=&\int_0^T\int_\Omega\left(\dot u(y,t)-\Delta u(y,t)\right)H_\Omega(x,y,T-t)d^nydt+\int_\Omega u(y,0)H_\Omega(x,y,T)d^ny\\
&-\int_0^T\int_{\partial\Omega}u(y,t)\nabla_yH_\Omega(x,y,T-t)\cdot Nd\sigma(y)dt
\end{aligned}$$

Lemma 5.3

For all $T > 0$ and $x, y \in \bar{\Omega}$: $H_\Omega(x, y, T) = H_\Omega(y, x, T)$

Proof.

$$
\begin{aligned}
&H_\Omega(x,y,T) - H_\Omega(y,x,T) = \lim_{t\searrow 0}(H_\Omega(x,\cdot,T) * \Phi(\cdot,t))(y) - \lim_{t\searrow 0}(H_\Omega(y,\cdot,T) * \Phi(\cdot,t))(x) \\
&= \int_\Omega H_\Omega(x,z,T) \lim_{t\searrow 0} \Phi(y-z,t) d^n z - \int_\Omega H_\Omega(y,z,T) \lim_{t\searrow 0} \Phi(x-z,t) d^n z \\
&= \int_\Omega H_\Omega(x,z,T) \lim_{t\searrow 0} H_\Omega(y,z,t) d^n z - \int_\Omega H_\Omega(y,z,T) \lim_{t\searrow 0} H_\Omega(x,z,t) d^n z \\
&= \int_\Omega H_\Omega(x,z,T) H_\Omega(y,z,0) d^n z - \int_\Omega H_\Omega(y,z,T) H_\Omega(x,z,0) d^n z \\
&= \int_0^T \int_\Omega H_\Omega(y,z,t) \left(\dot{H}_\Omega(x,z,T-t) + \Delta H_\Omega(x,z,T-t)\right) d^n z dt \\
&+ \int_0^T \int_\Omega \left(\dot{H}_\Omega(y,z,t) - \Delta H_\Omega(y,z,t)\right) H_\Omega(x,z,T-t) d^n z dt \\
&- \int_0^T \int_{\partial\Omega} (H_\Omega(y,z,t) \nabla H_\Omega(x,z,T-t) - \nabla H_\Omega(y,z,t) H_\Omega(x,z,T-t)) \cdot N d\sigma(z) dt \\
&= 0
\end{aligned}
$$

□

Because the solution only depends on the heat kernel, which in turn depends on the chosen Ω, the regularity of the solution only depends on the heat kernel and Ω, as the integral is linear.

Theorem 5.4

Consider the initial value problem

$$\dot{u} - \Delta u = f \text{ on } \Omega \times (0,T), \quad u = g \text{ on } \partial\Omega \times [0,T], \quad u(x,0) = h(x) \text{ on } \Omega,$$

where f is on $\Omega \times (0,T)$, g on $\partial\Omega \times [0,T]$ and h on Ω all have appropiate regularity, then the unique solution is given by

$$
\begin{aligned}
u(x,T) = &\int_0^T \int_\Omega f(y,t) H_\Omega(x,y,T-t) d^n y dt - \int_0^T \int_{\partial\Omega} g(z,t) \nabla_z H_\Omega(x,z,T-t) N d\sigma(z) dt \\
&+ \int_\Omega h(y) H_\Omega(x,y,T) d^n y.
\end{aligned}
$$

Proof. Because $(\partial_t - \Delta)(\cdot)$ is a linear operator, we can split the full problem into three simpler subproblems $u = u_f + u_h + u_g$ where

- u_f solves $\dot{u} - \Delta u = f$, $u(x,0) = 0$ and $u = 0$ on $\partial\Omega \times [0,T]$
- u_h solves $\dot{u} - \Delta u = 0$, $u(x,0) = h(x)$ and $u = 0$ on $\partial\Omega \times [0,T]$
- u_g solves $\dot{u} - \Delta u = 0$, $u(x,0) = 0$ and $u(x,t) = g(x,t)$ on $\partial\Omega \times [0,T]$.

This is is possible, because $(\partial_t - \Delta)(u_f + u_h + u_g) = f + 0 + 0 = f(x)$, $(u_f + u_h + u_g)(x, 0) = 0 + h(x) + 0 = h(x)$ and $(u_f + u_h + u_g)(x, t) = 0 + 0 + g(x) = g(x)$ on $\partial\Omega \times [0, T]$.
Remembering that we have shown that

$$\begin{aligned} u(x,T) = &\int_0^T \int_\Omega (\dot{u}(y,t) - \Delta u(y,t))\, H_\Omega(x,y,T-t) d^n y dt + \int_\Omega u(y,0) H_\Omega(x,y,T) d^n y \\ &- \int_0^T \int_{\partial\Omega} u(y,t) \nabla_y H_\Omega(x,y,T-t) \cdot N d\sigma(y) dt, \end{aligned}$$

we can plug in for the the three solutions in the above

$$\begin{aligned} u_f(x,T) &= \int_0^T \int_\Omega f(y,t) H_\Omega(x,y,T-t) d^n y dt \\ u_h(x,T) &= \int_\Omega h(y) H_\Omega(x,y,T) d^n y \\ u_g(x,T) &= -\int_0^T \int_{\partial\Omega} g(y) \nabla_y H_\Omega(x,y,T-t) \cdot N d\sigma(y) dt. \end{aligned}$$

Thus, $u = u_f + u_h + u_g$ yields the form of the theorem. □

Lemma 5.5

For all $\Omega \subset \mathbb{R}^n$ bounded connected open domain the corresponding heat kernel H_Ω is positive on the corresponding parabolic cyclinder, if it exists.

Proof. Define $w(x,t) := \Phi(x-y,t) - H_\Omega(x,y,t)$ on $\bar{\Omega} \times (0, T]$. Both summands solve the heat equation on $\Omega \times (0, T]$ and satisfy the weak maximum princple, i.e. the maximum is taken on the parabolic boundary. This is why we want to consider the difference on the parabolic boundary $\partial(\Omega \times (0,T]) = (\partial\Omega \times [0,T]) \cup (\bar{\Omega} \times \{0\})$.

- For $(x,t) \in \partial\Omega \times [0,T]$: We have $w(x,t) = \Phi(x-y,t) - 0$, because by definition H_Ω is on this set zero.
- For $\bar{\Omega} \times \{0\}$, it is by definition equal to zero, as $t \searrow 0$.

$w - \Phi$ is again a solution and with the above satisfies on $\partial(\Omega \times (0,T])$ that $w - \Phi \leq 0$. Thus, for $x \neq y$

$$w(x,t) - \Phi(x-y,t) \leq \max_{x \in \partial(\Omega \times (0,T])} (w(x,t) - \Phi(x-y,t)) = 0 \quad \Rightarrow \quad \forall (x,t) \in \Omega \times (0,T] : \quad w(x,t) \leq \Phi(x-y,t)$$

which is equivalent to $H_\Omega(x,y,t) \geq 0$. □

6 Spectral Theory and the Heat Equation

Here we solve the initial value problem

$$\dot{u} - \Delta u = 0, \text{ on } \Omega \times [0,T], \quad u = 0, \text{ on } \partial\Omega \times [0,T], \quad u = h, \text{ on } \Omega \times \{0\},$$

using the Laplace operator on Ω. Clearly, if h is an eigenfunction of the Laplace operator then

$$-\Delta h = \lambda h, \text{ on } \Omega \text{ and } h = 0 \text{ on } \partial\Omega \times [0,T].$$

Theorem 6.1

Let $T \in K(H)$ be a self adjoint compact linear operator, then there exists to every Eigenvalue problem

$$T(x) = \lambda x$$

finite or countably many Eigenvector and Eigenvalue pairs $(\lambda_j, x_j) \in \mathbb{R} \setminus \{0\} \times H, j \in \mathbb{N}$ such that

- $(x_j)_{j\in\mathbb{N}}$ is a orthogonal system
- $\forall x \in H$: $T(x) = \sum_{j\in\mathbb{N}} \lambda_j \langle x, x_j \rangle x_j$
- If there are infinite Eigenvalues then $\lambda_i \to 0, j \to \infty$.

Proof. See Functional Analyis □

The inverse of the Laplacian is a compact, self adjoint and positive linear operator on $L^2(\Omega)$. Using the Theorem above we get that there exist $(\phi_i, \mu_i)_{i\in\mathbb{N}}$ eigenfunctions and eigenvalues such that

$$(-\Delta)^{-1}\phi_i = \mu_i \phi_i, \quad i \in \mathbb{N} \quad \iff \quad -\Delta\phi_i = \frac{1}{\mu_i}\phi_i.$$

Thus the Eigenvalues $\lambda_i := \frac{1}{\mu_i}$ of the Laplacian go to infinity because $\mu_i \to 0$.
Further, because $(\phi_i)_{i\in\mathbb{N}}$ is a orthonormal basis of $L^2(\Omega)$, for any initial value we can write

$$h(x) = \sum_{i=1}^{\infty} \langle h, \phi_i \rangle \phi_i(x)$$

If we assume we can write the solution as $u(x,t) = \sum_{i=1}^{\infty} \varphi_i(t)\phi_i(x)$, where $\varphi_i(0) = \langle h, \phi_i \rangle$ we get that the heat equation is given by

$$\begin{aligned}
&\frac{\partial}{\partial t}u(x,t) - \Delta u(x,t) = 0 \iff \sum_{i=1}^{\infty} \dot{\varphi}_i(t)\phi_i(x) - \sum_{i=1}^{\infty} \varphi_i(t)\Delta\phi_i(x) = 0 \\
&\iff \forall i \in \mathbb{N} : \dot{\varphi}_i(t)\phi_i(x) + \varphi_i(t)\lambda_i\phi_i(x) = 0 \\
&\iff \forall i \in \mathbb{N} : \dot{\varphi}_i(t) = -\varphi_i(t)\lambda_i
\end{aligned}$$

The solution of these ODEs is given by $\varphi_i(t) = \varphi_i(0)e^{-\lambda_i(t-t_0)}$ for every $i \in \mathbb{N}$. With an initial value $\varphi_i(0) = \langle h, \phi_i \rangle$ we obtain the unique solution to the initial value problem of the heat equation as $u(x,t) = \sum_{i=1}^{\infty} e^{-\lambda_i t}\langle h, \phi_i \rangle \phi_i(x)$. This series exist, due to the fact that $\lambda_i \to \infty$ and $e^{-\lambda_i t} \to 0$ exponentially fast. However, this approach makes a strong assumption on the structure of the solution! We will now generalize via Fourie Transforms. Further, if $\Omega = \mathbb{R}^n$ then the eigenfunctions of the inverse Laplace operator do not exist, i.e. they are not in $L^2(\mathbb{R}^n)$. The Fourie transform gives us a way to integrate these Eigenfunctions.

We want to show that we can can translate $\int_{\mathbb{R}^n} e^{-(x-c)^2} d^n x$ by any constant $c \in \mathbb{C}$. It holds for any $x \in i\mathbb{R}^n$

Theorem 6.2

Let $A(x) := \sum_{n\in\mathbb{N}} a_n x^n$ and $B(x) := \sum_{n\in\mathbb{N}} b_n x^n$, be two power series functions, whose convergence radii $\frac{1}{\overline{lim}\sqrt[n]{|a_n|}}$ are larger than $r > 0$. If for $Q := \{x \in \mathbb{K} \mid |x| \leq r\}$, with $|Q| = \infty$ it holds that $\forall x \in Q: \quad P(x) = Q(x)$ then it follows that $\forall n \in \mathbb{N}: a_n = b_n$ i.e. $A = B$.

$$\pi^{n/2} e^{x^2} = \sum_{l=0}^{\infty} \underbrace{\frac{\pi^{n/2}}{l!}}_{=:a_l} |x|^{2l} = \int_{\mathbb{R}^n} e^{-(k-x)^2} d^n k \sum_{l=0}^{\infty} \frac{|x|^{2l}}{l!}$$

$$= \int_{\mathbb{R}^n} e^{-(k-x)^2+x^2} d^n k = \int_{\mathbb{R}^n} e^{-k^2+2kx} d^n k = \sum_{l=0}^{\infty} \int_{\mathbb{R}^n} e^{-k^2} \frac{(2kx)^l}{l!} d^n k = \sum_{l=0}^{\infty} \underbrace{\int_{\mathbb{R}^n} e^{-k^2} \frac{(2k)^{2l}}{(2l)!} d^n k}_{=:b_l} \, x^{2l}.$$

The above is for $x \in \mathbb{C}$ equivlanet to

$$\pi^{n/2} e^{x^2} = \int_{\mathbb{R}^n} e^{-(k-x)^2+x^2} d^n k \quad \Longleftrightarrow \quad \pi^{n/2} = \int_{\mathbb{R}^n} e^{-(k-x)^2} d^n k$$

Thus with the theorem above it follows forall

$$\frac{\pi^{n/2}}{l!} = \int_{\mathbb{R}^n} e^{-k^2} \frac{(2k)^{2l}}{(2l)!} d^n k.$$

Unclear??? it follows

$$\forall c \in \mathbb{C}: \quad \pi^{n/2} = \int_{\mathbb{R}^n} e^{-(z-c)^2} d^n z.$$

Using this we get

$$\Phi(x-y,t) = \frac{1}{(4\pi t)^{n/2}} e^{-\frac{-(x-y)^2}{4t}} = e^{-\frac{(x-y)^2}{4t}} \frac{\pi^{n/2}}{(4\pi^2 t)^{n/2}} = e^{-\frac{|x-y|^2}{4t}} \frac{1}{(2\pi\sqrt{t})^n} \pi^{n/2}$$

$$= e^{-\frac{|x-y|^2}{4t}} \frac{1}{(2\pi\sqrt{t})^n} \int_{\mathbb{R}^n} e^{-(z-c)^2} d^n z, \quad z := 2\pi\sqrt{t}k$$

$$= e^{-\frac{|x-y|^2}{4t}} \frac{(2\pi\sqrt{t})^n}{(2\pi\sqrt{t})^n} \int_{\mathbb{R}^n} e^{-(2\sqrt{t}\pi k - c)^2} d^n k, \quad c := i\frac{x-y}{2\sqrt{t}}$$

$$= e^{-\frac{|x-y|^2}{4t}} \int_{\mathbb{R}^n} e^{-(2\sqrt{t}\pi k - i\frac{x-y}{2\sqrt{t}})^2} d^n k$$

$$= e^{-\frac{|x-y|^2}{4t}} \int_{\mathbb{R}^n} e^{-\left(4\pi^2 k^2 t - 4\pi k\sqrt{t} i \frac{x-y}{2\sqrt{t}} + i^2 \frac{(x-y)^2}{4t}\right)} d^n k$$

$$= \int_{\mathbb{R}^n} e^{-4\pi^2 k^2 t + 2\pi k i(x-y)} d^n k = \int_{\mathbb{R}^n} e^{-4\pi^2 k^2 t} e^{2\pi k i(x-y)} d^n k$$

This then yields

$$\lim_{t\searrow 0} u(x,t) = \lim_{t\searrow 0} \int_{\mathbb{R}^n} \Phi(x-y,t)h(y)d^ny$$

$$= \lim_{t\searrow 0} \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} e^{-4\pi^2k^2t}e^{2\pi kix}e^{-2\pi kiy}h(y)d^nkd^ny$$

$$\overset{Fubini}{=} \lim_{t\searrow 0} \int_{\mathbb{R}^n} e^{-4\pi^2k^2t}e^{2\pi kix} \underbrace{\int_{\mathbb{R}^n} e^{-2\pi kiy}h(y)d^ny}_{=:\hat{h}(k)}\, d^nk$$

$$= \lim_{t\searrow 0} \int_{\mathbb{R}^n} e^{-4\pi^2k^2t}e^{2\pi kix}\hat{h}(k)d^nk = \int_{\mathbb{R}^n} e^{2\pi kix}\hat{h}(k)d^nk \overset{(*)}{=} h(x),$$

although we will now show in the following that this equality actually holds.

Definition 6.3

The Schwartz space $\mathcal{S}$ contains all smooth complex calued functions where all its partial derivatives are bounded, i.e.

$$\mathfrak{S} := \left\{ f \in C^\infty(\mathbb{R}^n;\mathbb{C}) \mid \forall l \in \mathbb{N}, \forall \alpha \in \mathbb{N}_0^n \exists C \geq 0 \forall x \in \mathbb{R}^n : \quad |x|^{2l}|\partial^\alpha f(x)| \leq C \right\}.$$

Definition 6.4

The Fourier transform is defined as

$$\mathcal{F} : \mathfrak{S} \to \mathfrak{S}, h \mapsto \mathcal{F}(h) := \hat{h}, \quad \hat{h}(x) := \int_{\mathbb{R}^n} e^{-2\pi i k^T y}h(y)d^ny.$$

Lemma 6.5

Let $(X, \|\cdot\|_X)$ be a normed space, $(Y, \|\cdot\|_Y)$ a Banach space, D a dense subspace of X and $T \in L(D,Y)$ a linear operator. Then there exists only one unique continuous continuation $\hat{T} \in L(X,Y)$ such that $\hat{T}\big|_D = T$ and $\|\hat{T}\| = \|T\|$.

Proof. See functional analysis □

Lemma 6.6

The inverse of the Fourier transform is given by

$$\mathcal{P} \circ \mathcal{F} : \mathfrak{S} \to \mathfrak{S}, \quad \hat{h} \mapsto h, \quad h(x) := \int_{\mathbb{R}^n} e^{2\pi i k^T x}\hat{h}(k)d^nk.$$

Proof. Consider the normed space $(C_0^\infty(\mathbb{R}^n,\mathbb{C}), \|\cdot\|_{L^1})$ and $(L^1(\mathbb{R}^n), \|\cdot\|_{L^1})$ and the Banach space $(C_b(\mathbb{R}^n,\mathbb{C}), \|\cdot\|_{L^\infty})$. Remember that $C_b(\mathbb{R}^n,\mathbb{C})$ is a closed subspace of $L^\infty(\mathbb{R}^n)$.
We begin by showing that the Fourie transform maps from the dense subspace $C_0^\infty(\mathbb{R}^n,\mathbb{C})$ of $L^1(\mathbb{R}^n)$ to $C_b(\mathbb{R}^n,\mathbb{C})$, i.e. $\mathcal{F}\big|_{C_0^\infty(\mathbb{R}^n,\mathbb{C})} : C_0^\infty(\mathbb{R}^n,\mathbb{C}) \to C_b(\mathbb{R}^n,\mathbb{C})$, exists. Then we can apply the Lemma above and

argue that the denseness implies the existsence of a unique continuoation $\mathcal{F}\Big|_{L^1(\mathbb{R}^n)}$. Because $v(y) = e^{-2\pi iky} = \cos(-2\pi ky) + i\sin(-2\pi ky)$ is inside the complex unit cyrcle, it follows that

$$|\hat{h}(k)| \leq \int_{\mathbb{R}^n} \underbrace{|v(y)|}_{\leq 1} |h(y)| d^n y \leq \int_{\mathbb{R}^n} |h(y)| d^n y = \|h\|_{L^1(\mathbb{R}^n)}.$$

As $h \in C_0^\infty(\mathbb{R}^n, \mathbb{C})$ was chosen arbitrary this inequality also holds for the supremum:

$$\|\hat{h}(k)\|_{L^\infty(\mathbb{R}^n)} \leq \|h\|_{L^1(\mathbb{R}^n)} < \infty.$$

The last norm is finite, because h has compact support. Additionally it is appearant that the Fourier transform $\hat{h}(k) = \int_{\mathbb{R}^n} e^{-2\pi iky} h(y) d^n y$ is continuos, as all integrants are continuous. In total we have now shown that $\mathcal{F}\Big|_{C_0^\infty(\mathbb{R}^n,\mathbb{C})} : C_0^\infty(\mathbb{R}^n, \mathbb{C}) \to C_b(\mathbb{R}^n, \mathbb{C})$ exists. Thus, the Fourier Transform can be uniquely contiued on $L^1(\mathbb{R}^n)$.
In order for the image to be in the Schwartz space, we need to show multiple things: To show: h decays faster than any inverse power $\Rightarrow$ Smoothness of $\hat{h}$. Smoothness of $h \Rightarrow \hat{h}$ decays faster than any inverse power.
Step 1: Smoothness of $h \Rightarrow \hat{h}$ decays faster than any inverse power.
Let $h \in C_0^\infty(\mathbb{R}^n, \mathbb{C})$, then using

$$\begin{aligned} &(\nabla v u)' = \nabla v \nabla u + \Delta v u, \quad (v \nabla u)' = \nabla v \nabla u + v \Delta u \\ &\implies v \Delta u = (v \nabla u)' - \nabla v \nabla u = (v \nabla u)' + (\nabla v u)' - \Delta v u, \end{aligned}$$

and choosing $v(y) := e^{-2\pi iky}$ the Fourie transform of Δh is given by

$$\begin{aligned} -\hat{\Delta h}(k) &= -\int_{\mathbb{R}^n} v(y) \Delta h(y) d^n y = \int_{\mathbb{R}^n} \Delta v(y) h(y) d^n y - \int_{\partial \mathbb{R}^n} v(z) \nabla h(z) + \nabla v(z) h(z) d\sigma(z) \\ &\overset{(*)}{=} \int_{\mathbb{R}^n} \Delta e^{-2\pi iky} h(y) d^n y = \int_{\mathbb{R}^n} 4\pi^2 k^2 e^{-2\pi iky} h(y) d^n y \\ &= 4\pi^2 k^2 \hat{h}(k). \end{aligned}$$

In $(*)$ we used that Schwartz functions decay faster than any inverse power, the boundary terms dissapear. Further, we used

$$\int_{\partial \mathbb{R}^n} (\cdots) d\sigma := \lim_{r \to \infty} \int_{\partial B_r(0)} (\cdots) d\sigma.$$

With the arguemnts above and the fact that the Laplacian is again in $C_0^\infty(\mathbb{R}^n, \mathbb{C})$ it follows

$$|\hat{\Delta h}(k)| \leq \|\Delta h\|_{L^1(\mathbb{R}^n)} =: C < \infty.$$

This yields

$$|\hat{h}(k)| = \frac{|\hat{\Delta h}(k)|}{4\pi^2 k^2} \leq \frac{C}{4\pi^2 k^2} \quad \implies \quad \hat{h}(k) \in \mathcal{O}(\frac{1}{k^2}).$$

Due to the fact that we can do this chain of arguments arbitrarly often, it follows that

$$\forall n \in \mathbb{N}: \quad \hat{h}(k) \in \mathcal{O}(k^{-n}).$$

Thus, the Fourier transform decays faster than any inverse power.

Step 2: h decays faster than any inverse power $\Rightarrow$ Smoothness of $\hat{h}$.
Because h decays faster than any inverse power, we can apply the DCT and get

$$\begin{aligned}
|\frac{\partial}{\partial k_i}\hat{h}(k)| &\leq \left|\frac{\partial}{\partial k_i}\int_{\mathbb{R}^n} e^{-2\pi iky}h(y)d^ny\right| \overset{DCT}{=} \left|\int_{\mathbb{R}^n}\frac{\partial}{\partial k_i}e^{-2\pi iky}h(y)d^ny\right| \\
&\leq \int_{\mathbb{R}^n}\left|(-2\pi i y_i)e^{-2\pi iky}h(y)\right|d^ny \\
&\leq \int_{\mathbb{R}^n}(-2\pi\underbrace{|i|}_{=1}|y_i|)\underbrace{\left|e^{-2\pi iky}\right|}_{\leq 1}|h(y)|d^ny \\
&\leq 2\pi\||y|h(y)\|_{L^1(\mathbb{R}^n)} < \infty,
\end{aligned}$$

which means that every partial derivative of $\hat{h}$ exists and thus, it is differentiable which implies continuity. Finally, we will now show that the inverse of the Fourier transform maps from $\mathfrak{S}$ to $\mathfrak{S}$. For that choose $f \in \mathfrak{S}$ arbitrary. Then with Theorem 1 we have that

$$\begin{aligned}
h(x) &:= \lim_{t\searrow 0}u(x,t) = \lim_{t\searrow 0}\int_{\mathbb{R}^n}\Phi(x-y,t)h(y)d^ny = \lim_{t\searrow 0}\int_{\mathbb{R}^n}e^{-4\pi^2k^2t}e^{2\pi kix}\hat{h}(k)d^nk \\
&= \lim_{t\searrow 0}\int_{\mathbb{R}^n}e^{-4\pi^2k^2t}e^{2\pi kix}\hat{h}(k)d^nk = \int_{\mathbb{R}^n}e^{2\pi kix}\hat{h}(k)d^nk. \\
&= \int_{\mathbb{R}^n}e^{-2\pi ki(-x)}(\mathcal{F}(h))(k)d^nk = ((P\circ\mathcal{F}\circ\mathcal{F})(h))(x).
\end{aligned}$$

This is equivalent to

$$h = P\circ\mathcal{F}\circ\mathcal{F}\circ h \quad\Longleftrightarrow\quad \mathbf{1}_{\mathfrak{S}} = P\circ\mathcal{F}\circ\mathcal{F} \quad\Longleftrightarrow\quad P = \mathcal{F}\circ\mathcal{F}.$$

Further, it also holds

$$\begin{aligned}
(P\circ\mathcal{F}\circ\hat{h})(x) &= \int_{\mathbb{R}^n}e^{2\pi ikx}\hat{h}(k)d^nk \\
&= \int_{\mathbb{R}^n}e^{-2\pi ik(-x)}\hat{h}(k)d^nk = (\mathcal{F}\circ P\circ\hat{h})(x)
\end{aligned}$$

which yields that

$$P\circ\mathcal{F} = \mathcal{F}\circ P.$$

Putting it all together yields

$$\mathcal{F}\circ P\circ\mathcal{F} = \mathcal{F}\circ\mathcal{F}\circ P = P\circ P = \mathbf{1}_{\mathfrak{S}} \quad\Longleftrightarrow\quad \mathcal{F}^{-1} = P\circ\mathcal{F}.$$

□

Remark 6.1

So we have now shown that

$$\lim_{t\searrow 0}u(x,t) = \lim_{t\searrow 0}\int_{\mathbb{R}^n}\Phi(x-y,t)h(y)d^ny = \int_{\mathbb{R}^n}e^{2\pi kix}\hat{h}(k)d^nk = h(x).$$

Up to now the Fourier tranform was only defined on the Schwartz space $\mathfrak{S}$. We will now show that on the Schwartz space, it does not destroy the geometry and that we can extend it to $L^2(\mathbb{R}^n)$.

Definition 6.7

The complex conjugate for a complex number $z := x + iy \in \mathbb{C}$ is defined as $\bar{z} := z - iy$.

It holds that

$$\overline{\hat{g}(k)} = \overline{\int_{\mathbb{R}^n} g(y)e^{-2\pi iky}d^ny} = \int_{\mathbb{R}^n} \overline{g(y)e^{-2\pi iky}}d^ny = \int_{\mathbb{R}^n} \overline{g(y)}e^{2\pi iky}d^ny.$$

Using Fubini and the above it follows for every $h, g \in \mathfrak{S}$

$$\begin{aligned}\int_{\mathbb{R}^n} \hat{h}(k)\overline{\hat{g}(k)}d^nk &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} \hat{h}(k)\overline{g(y)}e^{2\pi iky}d^nyd^nk \\ &= \int_{\mathbb{R}^n} \overline{g(y)}\int_{\mathbb{R}^n} \hat{h}(k)e^{2\pi iyk}d^nkd^ny = \int_{\mathbb{R}^n} h(k)\overline{g(y)}d^nk.\end{aligned}$$

For the hermitean skalar product

$$(f, g) := \int_{\mathbb{R}^n} f(x)\overline{g(x)}d^nx$$

it hold that for all $h, g \in \mathfrak{S}$

$$(h, g) = \int_{\mathbb{R}^n} h(y)\overline{g(y)}d^ny = \int_{\mathbb{R}^n} \hat{h}(k)\overline{\hat{g}(k)}d^nk = (\hat{h}, \hat{g}).$$

This means that the Fourier transform is an isometric isomporhism from $\mathfrak{S}$ to $\mathfrak{S}$. Because $\mathfrak{S}$ is dense in $L^2(\mathbb{R}^n)$ this property extends to $L^2(\mathbb{R}^n)$.
Note that for a hermitean scalar product it holds that

$$\overline{(f, f)} = (f, f).$$

We now transition from an analysis of the heat eqaution using the Fourier transform on $\mathbb{R}^n$ to bounded open domains Ω.

We will now view the heat equation as a linear ODE

$$\partial_t = \Delta u + f,$$

on some functional space. This functional space on which we define the Laplace operator will now be defined:

Definition 6.8

For any connected domain $\Omega \subset \mathbb{R}^n$ let $W_0^{2,2}(\Omega)$ be the closure of the Hilbert space

$$\left(C_0^\infty(\Omega), \langle f, g\rangle_{W_0^{2,2}} := \int_\Omega \Delta f\overline{\Delta g}d^nx + \int_\Omega f\overline{g}d^nx = \langle \Delta f, \Delta g\rangle_{L^2(\Omega)} + \langle f, g\rangle_{L^2(\Omega)}\right).$$

i.e.

$$W_0^{2,2}(\Omega) := \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{W_0^{2,2}}} = \left\{g \in L^2(\Omega) \mid \exists (g_n)_{n\in\mathbb{N}} \subset C_0^\infty(\Omega) : \quad g_n \to g \in L^2(\Omega) \text{ and } \Delta g_n \to v \in L^2(\Omega)\right\}$$

also called Sobolev Space.

Note that by letting the elements in $W_0^{2,2}(\Omega)$ vanish we encode the Dirichlet boundary into the space. Further, because $W_0^{2,2}(\Omega)$ is defined to be complete, these limits exist.
We now want to show that $-\Delta : W_0^{2,2}(\Omega) \to W_0^{2,2}(\Omega)$ is a linear, postive, closed, self-adjoint linear operator, for a spectral decomposition to exist.
Let $h \in C_0^\infty(\Omega)$ be arbitary, then

$$\langle \Delta h, \Delta h\rangle_{L^2(\Omega)} = \int_\Omega (\Delta h)\overline{\Delta h} d^n x \leq \int_\Omega \Delta h \overline{\Delta h} d^n x + \int_\Omega h\overline{h} d^n x = \langle h, h\rangle_{W_0^{2,2}(\Omega)}.$$

By definition any $h \in W_0^{2,2}(\Omega)$ the function $\Delta h \in L^2(\Omega)$. Using Cauchy Schwartz it follows

$$\begin{aligned}|\langle f, \Delta h\rangle_{L^2(\Omega)}| &= |\int_\Omega f\overline{\Delta h} d^n x| \leq (\int_\Omega |f|^2 d^n x)^{1/2} (\int_\Omega |\overline{\Delta h}|^2 d^n x)^{1/2} \\ &= \|f\|_{L^2(\Omega)} \overline{\|\Delta h\|_{L^2(\Omega)}} \\ &= \|f\|_{L^2(\Omega)} \|\Delta h\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|h\|_{W_0^{2,2}(\Omega)}.\end{aligned}$$

Definition 6.9

An operator $T : W \subset \mathcal{H} \to \mathcal{H}$ on the Hilbert space $\mathcal{H}$ is called closed if

$$\Gamma(T) := \{(x, Tx) \mid x \in W\}$$

is closed in $\mathcal{H} \times \mathcal{H}$, i.e.

$$\forall (x_n)_{n\in\mathbb{N}} \subset W \text{ with } x_n \to x \in \mathcal{H} \text{ and } Tx_n \to y \in \mathcal{H} \Rightarrow x \in W \text{ and } Tx = y.$$

We need to show that $x_n \to x$ and $-\Delta x_n \to y$ implies $-\Delta x = y$ and $x \in W_0^{2,2}(\Omega)$. This immediatly follows from the construction of $W_0^{2,2}(\Omega) := \overline{C_0^\infty(\Omega)}^{\|\cdot\|_{W_0^{2,2}}}$ as the closure.
The Laplacian is further self-adjoint, because it is symmetric due to Greens second identiy:

$$\langle -\Delta u, v\rangle_{L^2} = \int_\Omega \nabla u \overline{\nabla u} d^n x = \langle u, -\Delta v\rangle_{L^2} \text{ and } \langle -\Delta u, u\rangle_{L^2} = \|\nabla u\|^2 \geq 0.$$

This also shows positivity. Thus, it follows that $-\Delta$ has a spectral decomposition, i.e. $\exists(\phi_n, \lambda_n)_{n\in\mathbb{N}}$ such that

$$-\Delta\phi_n = \lambda_n \phi_n, \quad \forall n \in \mathbb{N}.$$

We define for all $h \in L^2(\Omega)$

$$e^{t\Delta} h := \sum_{n\in\mathbb{N}} e^{t\lambda_n} \langle h, \phi_n\rangle \phi_n,$$

which solves the heat equation:

$$\frac{d}{dt}\left(e^{t\Delta} h\right) = \frac{d}{dt}\left(\sum_{n\in\mathbb{N}} e^{-t\lambda_n}\langle h, \phi_n\rangle\phi_n\right) = \sum_{n\in\mathbb{N}} \lambda_n e^{-t\lambda_n}\langle h, \phi_n\rangle \phi_n = \Delta\left(e^{t\Delta} h\right),$$

with the boundary conditions:

$$u(x, 0) = h(x), \quad u(x, t) = 0, \quad x \in \partial\Omega.$$

Remember, the second conditions comes from the construction of the Sobolev space. Finally, because $(\phi_n)_{n\in\mathbb{N}}$ builds an othonormal basis of $L^2(\Omega)$ Parsevals identity applies:

$$\|x\| = \sum_{n\in\mathbb{N}} |\langle e_n, x\rangle|^2.$$

Using this yields

$$\begin{aligned}
\|e^{t\Delta}h\|_{L^2(\Omega)} &\overset{Parseval}{=} \sum_{n\in\mathbb{N}} |\langle e^{-t\lambda_n}h, \phi\rangle_{L^2(\Omega)}|^2 \\
&\leq \sum_{n\in\mathbb{N}} \underbrace{|e^{-t\lambda_n}|^2}_{\leq 1, \lambda_n \geq 0} |\langle h, \phi\rangle_{L^2(\Omega)}|^2 \\
&\overset{Parseval}{\leq} \|h\|_{L^2(\Omega)}.
\end{aligned}$$

This means that the solution of the form $u(x,t) = e^{t\Delta}h(x)$ does not increase in in total energy over time. The following remark connects this section to the sections beforehand.

Remark 6.2

The inhomogeneous heat equation $\dot{u} = \Delta u + f$ can be viewed as an ODE on an infinite functional space on which u is defined, i.e. we have the operator $H := \Delta$. Then the solution to this linear ODE is given by

$$\forall x \in \mathbb{R}^n : \quad u(x,t) = \int_0^t e^{(t-s)H} f(x,s)ds.$$

However, it is not quite clear what $e^{t\Delta}$ would mean. Remember that

$$\mathcal{F}(\Delta u)(k) = -4\pi^2|k|^2\hat{u}(k)$$

Using this we get

$$\begin{aligned}
\mathcal{F}(e^{t\Delta}f)(k) &= \mathcal{F}\left(\sum_{n\in\mathbb{N}} \frac{t^n}{n!}\Delta^n f\right)(k) = \sum_{n\in\mathbb{N}} \frac{t^n}{n!}\mathcal{F}\left(\Delta^n f\right)(k) \\
&= \sum_{n\in\mathbb{N}} \frac{t^n}{n!}(-4\pi^2|k|^2)^n \hat{f}(k) = e^{-4\pi^2|k|^2 t}\hat{f}(k)
\end{aligned}$$

which is equivalent to

$$(e^{t\Delta}f)(x) = \mathcal{F}^{-1}\left(e^{-4\pi^2|k|^2t}\hat{f}(k)\right) = \mathcal{F}^{-1}\left(e^{-4\pi^2|k|^2t}\right) * f(x)$$

and using

$$\mathcal{F}^{-1}\left(e^{-4\pi^2|k|^2t}\right)(x) = \int_{\mathbb{R}^n} e^{2\pi ikx}e^{-4\pi^2|k|^2t}d^nk \overset{???}{=} \frac{1}{(4\pi t)^{n/2}}e^{-\frac{-|x|^2}{4t}} = \Phi(x,t)$$

implying

$$(e^{t\Delta}f)(x) = (\Phi(\cdot,t) * h)(x),$$

which implies that

$$u(x,t) = \int_0^t e^{(t-s)H}f(s)ds = \int_0^t (\Phi(\cdot, t-s) * f)(x)ds = \int_0^t \int_{\mathbb{R}^n} \Phi(x-y, t-s)f(y)d^nyds.$$

Note that in analogy to $A = UDU^{-1}$ for a diagonalizable matrix A, we have here

$$\mathcal{F}(\Delta h) = 4\pi^2 k^2 \hat{h}(k) \quad \iff \quad \Delta h = \mathcal{F}^{-1}(4\pi^2 k^2 \mathcal{F}(h)).$$

7 Heat Kernel of S^1

We define the circle S^1 using the quotiient space $\mathbb{R}/\mathbb{Z} := \{[x] \mid x \in \mathbb{R}\}$, where $[x] := \{x + n \mid n \in \mathbb{Z}\}$. For $X := [0, 1) \subset \mathbb{R}$ there is a bijection $X \to \mathbb{R}/\mathbb{Z}, x \mapsto [x]$. Then, we have the relation

$$C(\mathbb{R}/\mathbb{Z}) \cong \{f \in C([0, 1]) \mid f(0) = \lim_{x\to 1} f(x)\}.$$

Remember that the set of polynomials are dense in the set of continuous functions and they are dense in L^2. We can write

$$e^{2\pi ikx} = \left(e^{2\pi ix}\right)^k,$$

i.e. as polynomials. Its linear span

$$\{\sum_{k=-N}^{N} c_k \left(e^{2\pi ix}\right)^k \mid c_{-N}, ..., c_N \in \mathbb{C}\}$$

forms an algebra on $\mathbb{C}$ (for $k > 0$ we have real values and for $k < 0$ complex ones) that seperates points and contains constants. Stone Weierstraß implies that this linear span is dense in $C(\mathbb{R}/\mathbb{Z})$. And with the above and transitivity it is also dense in $L^2(\mathbb{R}/\mathbb{Z})$. We have shown that this linear span lies dense in $L^2(\mathbb{R}/\mathbb{Z})$. The next step is to show it builds an orthogonal basis. Using that $e^{2\pi ikx}$ are eigenfunctions of $-\frac{d^2}{dx^2}$, due to

$$-\frac{d^2}{dx^2} e^{2\pi ikx} = 4\pi^2 k^2 e^{2\pi ikx},$$

with Eigenvalues $4\pi^2 k^2$, this immediatly follows.

$$\begin{aligned}
&\int_{\mathbb{R}/\mathbb{Z}} H_{\mathbb{R}/\mathbb{Z}}(x, y, t) h(y) dy \\
&= u(x, t) = \left(e^{-tH} h\right)(x) = \sum_{k\in\mathbb{Z}} c_k \left(e^{-tH} e^{2\pi ik\cdot}\right)(x) = \sum_{k\in\mathbb{Z}} c_k e^{2\pi ikx - t4\pi^2 k^2} \\
&= \sum_{k\in\mathbb{Z}} \int_{\mathbb{R}/\mathbb{Z}} h(y) e^{-2\pi iky} dy e^{2\pi ikx - t4\pi^2 k^2} \\
&= \int_{\mathbb{R}/\mathbb{Z}} \sum_{k\in\mathbb{Z}} h(y) e^{2\pi ik(x-y) - t4\pi^2 k^2} dy,
\end{aligned}$$

although here $He^{2\pi ikx} = 4\pi^2 k^2 e^{2\pi ikx}$. Because we chose $h \in L^2(\mathbb{R}/\mathbb{Z})$ arbitrary it follows that

$$H_{\mathbb{R}/\mathbb{Z}}(x, y, t) = \sum_{k\in\mathbb{Z}} e^{2\pi ik(x-y) - 4\pi^2 k^2 t} =: \Theta(x - y, 4\pi it).$$

We call $\Theta(x, \tau) := \sum_{k\in\mathbb{Z}} e^{2\pi ikx + \pi ik^2\tau}$ Jacobi's Theta function. It is finite, i.e. this sum converges, on the domain

$$(x, \tau) \in \mathbb{C} \times \{\tau \in \mathbb{C} \mid \mathfrak{J}(\tau) > 0\}, \quad \text{where } \mathfrak{J}(4\pi it) := 4\pi\mathcal{R}e(it).$$

The limit of the sum is a complex differntiable function (holomorphic).
The fact that $\cos(2\pi x) + i\sin(2\pi x) = e^{2\pi ix}$ will now be relevant.

Remark 7.1

Jacobi's Theta function Θ satisfies the following properties

- $\Theta(x+1,\tau)=\Theta(x,\tau)$
- $\Theta(x+\tau,\tau)=e^{-\pi i\tau-2\pi ix}\Theta(x,\tau)$
- $\Theta(\frac{1}{2}+\frac{1}{2}\tau,\tau)=0.$

The first property holds due to the fact that $e^{2\pi ikx}$ has period 1:

$$e^{2\pi ik(x+1)}=e^{2\pi ikx}e^{2\pi ik}=e^{2\pi ikx}(\cos(2\pi k)+i\sin(2\pi k))=e^{2\pi ikx}\cdot(1+0).$$

The second property follows from:

$$\begin{aligned}
\Theta(x+\tau,\tau)&=\sum_{k\in\mathbb{Z}}e^{2\pi ik(x+\tau)+\pi ik^2\tau}=\sum_{k\in\mathbb{Z}}e^{2\pi ikx+2\pi ik\tau+\pi ik^2\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi ikx+\pi i\tau+2\pi ik\tau+\pi ik^2\tau-\pi i\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi ikx+\pi i\tau(1+2k+k^2)-\pi i\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi ikx+\pi i\tau(k+1)^2-\pi i\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi i(k+1)x-2\pi i+\pi i\tau(k+1)^2-\pi i\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi i(k+1)x-2\pi ix+\pi i\tau(k+1)^2-\pi i\tau}\\
&=\sum_{k\in\mathbb{Z}}e^{2\pi i(k+1)x+\pi i\tau(k+1)^2}e^{-2\pi ix-\pi i\tau}\overset{IndexShift}{=}\Theta(x,\tau)e^{-2\pi ix-\pi i\tau}.
\end{aligned}$$

The last property from

$$\begin{aligned}
\Theta(\frac{1}{2}+\frac{\tau}{2},\tau)&=\sum_{k\in\mathbb{Z}}e^{2\pi ik(\frac{1}{2}+\frac{\tau}{2})+\pi ik^2\tau}=\sum_{k\in\mathbb{Z}}e^{\pi ik+\pi ik\tau+\pi ik^2\tau}\\
&=\sum_{k\in\mathbb{Z}}(\cos(\pi k)+i\sin(\pi k))e^{\pi i\tau(k+k^2)}=\sum_{k\in\mathbb{Z}}(-1)^k e^{\pi i\tau((k+\frac{1}{2})^2-\frac{1}{4})}\\
&=e^{-\frac{\pi i\tau}{4}}\sum_{k\in\mathbb{Z}}(-1)^k e^{\pi i\tau(k+\frac{1}{2})^2}
\end{aligned}$$

We can see that

$$(k+\frac{1}{2})^2=k^2+k+\frac{1}{4}=1+2k+k^2-k-1+\frac{1}{4}=(k+1)^2-(k+1)+\frac{1}{4}=(-(k+1)+\frac{1}{2})^2.$$

Using this and the fact that $|\mathbb{N}_0|=|\mathbb{Z}|$, it follows

$$\begin{aligned}
e^{-\frac{\pi i\tau}{4}}\sum_{k\in\mathbb{Z}}(-1)^k e^{\pi i\tau(k+\frac{1}{2})^2}&=e^{-\frac{\pi i\tau}{4}}\sum_{k\in\mathbb{N}_0}\Big((-1)^k e^{\pi i\tau(k+\frac{1}{2})^2}+(-1)^{-(k+1)}e^{\pi i\tau(-(k+1)+\frac{1}{2})^2}\Big)\\
&=e^{-\frac{\pi i\tau}{4}}\sum_{k\in\mathbb{N}_0}(1-1)^k e^{\pi i\tau(-(k+1)+\frac{1}{2})^2}=0.
\end{aligned}$$

8 Heat Kernel of $(0,1)$

Here we want to construct a basis for $L^2((0,1))$, i.e. we need some eigenfunctions of the Laplace operator, that vanish on 0 and 1, i.e. the boundary, so that the boundary conditions are satisfied. The Eigenfunction of $-\frac{d^2}{dx^2}$ on $(0,1)$ that satisfies the Dirichlet fboundary conditions, i.e. having their roots at $\partial(0,1)=\{0,1\}$, are given by $\sqrt{2}\sin(k\pi x), k\in\mathbb{N}$:

$$-\frac{d^2}{dx^2}\sqrt{2}\sin(k\pi x)=-\sqrt{2}\frac{d}{dx}\cos(k\pi x)k\pi=\sqrt{2}k^2\pi^2\sin(k\pi x),\ \text{i.e. } k^2\pi^2 \text{ are Eigenvalues.}$$

Due to $k,k'\in\mathbb{N}$:

$$\int_0^1\sqrt{2}\sin(k\pi x)\sqrt{2}\sin(k'\pi x)dx\overset{???}{=}\int_0^1\cos((k-k')\pi x)-\cos((k+k')\pi x)dx\overset{???}{=}\delta_{k,k'}$$

$(\sqrt{2}\sin(k\pi x))_{k\in\mathbb{N}}$ builds an orthonormal system of $C(0,1)$. Note that there exists a bijection between $(0,1)$ and $\mathbb{R}/2\mathbb{Z}$. Let f be continuous on $[0,1]$ with roots at $\partial(0,1)$. Define the the odd reflection

$$\hat{f}(x):=\begin{cases}f(x) & ,x\in(0,1)\\ -f(-x) & ,x\in(-1,0)\end{cases}$$

then $\hat{f}$ is continous on $(-1,1)$ and $\hat{f}(-1)=\hat{f}(0)=\hat{f}(1)=0$. If we repeat this pattern for $(1,2),(2,3),...$ we get

$$\tilde{f}(x):=\begin{cases}f(x-2n) & ,x\in[2n,2n+1],n\in\mathbb{Z}\\ -f(2n-x) & ,x\in[2n-1,2n],n\in\mathbb{Z}\end{cases}$$

is continuous on $\mathbb{R}$ and has roots on $\mathbb{Z}$ and period 2. Because $(e^{i\pi kx})_{k\in\mathbb{Z}}$ builds an ONB of $L^2(\mathbb{R}/2\mathbb{Z})$, we can write

$$\begin{aligned}\sum_{k\in\mathbb{Z}}c_ke^{i\pi kx}&=\tilde{f}(x)\\ &=-\tilde{f}(-x)=-\sum_{k\in\mathbb{Z}}c_ke^{i\pi k(-x)}=-\sum_{k\in\mathbb{Z}}c_{-k}e^{i\pi kx}\end{aligned}$$

which implies that $\forall k\in\mathbb{Z}:c_k=-c_{-k}$. This yields

$$\begin{aligned}\tilde{f}(x)&=\sum_{k\in\mathbb{N}}\left(c_ke^{i\pi kx}+c_{-k}e^{-i\pi kx}\right)=\sum_{k\in\mathbb{N}}c_k\left(e^{i\pi kx}-e^{-i\pi kx}\right)\\ &=\sum_{k\in\mathbb{N}}c_k2i\sin(\pi kx)=:\sum_{k\in\mathbb{N}}b_k\sin(\pi kx).\end{aligned}$$

Thus, the ONB $(\sin(\pi kx))_{k\in\mathbb{N}}$ lies dense in the set

$$\mathcal{A}:=\left\{\tilde{f}\in L^2(\mathbb{R}/2\mathbb{Z})\mid\tilde{f}(x)=\begin{cases}f(x-2n) & ,x\in[2n,2n+1],n\in\mathbb{Z}\\ -f(2n-x) & ,x\in[2n-1,2n],n\in\mathbb{Z}\end{cases}\right\}.$$

Due to the fact that

$$L^2((0,1))\to\mathcal{A},f\mapsto\frac{1}{\sqrt{2}}\tilde{f}$$

(we need to scale with $\frac{1}{\sqrt{2}}$ in order to get an isomporphism) satisfies

$$\langle \frac{1}{\sqrt{2}}\tilde{f}, \frac{1}{\sqrt{2}}\tilde{g}\rangle_{\mathcal{A}} = \frac{1}{2}\int_0^2 \tilde{f}(x)\tilde{g}(x)dx = \frac{1}{2}2\int_0^1 f(x)g(x)dx = \langle f, g\rangle_{L^2((0,1))},$$

i.e. this mapping is an isomporhism. This finally implies that $(\sqrt{2}\sin(\pi k x))_{k\in\mathbb{N}}$ builds an ONB of $L^2((0,1))$. Putting it all together, we can now write $\forall h \in L^2((0,1))$:

$$h(x) = \sum_{k\in\mathbb{N}} \underbrace{\langle \sqrt{2}\sin(k\pi y), h\rangle_{L^2}}_{=:a_k} \sqrt{2}\sin(k\pi x).$$

For the initial value problem of the Heat equation

$$\begin{aligned}
&\dot{u} + Hu = 0, \quad \text{on } (0,1)\times\mathbb{R}^+ \\
&u(x,0) = h(x), \quad x \in (0,1) \\
&u(0,t) = u(1,t) = 0, \quad t \in \mathbb{R}^+
\end{aligned}$$

where $H := -\frac{d^2}{dx^2}$, the solution is given by

$$\begin{aligned}
\int_{(0,1)} H_{(0,1)}(x,y,t)h(y)dy = u(x,t) &= \left(e^{-tH}h\right)(x) \\
&= \sum_{k\in\mathbb{N}} \langle \sqrt{2}\sin(\pi y, h)\rangle_{L^2} e^{-tk^2\pi^2}\sqrt{2}\sin(k\pi x) \\
&= \sum_{k\in\mathbb{N}} 2\int_0^1 \sin(k\pi y)h(y)dy e^{-tk^2\pi^2}\sin(k\pi x) \\
&= \int_0^1 \sum_{k\in\mathbb{N}} 2\sin(k\pi y)e^{-tk^2\pi^2}\sin(k\pi x)h(y)dy
\end{aligned}$$

which implies that

$$H_{(0,1)}(x,y,t) = \sum_{k\in\mathbb{N}} 2\sin(k\pi y)\sin(k\pi x)e^{-tk^2\pi^2}.$$

Using that $\sin(A)\sin(B) = \cos(A-B) - \cos(A+B)$ it follows that

$$H_{(0,1)}(x,y,t) = \sum_{k\in\mathbb{N}} e^{-\pi^2k^2t}\cos(k\pi(x-y)) - \sum_{k\in\mathbb{N}} e^{-\pi^2k^2t}\cos(k\pi(x+y)).$$

Remembering Jacobi's Theta function:

$$\begin{aligned}
\Theta(x,\tau) &:= \sum_{k\in\mathbb{Z}} e^{\pi i k^2/tau + 2\pi i k x} = e^0 + \sum_{k\in\mathbb{N}} e^{\pi i k^2\tau}\left(e^{2\pi i k x} + e^{-2\pi i k x}\right) \\
&= 1 + 2\sum_{k\in\mathbb{N}} e^{\pi i k^2\tau}\cos(2\pi k x) \\
\iff &\frac{1}{2}\Theta(x,\tau) - \frac{1}{2} = \sum_{k\in\mathbb{N}} e^{\pi i k^2\tau}\cos(2\pi k x),
\end{aligned}$$

leads to

$$\begin{aligned} H_{(0,1)}(x,y,t) &= \sum_{k\in\mathbb{N}} e^{-\pi^2k^2t}\cos(k\pi(x-y)) - \sum_{k\in\mathbb{N}} e^{-\pi^2k^2t}\cos(k\pi(x+y)) \\ &= \frac{1}{2}\Theta(\frac{x-y}{2}, it\pi) - \frac{1}{2}\Theta(\frac{x+y}{2}, it\pi). \end{aligned}$$

Interpretation: ???

Remark 8.1

Exercises:...

The heat kernel on the cartesian product of two domains can be easily be calculated as the heat kernels of both domains:

Lemma 8.1

If $\Omega \subset \mathbb{R}^m$ and $\Omega' \subset \mathbb{R}^n$ are two open, bounded and connected domains with heat kernels H_Ω and $H_{\Omega'}$ respectively, the heat kernel of $\Omega \times \Omega'$ is given

$$\forall (x,x'),(y,y') \in \bar{\Omega}\times\bar{\Omega}', t\in\mathbb{R}^+ : \quad H_{\Omega\times\Omega'}((x,x'),(y,y'),t) = H_\Omega(x,y,t)H_{\Omega'}(x',y',t),$$

Proof. We need to show that $H_{\Omega\times\Omega'}$ satisfies the definition of the heat kernel of $\Omega\times\Omega'$:

i) $\forall (x,x',t) \in \Omega\times\Omega'\times\mathbb{R}^+ : \quad (y,y') \mapsto H_{\Omega\times\Omega'}((x,x'),(y,y'),t)$ extends continuously to $\bar{\Omega}$ with 0 on $\partial(\Omega\times\Omega')$.

ii) For $(x,x') \in \Omega\times\Omega'$ the function $(y,y',t) \mapsto H_{\Omega\times\Omega'}((x,x'),(y,y'),t) - \Phi(x-y,t)\Phi(x'-y',t)$ solves the homogenuous heat equation and extends continuously to $\bar{\Omega}\times\bar{\Omega}'\times\mathbb{R}_0^+$ with 0 on $\bar{\Omega}\times\bar{\Omega}'\times\{0\}$.

i) Due to $\partial(\Omega\times\Omega') = (\partial\Omega\times\bar{\Omega}')\cup(\bar{\Omega}\times\partial\Omega')$ we have the two cases for $(\xi,\eta)\in\partial(\Omega\times\Omega')$:

- Case 1: $(\xi,\eta)\in\bar{\Omega}\times\partial\Omega'$:

$$\lim_{y'\to\eta} H_\Omega(x,y,t)H_{\Omega'}(x',y',t) = \lim_{y'\to\eta} H_\Omega(x,y,t)H_{\Omega'}(x',y',t) = 0$$

- Case 2: $(\xi,\eta)\in\partial\Omega\times\bar{\Omega}'$:

$$\lim_{y\to\xi} H_\Omega(x,y,t)H_{\Omega'}(x',y',t) = \lim_{y\to\xi} H_\Omega(x,y,t)H_{\Omega'}(x',y',t) = 0$$

The continuous extension is trivial due to the product.

ii) First, note that

$$\Delta_{\Omega\times\Omega'} = \sum_{j=1}^{n+m}\frac{\partial^2}{\partial x_j^2} = \sum_{i=1}^{n}\frac{\partial^2}{\partial x_i^2} + \sum_{k=1}^{m}\frac{\partial^2}{\partial x_k^2} = \Delta_\Omega + \Delta_{\Omega'}$$

leads to

$$\begin{aligned} &(\frac{\partial}{\partial t} - \Delta_{\Omega\times\Omega'})(H_\Omega H_{\Omega'} - \Phi'\Phi) = (\frac{\partial}{\partial t} - \Delta_\Omega - \Delta_{\Omega'})(H_\Omega H_{\Omega'} - \Phi'\Phi) \\ &= \dot{H}_\Omega H_{\Omega'} + H_\Omega \dot{H}_{\Omega'} - \dot{\Phi}\Phi' - \Phi\dot{\Phi}' - \big(\Delta_\Omega H_\Omega H_{\Omega'} + \Delta_{\Omega'} H_{\Omega'} H_\Omega - \Delta_\Omega \Phi\Phi' - \Delta_{\Omega'}\Phi'\Phi\big) \\ &= (\dot{H}_\Omega - \Delta_\Omega H_\Omega)H_{\Omega'} + (\dot{H}_{\Omega'} - \Delta_{\Omega'} H_{\Omega'})H_\Omega - \big((\dot{\Phi} - \Delta_\Omega\Phi)\Phi' + (\dot{\Phi}' - \Delta_{\Omega'}\Phi')\Phi\big) = 0 \end{aligned}$$

where Φ' is defined on Ω' and Φ on Ω.

Fix $x \in \Omega$. Define $W(x,y,t) := H_\Omega(x,y,t) - \Phi(x-y,t)$ then $(\partial_t - \Delta)W(x,y,t) = 0$ on $\Omega \times (0,T]$. By definition $W(x,y,t) = 0$ on $\bar{\Omega} \times \{0\}$. It holds that with using the first condition of the heat kernel that for $y \in \partial\Omega$ it holds $H_\Omega(x,y,t) = 0$ which implies that

$$\forall y \in \partial\Omega : \quad W(x,y,t) = -\Phi(x-y,t) \leq 0.$$

For $t \searrow 0$ both summands are the same delta distribution, i.e. $\lim_{t\searrow 0} W(x,y,0) = 0$ for $x,y \in \bar{\Omega}$. Because we can write the boundary parabolic cylinder as $\partial\Omega_T = (\partial\Omega \times [0,T]) \cup (\bar{\Omega} \times \{0\})$ we have just shown that $W(x,y,t) \leq 0$ on $\partial\Omega_T$. Using the maximum principle it follows that $\forall (y,t) \in \Omega_T : W(x,y,t) \leq 0$. This implies

$$\forall (y,t) \in \Omega_T : \quad H_\Omega(x,y,t) \leq \Phi(x-y,t) = \frac{1}{(4\pi t)^{n/2}} \underbrace{e^{-\frac{|x-y|^2}{4t}}}_{\leq 1} \leq \frac{1}{(4\pi t)^{n/2}}.$$

Thus, H_Ω is polynomially bounded $\mathcal{O}(t^{-n/2})$ for $t \to 0$.
Define $d_x := dist(x, \partial\Omega) > 0$. Then, for $t \in (0,T]$:

$$\begin{aligned}\sup_{y\in\partial\Omega} |W(x,y,t)| &= \sup_{y\in\partial\Omega} \left| \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x-y|^2}{4t}} \right| \\ &\leq \frac{1}{(4\pi t)^{n/2}} e^{-\frac{d_x^2}{4t}}.\end{aligned}$$

Because with the maximum principle the solution is in the interior smaller than on the boundary, it follows

$$\forall (y,t) \in \Omega_T : \quad |W(x,y,t)| \leq \frac{1}{(4\pi t)^{n/2}} e^{-\frac{d_x^2}{4t}}.$$

Thus, as $t \to 0$ it follows that $H_\Omega(x,y,t) - \Phi(x-y,t)$ vanishes exponentially fast.
Using this we finally have

$$\begin{aligned}&\lim_{t\searrow 0} (H_\Omega(x,y,t) H_{\Omega'}(x',y',t) - \Phi(x-y,t)\Phi'(x'-y',t)) \\ &= \lim_{t\searrow 0} \underbrace{(H_\Omega(x,y,t) - \Phi(x-y,t))}_{\in \mathcal{O}(e^{-\frac{1}{t}})} \underbrace{H_{\Omega'}(x',y',t)}_{\in \mathcal{O}(t^{-n/2})} + \underbrace{\Phi(x-y,t)}_{\in \mathcal{O}(t^{-n/2})} \underbrace{(H_{\Omega'}(x',y',t) - \Phi'(x'-y',t))}_{\in \mathcal{O}(e^{-\frac{1}{t}})} = 0.\end{aligned}$$

□

We now have a formula for heat kernels on all tori $(\mathbb{R}/\mathbb{Z})^n$ and on all cartesian products of $(0,1)^n$. However, the boundary $\partial(0,1)^n$ is not a continuously differentiable submanifold of $\mathbb{R}^n$. The faces are $n-1$ dimensional submanifolds, the edges and corners are $n-2$ dimionsal and $n-3$ dimensional submanifolds respectively. Thus, when applying the divergence theorem, we get additional boundary terms. Luckily, these additional terms vanish, due to the fact that they are zero sets on the $\mathbb{R}^{n-1}$ Lesbesgue measure. Thus, the divergence theorem still holds and we can use all previous results. Hence, for the initial value Problem

$$\dot{u} - \Delta u = 0, \quad \text{on } (0,1)^n \times (0,T], \quad u(x,0) = h(x), \quad x \in (0,1)^n, \quad u(x,t) = 0, \quad x \in \partial(0,1)^n, t \in (0,T],$$

the heat kernel is given by

$$H_{(0,1^n)}(x,y,t) = \prod_{i=1}^{n} H_{(0,1)}(x_i, y_i, t),$$

thus for $y := (y_1, ..., y_n) \in (0,1)^n$ and $x = (x_1, ..., x_n) \in (0,1)^n$ the solution is given by

$$u(x,t) = \int_{(0,1)^n} H_{(0,1)^n}(x,y,t)h(y)d^ny = \int_{(0,1)^n} \prod_{i=1}^n H_{(0,1)}(x_i,y_i,t)h(y)d^ny.$$

Due to

$$\Phi(x-y,t) = \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x-y|^2}{4t}} = \frac{1}{r^n}\frac{1}{(4\pi\frac{t}{r^2})^{n/2}} e^{-\frac{|\frac{x-y}{r}|^2}{4\frac{t}{r^2}}} = \frac{1}{r^n}\Phi\left(\frac{x-y}{r}, \frac{t}{r^2}\right)$$

it follows

$$\begin{aligned} H_{(0,r)}(x,y,t) &= \sum_{k\in\mathbb{Z}} \left(\Phi(x-y+2kr,t) - \Phi(x+y+2kr,t)\right) \\ &= \sum_{k\in\mathbb{Z}} \frac{1}{r}\left(\Phi(\frac{x-y+2kr}{r}, t/r^2) - \Phi(\frac{x+y+2kr}{r}, t/r^2)\right) \\ &= \sum_{k\in\mathbb{Z}} \frac{1}{r}\left(\Phi(\frac{x-y}{r}+2k, t/r^2) - \Phi(\frac{x+y}{r}+2k, t/r^2)\right) \\ &= \frac{1}{r}H_{(0,1)}\left(\frac{x}{r}, \frac{y}{r}, \frac{t}{r^2}\right). \end{aligned}$$

This leads to

$$H_{(0,r)^n}(x,y,t) = \prod_{i=1}^n \frac{1}{r}H_{(0,1)}\left(\frac{x_i}{r}, \frac{y_i}{r}, \frac{t}{r^2}\right) = \frac{1}{r^n}H_{(0,1)^n}\left(\frac{x}{r}, \frac{y}{r}, \frac{t}{r^2}\right).$$

Corollary 8.2

Any u solution of the homogeneous heat equation on $[0,r]^n \times [0,T] \subset \mathbb{R}^n \times \mathbb{R}$ satisfies

$$u(x,T) = \underbrace{-\int_0^T \int_{\partial[0,r]^n} u(z,t)\nabla_z H_{[0,r]^n}(x,z,T-t)\cdot N(z)d\sigma(z)dt}_{=0} + \int_{[0,r]^n} \underbrace{u(y,0)}_{=h(y)} H_{[0,r]^n}(x,y,T)d^ny.$$

Corollary 8.3

Any solution u of the homogenuous heat equation on an open domain $\Omega_T \subset \mathbb{R}^n \times \mathbb{R}$ is smooth and for fixed t it is analyitic wrt. x.

Proof.

- To show $u \in C^\infty(\Omega \times (0,T])$.
  Because Φ is smooth and $y \mapsto H_\Omega(x,y,t) - \Phi(x-y,t)$ extends continuously to $\bar{\Omega} \times \mathbb{R}^+$ it follows that H_Ω is smooth wrt. x and y (symmetry of heat kernel) on $\Omega \times \Omega \times \mathbb{R}^+$. Thus, u from the corollary above is in $C^\infty(\Omega \times (0,T])$, because the integrant is smooth wrt. x and y and the integration is over y.

- To show that $u(\cdot,t)$ is analyitic for fixed t.
  For $y \in \partial\Omega$ it follows that $\Phi(x-y,t) \to 0$, $t \searrow 0$ and all partial derivatives of Φ also satisfy this property, i.e. $D^\alpha\Phi(x-y,t) \to 0$ for $t \searrow 0$. H_Ω inherits all of these property by definition of the heat kernel ii). Thus,

$$\left|\frac{H_\Omega^{(n)}(x,\xi,t)}{n!}(x-y)^n\right| \to 0, \quad \forall \xi \in (x,y), \quad t \searrow 0.$$

Thus, for fixed t the Taylor series of H_Ω around x converges $\forall y \in \Omega$. Because the integration and differentiation can be interchanged (due to smoothness of H_Ω wrt. x and y) it follows that the Taylor series of $u(\cdot,t)$ around x converges $\forall x \in \Omega$. Thus, $u(\cdot,t)$ is analytic.

??? $\square$

9 Summary

We first noticed that scaling invariance of the soltuion to the heat equation on $\mathbb{R}^n$, where $u(x,t)$ is a solution $\iff$ $u(\lambda x, \lambda^2 t)$ is also a solution. This suggests the self-similar Ansatz $u(x,t) = v(\frac{x}{\sqrt{t}})$ leading to the fundamental solution Φ. This fundamental solution is non negative, its integral is one, i.e. it is similar to mollifiers. I.e. we get something like an approximate identity if we take the convolution of Φ with the initial value as $t \searrow 0$. Also the solution to the homogeneuous initial value problem on $\mathbb{R}^n$ is the convolution $u = \Phi * h$ for $t > 0$. As the solution depends on the entire $\mathbb{R}^n$ the speed of information propagation is infinite. Next, we showed that we can view the inhomogeneous heat equation as an ODE on an infinite dimensional functional space and showed that the solution is given by

$$\int_0^t \Phi(\cdot, t-s) * f(\cdot, s) ds.$$

Further, the inhomogeneous heat equation can be obtained from choosing the initial value as $f(\cdot,t)$ and integrating over the solution of the homogeneous initial value problem. The solution to the Cauchy problem is then obtained by splitting the solution $u = v + w$ into the inhomogeneous and homogeneous initial value problem (superposition).

Unlike the Laplace equatin the solution to the heat equation satisfies a mean value property on a space time region (not a ball) called heat balls $E(x,t,r)$, which are defined through Φ.

The maximums princpiles are defined on Ω_T, where the parabolic boundary is $\partial\Omega_T = (\partial\Omega \times (0,T]) \cup (\Omega \times \{0\})$. If $u \in C^2$ is a solution of the heat equation on Ω_T, which is path connected, and extends continuously to $\overline{\Omega_T}$ and takes its maximum in Ω_T, then it is constant on Ω_T.

If Ω is open and boundad and $u \in C^2$ is the solution to the heat equation extending constinuously to $\bar{\Omega}_T$, it follows that the maximum is on $\partial\Omega_T$. Using these we can show the existence and uniqueness of the Cauchy problem on an open and bounded domain.

On $\mathbb{R}^n$ the uniqueness is more constrained. A counterexample shows this. To show the uniqueness here, we first define the maximums principle for the Cauchy problem on $\mathbb{R}^n$ for bounded solutions

$$u(x,t) \leq Ae^{a|x|^2}.$$

This is then used to show the existence and uniqueness of such solutions on $\mathbb{R}^n$.

Another approach to show uniquness is the usage of energy functions

$$e(t) := \int_\Omega u^2(x,t) d^n x.$$

Up to now we have only found explicit forms for the solutions on the interior of the domain, by ignoring the boundary. Here heat kernels come into play. They are symmetric, positive and give us a form for the solution on $\bar{\Omega}$. Using spectral thoery we can show the existence of these heat kernels by deriving explicit formulas on S^1 and $(0,1)$ and finally generalizing to any $[0,r]^n \times [0,T]$. Further, this section shows that convolution with Φ is analougously to summation of eigenfunctions and gives us a justification to treat the heat equation as an ODE.

CHAPTER 5

WAVE EQUATION

We consider the homogeneous and inhomogenuous wave equation $\frac{\partial^2}{\partial t^2}u - \Delta u = 0$ and $\frac{\partial^2}{\partial t^2}u - \Delta u = f$ respectively on open subsets of $\mathbb{R}^n \times \mathbb{R}$. It is a linear second order PDE. The matrix for the coefficients of the differential operator has one positive and n negative eigenvalues, i.e. it is neither definite or semi definite. The theory to derive a solution in this case is completely different to the chapters before. We have information that propagates with finite speed through space. See electromagnetic waves.

1 D'Alemberts Formula $n = 1$

Consider the initial value proplem

$$\begin{aligned} &\frac{\partial^2}{\partial t^2}u - \frac{\partial^2}{\partial x^2}u = 0, \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+, \\ &u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}. \end{aligned}$$

We chose these initial conditions, because we interpret this PDE as an ODE on the vector space of smooth functions on $\mathbb{R}$, i.e. $C^\infty(\mathbb{R})$, then ODE theory suggests to fix an initial value and an initial velocity (first derivative) at an initial time t_0. We set $t_0 = 0$. We clearly want to have uniqueness of the solution.
Note that for $n = 1$ we have

$$\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} = \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right).$$

In the homogenuous case we know that if u solves the homogeneous wave equation, then $v(x,t) := \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right)u(x,t)$ solves the transport equation $\frac{\partial}{\partial t}v + \frac{\partial}{\partial x}v = 0$. Using the method of characteristics, the unique solution of this transport equation with initial value $v(x,0) = a(x)$ is given by

$$v(x,t) = a(x-t).$$

Thus, we have

$$\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right)u(x,t) = a(x-t).$$

This is an inhomogeneous transport equation with constant coefficients. Because $f'(u(x,s)) = -u(x,s)$ and choosing $x_0 = x + t$ the characteristic $X(s) := x + t - s$ and the solution $U(s) := u(X(s), s)$ along it then yield

$$\frac{d}{ds}U(X(s),s) = \frac{\partial}{\partial t}u(X(s),s) + \frac{\partial}{\partial x}u(X(s),s)\underbrace{\frac{dX}{ds}}_{=-1} = \frac{\partial}{\partial t}u(X(s),s) - \frac{\partial}{\partial x}u(X(s),s) = a(X(s)-s) = a(x+t-2s)$$

is an ODE, whose solution is given by

$$U(t) - U(0) = \int_0^t a(x+t-2s)ds.$$

Thus, we have for $b(x+t) := u(\underbrace{x+t}_{=X(0)}, 0)$ that

$$u(X(t),t) = \int_0^t a(x+t-2s)ds + b(x+t) \overset{y=x+t-2s,}{=}{}^{dy=(-2)ds} \int_{x+t-2\cdot 0}^{x+t-2t} (-\frac{1}{2})a(y)dy + b(x+t) = \frac{1}{2}\int_{x-t}^{x+t} a(y)dy + b(x+t).$$

Inserting the initial values, i.e.

$$u(x,0) = b(x) = g(x), \quad a(x) = v(x,0) = \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right) u(x,0) = h(x) - g'(x),$$

we get

$$\begin{aligned} u(x,t) &= \frac{1}{2}\int_{x-t}^{x+t} h(y) - g'(y)dy + g(x+t) \\ &= -\frac{1}{2}(g(x+t) - g(x-t)) + g(x+t) + \frac{1}{2}\int_{x-t}^{x+t} h(y)dy \\ &= \frac{1}{2}(g(x+t) + g(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} h(y)dy. \end{aligned}$$

Thus we have used the homoegenous and the inhomogenous transport equation to derive

Theorem 1.1: D'Alemberts Formula

If $g : \mathbb{R} \to \mathbb{R}$ is twice continuously differentiable and $h : \mathbb{R} \to \mathbb{R}$ is continuously differentiable, then

$$u(x,t) = \frac{1}{2}(g(x+t) + g(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} h(y)dy$$

is twice continuously differentiable and the unique solution of the initial value problem

$$\begin{aligned} &\frac{\partial^2}{\partial t^2}u - \frac{\partial^2}{\partial x^2}u = 0, \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+, \\ &u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}. \end{aligned}$$

Thus, we can write

$$u(x,t) = \frac{1}{2}(g(x+t)+g(x-t)) + \frac{1}{2}(H(x+t) - H(x-t)) = \underbrace{\frac{1}{2}(g(x+t) + H(x+t))}_{=:F(x+t)} + \underbrace{\frac{1}{2}(g(x-t) - H(x-t))}_{=:G(x-t)}.$$

It follows

$$\left(\frac{\partial}{\partial t} - \frac{\partial}{\partial x}\right) F(x+t) = F'(x+t)\cdot 1 - F'(x+t)\cdot 1 = 0,$$

and

$$\left(\frac{\partial}{\partial t}+\frac{\partial}{\partial x}\right)G(x-t)=G'(x-t)\cdot(-1)+G'(x+t)\cdot 1=0,$$

which implies

$$\left(\frac{\partial^2}{\partial t^2}-\frac{\partial^2}{\partial x^2}\right)u(x,t)=\left(\frac{\partial}{\partial t}-\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t}+\frac{\partial}{\partial x}\right)(F(x+t)+G(x-t))=0,$$

if F and G are twice differentiable. We have shown that every function of the form $u(x,t)=F(x+t)+G(x-t)$ solves the homogeneous wave equation.

We interpret the fact, that the value of the solution at (x,t) depends only on the values of F and G at $x+t$ and $x-t$ respectively, i.e. $u(x,t)$ depends excusively on the intervall $[x-t,x+t]$, which means information speed spreads with speed 1 in both directions.

Further, in the heat equation we had the property that the solution increases in regularity over time. Here, the regularity stays the same, i.e. if H and G are k-times differentiable, then u is also k-times differentiable.

In order to define the wave equation on $[0,\infty)$ the following approach will be of importance. Consider the initial value problem

$$\begin{aligned}&\frac{\partial^2}{\partial t^2}u(x,t)-\frac{\partial^2}{\partial x^2}u(x,t)=0, \quad (x,t)\in\mathbb{R}^+\times\mathbb{R}^+\\ &u(x,0)=g(x), \quad \frac{\partial}{\partial t}u(x,0)=h(x), \quad x\in\mathbb{R}^+, \quad u(0,t)=0, t\in\mathbb{R}_0^+.\end{aligned}$$

Define the point reflection of u,g,h as

$$\tilde{u}(x,t)=\begin{cases}u(x,t) & ,x\geq 0\\ -u(-x,t) & ,x\leq 0\end{cases}, \quad \tilde{g}(x)=\begin{cases}g(x) & ,x\geq 0\\ -g(-x) & ,x\leq 0\end{cases}, \quad \tilde{h}(x)=\begin{cases}h(x) & ,x\geq 0\\ -h(-x) & ,x\leq 0\end{cases}$$

then $\tilde{u},\tilde{g},\tilde{h}$ are defined on the entire space $\mathbb{R}^+\times\mathbb{R}_0^+$. Next, we see that for the solution $\tilde{u}$ of the initial value problem

$$\begin{aligned}&\frac{\partial^2}{\partial t^2}\tilde{u}(x,t)-\frac{\partial^2}{\partial x^2}\tilde{u}(x,t)=0, \quad (x,t)\in\mathbb{R}\times\mathbb{R}^+\\ &\tilde{u}(x,0)=\tilde{g}(x), \quad \frac{\partial}{\partial t}\tilde{u}(x,0)=\tilde{h}(x), \quad x\in\mathbb{R}\end{aligned}$$

the function $(x,t)\mapsto -\tilde{u}(-x,t)$ also solves the same initial value problem. By uniqueness (under sufficient regularity) of the solution it follows that both solution coincide: $\tilde{u}(x,t)=-\tilde{u}(-x,t)$.

- For $t\leq x$ the solution $u(x,t)$ depends on the domain $[x-t,x+t]\subset\mathbb{R}_0^+$, i.e. only the non negative part of the real axis. Using D'Alemberts formula it follows

$$u(x,t)=\frac{1}{2}(g(x+t)+g(x-t))+\frac{1}{2}\int_{x-t}^{x+t}h(y)dy.$$

- However, for $x \leq t$ the solution depends on the domain $[x-t, x+t]$ which contains negative values. Using the point reflection we get

$$\begin{aligned}
\tilde{u}(x,t) &= \frac{1}{2}(\underbrace{\tilde{g}(x+t)}_{=g(x+t)} + \underbrace{\tilde{g}(x-t)}_{=-g(t-x)}) + \frac{1}{2}\int_{x-t}^{x+t} \tilde{h}(y)dy \\
&= \frac{1}{2}(g(x+t) - g(t-x)) + \frac{1}{2}\left(\int_{x-t}^{t-x} \tilde{h}(y)dy + \int_{t-x}^{x+t} \tilde{h}(y)dy\right) \\
&= \frac{1}{2}(g(x+t) - g(t-x)) + \frac{1}{2}\int_{t-x}^{x+t} h(y)dy
\end{aligned}$$

Although we used in the last step that an integral over an odd function on a symmetric intervall vanishes.

Thus, it total

$$u(x,t) = \begin{cases} \frac{1}{2}(g(x+t) + g(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} h(y)dy & ,0 \leq t \leq x \\ \frac{1}{2}(g(x+t) - g(t-x)) + \frac{1}{2}\int_{t-x}^{x+t} h(y)dy & ,0 \leq x \leq t \end{cases}.$$

This describes waves moving on $[0,\infty)$ where waves moving towards $x=0$ get reflected. This is prescribed using point reflection, i.e. the waves moves through $x=0$ and gets inverted.

2 Spherical Means of the Wave equation

The methods of spherical means is a technique that transforms the initial value problem of the wave equation on $\mathbb{R}^n$ into a lower dimensional problem by using the translation invariance and local rotaional averaging. Define $x \in \mathbb{R}^n, t \geq 0, r > 0$:

$$U(x,r,t) := \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u(y,t)d\sigma(y) =: \fint_{\partial B_r(x)} u(y,t)d\sigma(y),$$

where $\fint$ denotes the average value and ω_n the volume of the n-dimensional unit ball. Analogously we define

$$G(x,r) := \fint_{\partial B_r(x)} g(y)d\sigma(y), \quad H(x,r) := \fint_{\partial B_r(x)} h(y)d\sigma(y).$$

Lemma 2.1

If $u \in C^m(\mathbb{R}^n \times \mathbb{R}_0^+)$ is a m-times continuously differentiable solution of the initial value problem

$$\frac{\partial^2}{\partial t^2}u - \Delta u = 0, \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+, \quad u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}^n,$$

then the spherical mean $(r,t) \mapsto U(x,r,t)$ for fixed $x \in \mathbb{R}^n$ is a m-times differntiable function that satisfies the initial value problem (Euler-Poisson-Darboux equation):

$$\begin{aligned}
&\frac{\partial^2}{\partial t^2}U(x,r,t) - \frac{\partial^2}{\partial r^2}U(x,r,t) - \frac{n-1}{r}\frac{\partial}{\partial r}U(x,r,t) = 0, \quad (r,t) \in \mathbb{R}^+ \times \mathbb{R}^+, \\
&U(x,r,0) = G(x,r), \quad \frac{\partial}{\partial t}U(x,r,0) = H(x,r), \quad r \in \mathbb{R}^+.
\end{aligned}$$

Proof. We want to show that the equation for u holds, if u satisfies this initial value problem. Further, we need to show that all summands exist if $r \to 0$.

- We begin with

$$\begin{aligned} U(x,r,t) &= \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u(y,t)d\sigma(y) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_1(0)} u(x+rz,t)r^{n-1}d\sigma(z) \\ &= \frac{1}{n\omega_n} \int_{\partial B_1(0)} u(x+rz,t)d\sigma(z). \end{aligned}$$

where we see that the integral becomes independent of t are r. It follows

$$\begin{aligned} \frac{\partial}{\partial r} U(x,r,t) &= \frac{1}{n\omega_n} \int_{\partial B_1(0)} \frac{\partial}{\partial r} u(x+ry,t)d\sigma(y) = \frac{1}{n\omega_n} \int_{\partial B_1(0)} \nabla_x u(x+ry,t) \cdot y d\sigma(y), \\ &= \frac{1}{n\omega_n} \frac{r^n}{r^n} \int_{B_1(0)} \nabla \cdot \nabla u(x+ry,t)d^n y = \frac{r^n}{n} \fint_{B_1(0)} \nabla \cdot \nabla u(x+ry,t)d^n y \\ &= \frac{r}{n} \fint_{B_r(x)} \Delta u(y,t)d^n y. \end{aligned}$$

The summand in the equation exists, due to

$$\lim_{r\to 0} \frac{n-1}{r} \frac{\partial}{\partial r} U(x,r,t) = \lim_{r\to 0} \frac{n-1}{n} \fint_{B_r(x)} \Delta u(y,t)d^n y = \frac{n-1}{n} \Delta u(x,t).$$

- Next, we have

$$\begin{aligned} \frac{\partial^2}{\partial r^2} U(x,r,t) &= \frac{\partial}{\partial r} \frac{r}{n} \fint_{B_r(x)} \Delta u(y,t)d^n y \\ &= \frac{\partial}{\partial r} \frac{r}{n\omega_n r^n} \int_{B_r(x)} \Delta u(y,t)d^n y = \frac{\partial}{\partial r} \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u(y,t)d^n y \\ &= \frac{-(n-1)}{n\omega_n r^n} \int_{B_r(x)} \Delta u(y,t)d^n y + \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \frac{\partial}{\partial r} \Delta u(y,t)d^n y \\ &= \frac{1-n}{n\omega_n r^n} \int_{B_r(x)} \Delta u(y,t)d^n y + \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} \Delta u(y,t)d\sigma(y) \\ &= \left(\frac{1}{n} - 1\right) \fint_{B_r(x)} \Delta u(y,t)d^n y + \fint_{\partial B_r(x)} \Delta u(y,t)d\sigma(y). \end{aligned}$$

And in the limit we have

$$\begin{aligned} &\lim_{r\to 0} \frac{\partial^2}{\partial r^2} U(x,r,t) = \lim_{r\to 0} \left(\left(\frac{1}{n} - 1\right) \fint_{B_r(x)} \Delta u(y,t)d^n y + \fint_{\partial B_r(x)} \Delta u(y,t)d\sigma(y) \right) \\ &= \left(\frac{1}{n} - 1\right) \Delta u(x,t) + \Delta u(x,t) = \frac{1}{n} \Delta u(x,t), \end{aligned}$$

which exists.

- Finally, due to the fact, that u is the solution of the initial value problem stated on $\mathbb{R}^n \times \mathbb{R}^+$ in the theorem, we have

$$\begin{aligned}
\frac{\partial}{\partial r} r^{n-1} \frac{\partial}{\partial r} U(x,r,t) &= \frac{\partial}{\partial r} r^{n-1} \frac{r}{n} \fint_{B_r(x)} \Delta u(y,t) d^n y \\
&= \frac{\partial}{\partial r} \frac{r^n}{n\omega_n r^n} \int_{B_r(x)} \Delta u(y,t) d^n y = \frac{\partial}{\partial r} \frac{1}{n\omega_n} \int_{B_r(x)} \Delta u(y,t) d^n y \\
&= \frac{1}{n\omega_n} \int_{\partial B_r(x)} \Delta u(y,t) d\sigma(y) \\
&= \frac{1}{n\omega_n} \int_{\partial B_r(x)} \frac{\partial^2}{\partial t^2} u(y,t) d\sigma(y) = r^{n-1} \frac{\partial^2}{\partial t^2} \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u(y,t) d\sigma(y) \\
&= r^{n-1} \frac{\partial^2}{\partial t^2} U(x,r,t).
\end{aligned}$$

This is equivalent to

$$\begin{aligned}
& r^{n-1} \frac{\partial^2}{\partial t^2} U(x,r,t) = (n-1) r^{n-2} \frac{\partial}{\partial r} U(x,r,t) + r^{n-1} \frac{\partial^2}{\partial r^2} U(x,r,t) \\
\iff \quad & \frac{\partial^2}{\partial t^2} U(x,r,t) = \frac{n-1}{r} \frac{\partial}{\partial r} U(x,r,t) + \frac{\partial^2}{\partial r^2} U(x,r,t).
\end{aligned}$$

□

3 Solution in 3 Dimensions

We will see that the wave equation can be transformed into a one dimensional wave equation, if the dimension is odd. However, if the dimension is even, this is not possible. We start with the three dimensional case $n = 3$. Fix $x \in \mathbb{R}^3$. Consider the initial value problem for the spherical mean

$$\begin{aligned}
& \frac{\partial^2}{\partial t^2} U(x,r,t) - \frac{\partial^2}{\partial r^2} U(x,r,t) - \frac{2}{r} \frac{\partial}{\partial r} U(x,r,t) = 0, \quad (r,t) \in \mathbb{R}^+ \times \mathbb{R}^+, \\
& U(x,r,0) = G(x,r), \quad \frac{\partial}{\partial t} U(x,r,0) = H(x,r), \quad r \in \mathbb{R}^+.
\end{aligned}$$

By substiuting $\tilde{U}(x,r,t) := rU(x,r,t)$ we get

$$\begin{aligned}
& \frac{\partial^2}{\partial t^2} \frac{\tilde{U}}{r} - \frac{\partial^2}{\partial r^2} \frac{\tilde{U}}{r} - \frac{2}{r} \frac{\partial}{\partial r} \frac{\tilde{U}}{r} = \frac{1}{r} \frac{\partial^2}{\partial t^2} \tilde{U} - \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} \tilde{U} - \frac{\tilde{U}}{r^2} \right) - \frac{2}{r} \left(\frac{1}{r} \frac{\partial}{\partial r} \tilde{U} - \frac{\tilde{U}}{r^2} \right) \\
&= \frac{1}{r} \frac{\partial^2}{\partial t^2} \tilde{U} - \left(\frac{-1}{r^2} \frac{\partial}{\partial r} \tilde{U} + \frac{1}{r} \frac{\partial^2}{\partial r^2} \tilde{U} - \frac{1}{r^2} \frac{\partial}{\partial r} \tilde{U} + \frac{2}{r^3} \tilde{U} \right) + \frac{2}{r^3} \tilde{U} - \frac{2}{r^2} \frac{\partial}{\partial r} \tilde{U} \\
&= \frac{1}{r} \left(\frac{\partial^2}{\partial t^2} \tilde{U} - \frac{\partial^2}{\partial r^2} \tilde{U} \right) = 0 \\
\iff \quad & \frac{\partial^2}{\partial t^2} \tilde{U} - \frac{\partial^2}{\partial r^2} \tilde{U} = 0
\end{aligned}$$

Due to $\lim_{r \to 0} U(x,r,t) = u(x,t)$ it follows

$$\tilde{U}(x,0,t) = \lim_{r \to 0} rU(x,r,t,) = 0 \cdot u(x,t) = 0.$$

We now get an additional condition for the transformed initial value problem:

$$\frac{\partial^2}{\partial t^2}\tilde{U} - \frac{\partial^2}{\partial r^2}\tilde{U} = 0, \quad \text{on } \{x\} \times \mathbb{R}^+ \times \mathbb{R}^+, \quad \tilde{U}(x,0,t) = 0, \quad t \in \mathbb{R}_0^+,$$
$$\tilde{U}(x,r,0) = rG(x,r) =: \tilde{G}(x,r), \quad \frac{\partial}{\partial t}\tilde{U}(x,r,0) = rH(x,r) =: \tilde{H}(x,r), \quad r \in \mathbb{R}^+.$$

This is the initial value problem of the last section whoose solution is given by

$$\tilde{U}(x,r,t) = \frac{1}{2}(\tilde{G}(x,r+t) - \tilde{G}(x,t-r)) + \frac{1}{2}\int_{t-r}^{r+t} \tilde{H}(x,s)ds, \quad 0 \leq r \leq t.$$

Note that we only consider the case $0 \leq r \leq t$, because we are interested in $r \searrow 0$, due to the following: Using the continuity of $r \mapsto \tilde{U}(x,r,t)$ we get

$$u(x,t) = \lim_{r \searrow 0} U(x,r,t) = \lim_{r \searrow 0} \frac{\tilde{U}(x,r,t)}{r}.$$

Thus we can recover the solution for the 3D problem as the limit $r \searrow 0$: For all $x \in \mathbb{R}^3, t > 0$:

$$\begin{aligned}
u(x,t) &= \lim_{r \searrow 0} \frac{1}{2}\left(\frac{\tilde{G}(x,t+r) - \tilde{G}(x,t-r) + \tilde{G}(x,t) - \tilde{G}(x,t)}{r}\right) + \lim_{r \searrow 0} \frac{1}{2r}\int_{t-r}^{t+r} \tilde{H}(x,s)ds \\
&= \lim_{r \searrow 0} \frac{1}{2}\left(\frac{\tilde{G}(x,t+r) - \tilde{G}(x,t)}{r} - \frac{\tilde{G}(x,t-r) - \tilde{G}(x,t)}{r}\right) + \lim_{r \searrow 0} \frac{1}{2r}\int_{t-r}^{t+r} \tilde{H}(x,s)ds \\
&= \frac{\partial}{\partial t}\tilde{G}(x,t) + \tilde{H}(x,t) \\
&= \frac{\partial}{\partial t}(tG(x,t)) + tH(x,t) \\
&= G(x,t) + t\frac{\partial}{\partial t}G(x,t) + tH(x,t) \\
&= \fint_{\partial B_t(x)} g(y)d\sigma(y) + t\frac{\partial}{\partial t}\fint_{\partial B_t(x)} g(y)d\sigma(y) + t\fint_{\partial B_t(x)} h(y)d\sigma(y) \\
&= t\frac{\partial}{\partial t}\fint_{\partial B_1(0)} g(x+tz)d\sigma(z) + \fint_{\partial B_t(x)} (th(y) + g(y))d\sigma(y) \\
&= t\fint_{\partial B_1(0)} \nabla_x g(x+tz) \cdot z d\sigma(z) + \fint_{\partial B_t(x)} (th(y) + g(y))d\sigma(y) \\
&= t\fint_{\partial B_t(x)} \nabla_z g(z) \cdot \frac{z-x}{t} d\sigma(z) + \fint_{\partial B_t(x)} (th(y) + g(y))d\sigma(y) \\
&= \fint_{\partial B_t(x)} th(y) + g(y) + \nabla_y g(y) \cdot (y-x)d\sigma(y).
\end{aligned}$$

Also called Kirchhoffs formula. Note that the solution at (x,t) depends only on the values of g and h on the sphere $\partial B_t(x)$, i.e. information propagates with speed 1 in all directions.

4 Solution in 2 Dimensions

In two dimensions the Euler-Poisson-Darboux equation cannot be transferred into the one dimensional wave equation. We present a method to transfer the two dimensional initial value problem into a three dimensional

one, by choosing initial values that only depend on x_1 and x_2 but not on x_3.
Let $\bar{u}$ be on $\mathbb{R}^3 \times \mathbb{R}_0^+$ a solution of the initial value problem

$$\frac{\partial^2}{\partial t^2}\bar{u} - \Delta \bar{u} = 0, \quad \text{on } \mathbb{R}^3 \times \mathbb{R}^+,$$
$$\bar{u}(x,0) = g(\bar{x}), \quad \frac{\partial}{\partial t}\bar{u}(x,0) = h(\bar{x}), \quad x \in \mathbb{R}^3,$$

where $\bar{x} := (x_1, x_2) \in \mathbb{R}^2$.
Note that for a arbitary function f that only depends on $\bar{x}$ it holds

$$\frac{\partial}{\partial x_3} \fint_{\partial B_r(x)} f(y) d\sigma(y) = \frac{\partial}{\partial x_3} \fint_{\partial B_r(0)} f(y+x) d\sigma(y) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(0)} \underbrace{\frac{\partial}{\partial x_3} f(y+x)}_{=0} d\sigma(y) = 0,$$

i.e. the mean over $\partial B_r(x)$ of f is only dependent on $\bar{x}$ and r.
Because $\bar{u}$ is by definition independent of x_3 we have

$$\Delta_{\mathbb{R}^3}\bar{u} = \frac{\partial^2}{\partial x_1^2}\bar{u} + \frac{\partial^2}{\partial x_2^2}\bar{u} + \underbrace{\frac{\partial^2}{\partial x_3^2}\bar{u}}_{=0} = \Delta_{\mathbb{R}^2}\bar{u},$$

Thus $\frac{\partial^2}{\partial t^2}\bar{u} - \Delta_{\mathbb{R}^3}\bar{u} = 0 = \frac{\partial^2}{\partial t^2}\bar{u} - \Delta_{\mathbb{R}^2}\bar{u}$. Thus, if u is the solution of the two dimensional initial value problem, then $\bar{u}(x,t) = u(\bar{x},t)$, $x \in \mathbb{R}^3$. In particular for $x := (\bar{x}, 0)$ we have this equality. Using Kirchhoff's Formula it holds

$$\bar{u}(\underbrace{(\bar{x},0)}_{=x}, t) = \fint_{\partial B_t(x)} th(y) + g(y) d\sigma(y) + \fint_{\partial B_t(x)} \nabla_y g(y) \cdot (y-x) d\sigma(y).$$

We now want to transform this over to integrals on the two dimensional disc. For the Jacobi transformation, define the two parametrizations

$$\phi_+ : B_r^2(\bar{x}) \to \mathbb{R}^3, \quad z \mapsto (z, \gamma(z))^T := \begin{pmatrix} z_1 \\ z_2 \\ \sqrt{r^2 - |z-\bar{x}|^2} \end{pmatrix}$$
$$\phi_- : B_r^2(\bar{x}) \to \mathbb{R}^3, \quad z \mapsto (z, -\gamma(z))^T := \begin{pmatrix} z_1 \\ z_2 \\ -\sqrt{r^2 - |z-\bar{x}|^2} \end{pmatrix}.$$

The equator $\{y \in B_r^3(x) \mid y_3 = 0\}$ has measure zero and can thus be neglected. The Jacobi Matrix is

$$D\phi_+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -\frac{1}{2}2(z_1 - x_1)(r^2 - |z-\bar{x}|^2)^{-1/2} & -\frac{1}{2}2(z_2 - x_2)(r^2 - |z-\bar{x}|^2)^{-1/2} \end{pmatrix}$$

and

$$D\phi_- = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{1}{2}2(z_1 - x_1)(r^2 - |z-\bar{x}|^2)^{-1/2} & \frac{1}{2}2(z_2 - x_2)(r^2 - |z-\bar{x}|^2)^{-1/2} \end{pmatrix}$$

then

$$(D\phi_+)^T D\phi = \begin{pmatrix} 1 + (z_1 - x_1)^2 (r^2 - |z-\bar{x}|^2)^{-1} & (z_1 - x_1)(z_2 - x_2)(r^2 - |z-\bar{x}|^2)^{-1} \\ (z_1 - x_1)(z_2 - x_2)(r^2 - |z-\bar{x}|^2)^{-1} & 1 + (z_2 - x_2)^2 (r^2 - |z-\bar{x}|^2)^{-1} \end{pmatrix} = (D\phi_-)^T D\phi_-,$$

which leads to

$$\det((D\phi_+)^T D\phi) = \left(1 + \frac{(z_1 - x_1)^2}{r^2 - |z - \bar{x}|^2}\right)\left(1 + \frac{(z_2 - x_2)^2}{r^2 - |z - \bar{x}|^2}\right) - \left(\frac{(z_1 - x_1)^2 (z_2 - x_2)^2}{(r^2 - |z - \bar{x}|^2)^2}\right)$$
$$= 1 + \frac{(z_1 - x_1)^2 + (z_2 - x_2)^2}{r^2 - |z - \bar{x}|^2} = \frac{r^2}{r^2 - |z - \bar{x}|^2}.$$

This now implies that for any f

$$\begin{aligned}
&⨍_{\partial B_t^3(x)} f(z)d\sigma(z) = ⨍_{\partial B_t^3(x)\{y>0\}} f(z)d\sigma(z) + ⨍_{\partial B_t^3(x)\{y<0\}} f(z)d\sigma(z) \\
&= \frac{1}{4\pi t^2}\int_{B_t^2(x)} f(z)\sqrt{\det((D\phi_+)^T D\phi_+)}d^2z + \frac{1}{4\pi t^2}\int_{B_t^2(x)} f(z)\sqrt{\det((D\phi_-)^T D\phi_-)}d^2z \\
&= 2\frac{1}{4\pi t^2}\int_{B_t^2(x)} f(z)\frac{t}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z = \frac{1}{2\pi t^2}\int_{B_t^2(x)} f(z)\frac{t}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z \\
&= \frac{1}{2\pi t}\int_{B_t^2(x)} f(z)\frac{1}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z = \frac{t}{2}⨍_{B_t^2(x)} f(z)\frac{1}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z
\end{aligned}$$

Applying this now to the Kirchhoff's Formula we get for $x := (\bar{x}, 0)$

$$\begin{aligned}
u(\bar{x}, t) = \bar{u}(x, t) &= ⨍_{\partial B_t^3(x)} th(y) + g(y)d\sigma(y) + ⨍_{\partial B_t^3(x)} \nabla_y g(y)\cdot(y - x)d\sigma(y) \\
&= \frac{t}{2}⨍_{B_t^2(\bar{x})} \frac{th(z) + g(z)}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z + \frac{t}{2}⨍_{B_t^2(\bar{x})} \frac{\nabla_z g(z)\cdot(z - \bar{x})}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z \\
&= \frac{t}{2}⨍_{B_t^2(\bar{x})} \frac{th(z) + g(z) + \nabla_z g(z)\cdot(z - \bar{x})}{\sqrt{t^2 - |z - \bar{x}|^2}}d^2z
\end{aligned}$$

In three dimensions we had that the solution only depended on $\partial B_t(x)$. This implied that information triversion speed is 1. In the two dimensional case we now have the solution depending on the entire ball $B_t(\bar{x})$, which yields that the information speed is everything from 0 to 1.

Further, we call the method of deriving the solution of an initial value problem, by transforming the problem onto an initial value problem in a higher dimensional space, method of descent. In that higher dimensional space the initial values do not depend on all coordinates. We need to show that the solution also deos not depend on these coordinates. We just did this in two dimensions, i.e. $\bar{u}(x, t) = u(\bar{x}, t)$.
A natural question is weather we can apply this method to the solution of the initial value problem in one dimension.

5 Solution in odd Dimensions

In any odd dimension we can transform the Euler-Poisson-Darboux equation into the one dimensional wave equation.

Lemma 5.1

Let $\phi : \mathbb{R} \to \mathbb{R}$ be a $k+1$ times continuously differentiable function, then $\forall k \in \mathbb{N}$ it holds

$$i) \left(\frac{d}{dr}\right)^2 \left(\frac{1}{r}\frac{d}{dr}\right)^{k-1} r^{2k-1}\phi(r) = \left(\frac{1}{r}\frac{d}{dr}\right)^{k} r^{2k}\frac{d}{dr}\phi(r)$$

$$ii) \left(\frac{1}{r}\frac{d}{dr}\right)^{k-1} r^{2k-1}\phi(r) = \sum_{j=0}^{k-1} \beta_{k,j} r^{j+1}\frac{d^j}{dr^j}\phi(r), \quad \text{where } \beta_{k,j} \text{ do not depend on } \phi$$

$$iii) \beta_{k,0} = 1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2k-1) = \frac{(2k-1)!}{2 \cdot 4 \cdot \ldots \cdot (2k-2)} = \frac{(2k-1)!}{2^{k-1}(k-1)!}$$

Proof. Skipped □

Let the dimension be $n = 2k+1 \geq 3$ and $u \in C^{k+1}(\mathbb{R}^n \times \mathbb{R}_0^+)$ be the solution of the initial value problem

$$\frac{\partial^2}{\partial t^2}u - \Delta u = 0, \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+$$
$$U(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}^n.$$

Define

$$\tilde{U}(x,r,t) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}U(x,r,t)$$
$$\tilde{G}(x,t) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}G(x,r), \quad \tilde{H}(x,r) := \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}H(x,r).$$

Lemma 5.2

If $u \in C^{k+1}(\mathbb{R}^{2k+1} \times \mathbb{R}_0^+)$ solves the initial value problem of the wave equation, then $\tilde{U}(x,r,t)$ solves for all fixed $x \in \mathbb{R}^{2k+1}$ the following initial value problem

$$\frac{\partial^2}{\partial t^2}\tilde{U} - \frac{\partial^2}{\partial r^2}\tilde{U} = 0, \quad \text{on } \{x\} \times \mathbb{R}^+ \times \mathbb{R}^+, \quad \tilde{U}(x,0,t) = 0, \quad t \in \mathbb{R}_0^+$$
$$\tilde{U}(x,r,0) = \tilde{G}(x,r), \quad \frac{\partial}{\partial t}\tilde{U}(x,r,0) = \tilde{H}(x,r), \quad r \in \mathbb{R}^+.$$

Proof. Let U be the solution of the Euler-Poisson-Darboux equation in dimension $n = 2k+1$, then we have

$$\begin{aligned}
\frac{\partial^2}{\partial r^2}\tilde{U}(x,r,t) &\overset{Def.}{=} \frac{\partial^2}{\partial r^2}\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}U(x,r,t) \\
&\overset{i)}{=} \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k} r^{2k}\frac{\partial}{\partial r}U(x,r,t) \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} \left(\frac{2kr^{2k-1}}{r}\frac{\partial}{\partial r}U(x,r,t) + \frac{1}{r}r^{2k}\frac{\partial^2}{\partial r^2}U(x,r,t)\right) \\
&= \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}\left(\frac{2k}{r}\frac{\partial}{\partial r}U(x,r,t) + \frac{\partial^2}{\partial r^2}U(x,r,t)\right) \\
&\overset{(*)}{=} \left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}\frac{\partial^2}{\partial t^2}U(x,r,t) = \frac{\partial^2}{\partial t^2}\tilde{U}(x,r,t),
\end{aligned}$$

altough we used in $(*)$ that $2k = n-1$ by definition of the dimension. Finally, we need to check the boundary condition,i.e. to show that $\tilde{U}$ vanishes as $r \to 0$:

$$\begin{aligned}
\lim_{r\to 0}\tilde{U}(x,r,t) = \lim_{r\to 0}\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{k-1} r^{2k-1}U(x,r,t) &\overset{ii)}{=} \lim_{r\to 0}\sum_{j=0}^{k-1}\beta_{k,j}r^{j+1}\frac{\partial^j}{\partial r^j}U(x,r,t) \\
&= \sum_{j=0}^{k-1}\beta_{k,j}\cdot 0\cdot \underbrace{\frac{\partial^j}{\partial r^j}U(x,r,t)\big|_{r=0}}_{<\infty} = 0.
\end{aligned}$$

□

Thus, we have reduced the odd dimensional initial value problem into a one dimensional initial value problem on the intervall $[0,\infty)$. As we have already considered this problem, we know its solution

$$\tilde{U}(x,r,t) = \frac{1}{2}(\tilde{G}(x,t+r) - \tilde{G}(x,t-r)) + \frac{1}{2}\int_{t-r}^{t+r}\tilde{H}(x,s)ds.$$

With the relation

$$\begin{aligned}
\tilde{U}(x,r,t) = \sum_{j=0}^{k-1}\beta_{k,j}r^{j+1}\frac{\partial^j}{\partial r^j}U(x,r,t) &= \beta_{k,0}rU(x,r,t) + \mathcal{O}(r^2) \\
\iff \frac{\tilde{U}(x,r,t)}{\beta_{k,0}r} &= U(x,r,t) + \mathcal{O}(r) \\
\iff \lim_{r\to 0}\frac{\tilde{U}(x,r,t)}{\beta_{k,0}r} &= \lim_{r\to 0}U(x,r,t) = u(x,t),
\end{aligned}$$

which finally implies that

$$\begin{aligned}
u(x,t) = \frac{1}{\beta_{k,0}}\lim_{r\to 0}\frac{\tilde{U}(x,r,t)}{r} &= \frac{2^{k-1}(k-1)!}{(2k-1)!}\lim_{r\to 0}\left(\frac{1}{2r}(\tilde{G}(x,t+r) - \tilde{G}(x,t-r)) + \frac{1}{2r}\int_{t-r}^{t+r}\tilde{H}(x,s)ds\right) \\
&= \frac{2^{k-1}(k-1)!}{(2k-1)!}\left(\frac{\partial}{\partial t}\tilde{G}(x,t) + \tilde{H}(x,t)\right).
\end{aligned}$$

Theorem 5.3

For odd $n \geq 3$ the solution of the initial value problem

$$\frac{\partial^2}{\partial t^2}u - \Delta u = 0, \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+, \quad u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), x \in \mathbb{R}^n,$$

with $g \in C^{\frac{n+3}{2}}(\mathbb{R}^n)$, $h \in C^{\frac{n+1}{2}}(\mathbb{R}^n)$ the solution is given by

$$u(x,t) = \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}\left(\frac{\partial}{\partial t}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} t^{n-2} \fint_{\partial B_t(x)} g(y)d\sigma(y) + \left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} t^{n-2} \fint_{\partial B_t(x)} h(y)d\sigma(y)\right).$$

Proof. We can split the above initial value problem into the following subproblems

$$\begin{cases} \frac{\partial^2}{\partial t^2}u - \Delta u = 0, & \text{on } \mathbb{R}^n \times \mathbb{R}^+ \\ u(x,0) = 0, & x \in \mathbb{R}^n, \\ \frac{\partial}{\partial t}u(x,0) = h(x), x \in \mathbb{R}^n, \end{cases} \quad \text{and} \quad \begin{cases} \frac{\partial^2}{\partial t^2}u - \Delta u = 0, & \text{on } \mathbb{R}^n \times \mathbb{R}^+ \\ u(x,0) = h(x), & x \in \mathbb{R}^n, \\ \frac{\partial}{\partial t}u(x,0) = 0, x \in \mathbb{R}^n, \end{cases}$$

I.e. if u and v are the solutions to the first and second initial value problem respectively, then their sum $u+v$ is the solution to the initial value problem in the theorem.
Note that if v is the solution to the first problem, then $u(x,t) := \frac{\partial}{\partial t}v(x,t)$ is the solution to the second problem:

$$\begin{aligned} \frac{\partial^2}{\partial t^2}u - \Delta u &= \frac{\partial}{\partial t}\left(\frac{\partial^2}{\partial t^2}v - \Delta v\right) = 0 \\ u(x,0) &= \frac{\partial}{\partial t}v(x,0) = h(x) \\ \frac{\partial}{\partial t}u(x,0) &= \frac{\partial^2}{\partial t^2}v(x,0) = \Delta v(x,0) = 0. \end{aligned}$$

Thus, we only need to focus on the first initial value problem. Using that $n-1 = 2k \iff k-1 = \frac{n-3}{2}$, we get with $g = 0$:

$$\begin{aligned}
\frac{\partial^2}{\partial t^2}u(x,t) &= \frac{\partial^2}{\partial t^2}\frac{2^{k-1}(k-1)!}{(2k-1)!}\tilde{H}(x,t) = \frac{\partial^2}{\partial t^2}\frac{2^{k-1}(k-1)!}{(2k-1)!}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{k-1} t^{2k-1}H(x,t) \\
&\overset{i)}{=} \frac{2^{k-1}(k-1)!}{(2k-1)!}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{k} t^{2k}\frac{\partial}{\partial t}H(x,t) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} t^{n-1}\frac{\partial}{\partial t}\frac{1}{n\omega_n t^{n-1}}\int_{\partial B_t(x)} h(y)d\sigma(y) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} t^{n-1}\frac{\partial}{\partial t}\frac{1}{n\omega_n t^{n-1}}\int_{\partial B_1(0)} h(x+tz)t^{n-1}d\sigma(z) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} t^{n-1}\frac{1}{n\omega_n}\int_{\partial B_1(0)} \frac{\partial}{\partial t}h(x+tz)d\sigma(z) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} t^{n-1}\frac{1}{n\omega_n t^{n-1}}\int_{\partial B_1(0)} \nabla_x t^{n-1}h(x+tz)\cdot z d\sigma(z) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} t^{n-1}\frac{1}{n\omega_n t^{n-1}}\int_{\partial B_t(x)} \nabla h(z)\cdot \underbrace{\frac{z-x}{t}}_{=N(z)} d\sigma(z) \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} \left(\frac{1}{t}\frac{\partial}{\partial t}\right)\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{-1}\frac{1}{n\omega_n}\int_{B_t(x)} \Delta h(z)d^n z \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} \frac{1}{n\omega_n t}\frac{\partial}{\partial t}\int_{B_t(x)} \Delta h(z)d^n z \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} \frac{1}{n\omega_n t}\int_{\partial B_t(x)} \Delta h(z)d^n z \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} t^{n-2}\frac{1}{n\omega_n t^{n-1}}\int_{\partial B_t(0)} \Delta h(z+x)d^n z \\
&= \frac{2^{\frac{n-3}{2}}(\frac{n-3}{2})!}{(n-2)!}t\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} t^{n-2}\fint_{\partial B_t(0)} \Delta h(z+x)d^n z = \Delta u(x,t).
\end{aligned}$$

Finally, it only remains to show the boundary limit for $t \to 0$. First, notice that

$$\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-3}{2}} t^{n-2} = \left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-1}{2}} (n-2)t^{n-4} \sim t^{n-2-2\frac{n-3}{2}} = t^1.$$

Using this we get that the solution is asymptotically equivalent wrt. t to :

$$u(x,t) \sim \frac{\partial}{\partial t}t\fint_{\partial B_t(x)} g(y)d\sigma(y) + t\fint_{\partial B_t(x)} h(y)d\sigma(y).$$

Thus we get

$$\begin{aligned}
u(x,0) &= \lim_{t\to 0}\left(\frac{\partial}{\partial t} t \fint_{\partial B_t(x)} g(y)d\sigma(y) + t\fint_{\partial B_t(x)} h(y)d\sigma(y)\right) = g(x)\\
\frac{\partial}{\partial t}u(x,0) &= \lim_{t\to 0}\frac{\partial}{\partial t}\left(\frac{\partial}{\partial t} t \fint_{\partial B_t(x)} g(y)d\sigma(y) + t\fint_{\partial B_t(x)} h(y)d\sigma(y)\right)\\
&= \lim_{t\to 0}\left(\frac{\partial^2}{\partial t^2} t \fint_{\partial B_t(x)} g(y)d\sigma(y) + \frac{\partial}{\partial t} t\fint_{\partial B_t(x)} h(y)d\sigma(y)\right)\\
&= 0 + \lim_{t\to 0}\left(t \fint_{\partial B_1(x)} \frac{\partial^2}{\partial t^2} g(ty)d\sigma(y) + \frac{\partial}{\partial t} t\fint_{\partial B_t(x)} h(y)d\sigma(y)\right)\\
&= \lim_{t\to 0}\left(t \fint_{\partial B_t(x)} \Delta g(y)d\sigma(y) + \frac{\partial}{\partial t} t\fint_{\partial B_t(x)} h(y)d\sigma(y)\right) = h(x).
\end{aligned}$$

□

It is visible that the solution only depends on the boundary $\partial B_t(x)$, i.e. the speed of propagation is 1. Additionally, u is at (x,t) $\frac{n-1}{2}$ times less differentiable than g and $\frac{n-3}{2}$ less differentiable than h.
In contrast the heat equation has infinite speed and regardless of the regularity of the initial value, the solution is for some $t > 0$ smooth.

6 Solution in even Dimension

Again, we use the method of descent meaning we formulate the n dimensional problem into a $n+1$ dimensional problem with initial values g and h on $\mathbb{R}^{n+1}$ not depending on x_{n+1}. This immediatly implies that

$$\frac{\partial}{\partial x_{n+1}} g(x) = 0, \quad \frac{\partial}{\partial x_{n+1}} h(x) = 0.$$

Their partial derivates of the integrals on the boundary vanish as well as a result of that:

$$\frac{\partial}{\partial x_{n+1}} \fint_{\partial B_t(x)} g(y)d\sigma(y) = \fint_{B_t(0)} \frac{\partial}{\partial x_{n+1}} g(x+y)d\sigma(y) = 0.$$

Because the above holds for all $t > 0$ it follows that

$$\frac{\partial}{\partial t}\frac{\partial}{\partial x_{n+1}} \fint_{\partial B_t(x)} g(y)d\sigma(y)$$

Let u be the solution of the $2k$ dimensional problem and v the solution to the $2k+1$ dimensional problem, then with the mapping $i : \mathbb{R}^n \to \mathbb{R}^{n+1}, x \mapsto (x,0)$ we have

$$u(x,t) \overset{independence}{=} (v\circ i)(x,t) = v(i(x),t) = v((x,0),t),$$

i.e. we can obtain the solution in $2k$ dimensions by solving the $2k+1$ problem and evaluating it at $x_{n+1}=0$. Similar to before, define the parametrization

$$\phi_+ : B_r^n(\bar{x}) \to \mathbb{R}^{n+1}, \quad z \mapsto (z, \gamma(z))^T := \begin{pmatrix} z_1 \\ \dots \\ z_n \\ \sqrt{r^2 - |z-\bar{x}|^2} \end{pmatrix}$$

$$\phi_- : B_r^n(\bar{x}) \to \mathbb{R}^{n+1}, \quad z \mapsto (z, -\gamma(z))^T := \begin{pmatrix} z_1 \\ \dots \\ z_n \\ -\sqrt{r^2 - |z-\bar{x}|^2} \end{pmatrix}.$$

Its Jacobian is then

$$D\phi_+ = \begin{pmatrix} & Id_{n\times n} & \\ (x_1-z_1)\gamma(z)^{-1} & \cdot & (x_n - z_n)\gamma(z)^{-1} \end{pmatrix}$$

and $\det((D\phi_+)^T D\phi_+) = 1 + \gamma(z)^{-2}\sum_{i=1}^n (z_i - x_i)^2 = \frac{r^2}{\gamma(z)^2}$.
For $\bar{f} : \mathbb{R}^{n+1} \to \mathbb{R}$ and $f := \bar{f} \circ \phi_\pm : \mathbb{R}^n \to \mathbb{R}$ we have

$$\begin{aligned}
&\fint_{\partial B_t^{n+1}(x)} \bar{f}(y)d\sigma(y) = \frac{1}{(n+1)\omega_{n+1}t^n}\int_{\partial B_t^{n+1}(x)} \bar{f}(y)d\sigma(y) \\
&= \frac{1}{(n+1)\omega_{n+1}t^n}\left(\int_{B_t^n(x)\cap\{y_{n+1}>0\}} f(y)\sqrt{\det(D\phi_+)^T D\phi_+)}d^n y + \int_{B_t^n(x)\cap\{y_{n+1}<0\}} f(y)\sqrt{\det(D\phi_-)^T D\phi_-)}d^n y\right) \\
&= \frac{1}{(n+1)\omega_{n+1}t^n}\left(\int_{B_t^n(x)} f(y)\sqrt{\det(D\phi_+)^T D\phi_+)}d^n y + \int_{B_t^n(x)} f(y)\sqrt{\det(D\phi_-)^T D\phi_-)}d^n y\right) \\
&= \frac{2}{(n+1)\omega_{n+1}t^n}\int_{B_t^n(x)} f(y)\frac{t}{\gamma(y)}d^n y \\
&= \frac{2\omega_n t^n}{(n+1)\omega_{n+1}t^n}\fint_{B_t^n(x)} f(y)\frac{t}{\gamma(y)}d^n y \\
&= \frac{2\omega_n t}{(n+1)\omega_{n+1}}\fint_{B_t^n(x)} f(y)\frac{1}{\sqrt{t^2-|y-x|^2}}d^n y
\end{aligned}$$

Theorem 6.1

Let n be positive and even, $g \in C^{\frac{n+4}{2}}(\mathbb{R}^n)$ and $h \in C^{\frac{n+2}{2}}(\mathbb{R}^n)$, then the solution ot the initial value problem

$$\begin{aligned}
&\frac{\partial^2}{\partial t^2}u - \Delta u = 0, \quad \mathbb{R}^n \times \mathbb{R}^+, \\
&u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}^n,
\end{aligned}$$

is given at $(x,t) \in \mathbb{R}^n \times \mathbb{R}_0^+$ by

$$u(x,t) = \frac{1}{2^{n/2}\frac{n}{2}!}\left(\frac{\partial}{\partial t}\left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}} t^n \fint_{B_t(x)} \frac{g(z)}{\sqrt{t^2-|z-x|^2}} d^n z + \left(\frac{1}{t}\frac{\partial}{\partial t}\right)^{\frac{n-2}{2}} t^n \fint_{B_t(x)} \frac{h(z)}{\sqrt{t^2-|z-x|^2}} d^n z\right).$$

Proof. Unclear ??? □

7 Inhomogeneous Wave Equation

We interpret in the inhomogeneous wave equation f as an external force:

$$\frac{\partial^2}{\partial t^2}u - \Delta u = f, \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+$$
$$u(x,0) = 0, \quad \frac{\partial}{\partial t}u(x,0) = 0, \quad x \in \mathbb{R}^n.$$

We can interpret the solution to the inhomogeneous wave eqation as the integral of the solution to the homogeneous wave eqaution with the external force reflected in the initial condition:

$$\frac{\partial^2}{\partial t^2}v_s - \Delta v_s = 0, \quad \text{on } \mathbb{R}^n \times [s,\infty)$$
$$v_s(x,s) = 0, \quad \frac{\partial}{\partial t}v_s(x,s) = f(x,s), \quad x \in \mathbb{R}^n$$

and $u(x,t) = \int_0^t v_s(x,t)ds$. This true because with the Leipniz rule:

$$\frac{\partial^2}{\partial t^2}u(x,t) = \frac{\partial}{\partial t}\left(\underbrace{v_t(x,t)}_{=0} + \int_0^t \frac{\partial}{\partial t}v_s(x,t)ds\right) = \frac{\partial}{\partial t}v_t(x,t) + \int_0^t \frac{\partial^2}{\partial t^2}v_s(x,t)ds$$
$$= f(x,t) + \int_0^t \Delta v_s(x,t) = f(x,t) + \Delta u(x,t).$$

The sum of the solutions to the homoegenous and inhmomogenous wave equation solve by linearity of the differential operator the following initial value problem

$$\frac{\partial^2}{\partial t^2}u - \Delta u = f, \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+$$
$$u(x,0) = g(x), \quad \frac{\partial}{\partial t}u(x,0) = h(x), \quad x \in \mathbb{R}^n.$$

When determining the past from the present, define $v(x,\tau) := u(x,-t)$, where $\tau = -t$. Then

$$\frac{\partial}{\partial \tau}v(x,\tau) = -\frac{\partial}{\partial t}v(x,\tau), \quad \frac{\partial^2}{\partial \tau^2}v(x,\tau) = \frac{\partial^2}{\partial t^2}v(x,\tau).$$

This means that the PDE does not change, but the initial velocity does:

$$\frac{\partial^2}{\partial \tau^2}v - \Delta v = f(x,-\tau) \quad \text{on } \mathbb{R}^n \times \mathbb{R}^+$$
$$v(x,0) = g(x), \quad \frac{\partial}{\partial \tau}v(x,0) = -h(x).$$

8 Energy Methods

Hyperbolic PDEs do not satisfy maximum principle???
The theorem for the maximum principle for elliptic second order PDEs applies to degenerate elliptic PDEs???

Theorem 8.1

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain, then the initial value problem

$$\begin{aligned} &\frac{\partial^2}{\partial t^2}u - \Delta u = f, \qquad \text{on } \Omega \times (0,T) \\ &u(x,t) = g(x,t), \qquad \text{on } \Omega \times \{0\} \text{ and } \partial\Omega \times (0,T) \\ &\frac{\partial}{\partial t}u(x,t) = h(x), \qquad \text{on } \Omega \times \{0\}, \end{aligned}$$

has a unique solution in $C^2(\Omega \times (0,T))$ and

$$\forall |\alpha| \leq 2: \quad \partial^\alpha u \text{ extends continuously to } \bar{\Omega} \times [0,T].$$

Proof. Let u be the solution of the initial value problem with $f = g = h = 0$ and define

$$e(t) := \frac{1}{2}\int_\Omega \left(\frac{\partial}{\partial t}u(x,t)\right)^2 + (\nabla u(x,t))^2 \, d^n x.$$

Then it follows

$$\begin{aligned} \frac{d}{dt}e(t) &= \int_\Omega \frac{\partial^2}{\partial t^2}u(x,t)\frac{\partial}{\partial t}u(x,t) + \underbrace{\frac{\partial}{\partial t}\nabla u(x,t)\nabla u(x,t)}_{=\nabla\partial_t u\nabla u = (\partial_t u \nabla u)' - \partial_t u \Delta u} \; d^n x \\ &= \int_\Omega \frac{\partial^2}{\partial t^2}u(x,t)\frac{\partial}{\partial t}u(x,t) + \frac{\partial}{\partial t}u(x,t)\Delta u(x,t) d^n x \\ &= \int_\Omega \frac{\partial}{\partial t}u(x,t)\left(\frac{\partial^2}{\partial t^2}u(x,t) + \Delta u(x,t)\right) d^n x = 0. \end{aligned}$$

Because we set $e(0) = 0$ and there is no change, it stays zero for all $t > 0$. □

9 Summary

We begin with the one dimensional case, wehre the differential operator factorizes itno the two transport operators $(\frac{\partial}{\partial t} - \frac{\partial}{\partial x})(\frac{\partial}{\partial t} + \frac{\partial}{\partial x})$. This implies that the solution is the sum of a wave moving to the right and one to the left, $F(x+t) + G(x-t)$. Adding n initial position and velocity to the problem yields the D'Alembert's formula. It shows the domain of dependence is $[x-t, x+t]$ confirming finite speed of propagation. We also extend this method to the half-line problem $\mathbb{R}^+$ using the method of reflection and extending u, g, h to odd functions.
Defining the wave eqaution in terms of spherical means establishes the Euler-Poisson-Darboux equation

$$\frac{\partial^2}{\partial t^2}U - \frac{\partial^2}{\partial r^2}U - \frac{n-1}{r}\frac{\partial}{\partial r}U = 0,$$

whcih transforms the PDE into a one dimensional one. In $n = 3$ substuting $\tilde{U} := rU$ transforms the Euler-Poisson-Darboux equation the one dimensional wave equation $\frac{\partial^2}{\partial t^2}\tilde{U} - \frac{\partial^2}{\partial r^2}\tilde{U} = 0$. Solving this using D'Alembert's formula and taking the limit $r \to 0$ yields Kirchhoffs formula. In this case ($n = 3$) the solution only depends on the sphere $\partial B_t(x)$, which means that in $3D$ propagation is on sharp wavefronts without a "wake".
For $n = 2$ we use the method of descent, meaning we formulate the $2D$ problem as a special case of the $3D$ problem, by setting the initial data nad solution such that is is independend from the third coordinate x_3. Using Kirchhoffs formula in $3D$ and transforming the integrals to $2D$ yields a formolation for the solution. However, it depends on the entire ball $B_t(x)$, i.e. propagation occurs including wakes.
Generalizing for odd dimensions, multipliying the spherical means by $\left(\frac{1}{r}\frac{\partial}{\partial r}\right)^{n-1} r^{2k-1}$ a specific form is derived for the solution.
Generalizing fir even dimensions, we do the same appraoch by setting initial data nad solution such taht it is independend from the $n + 1^{th}$ coordinate, giving a formula for the solution.
???.